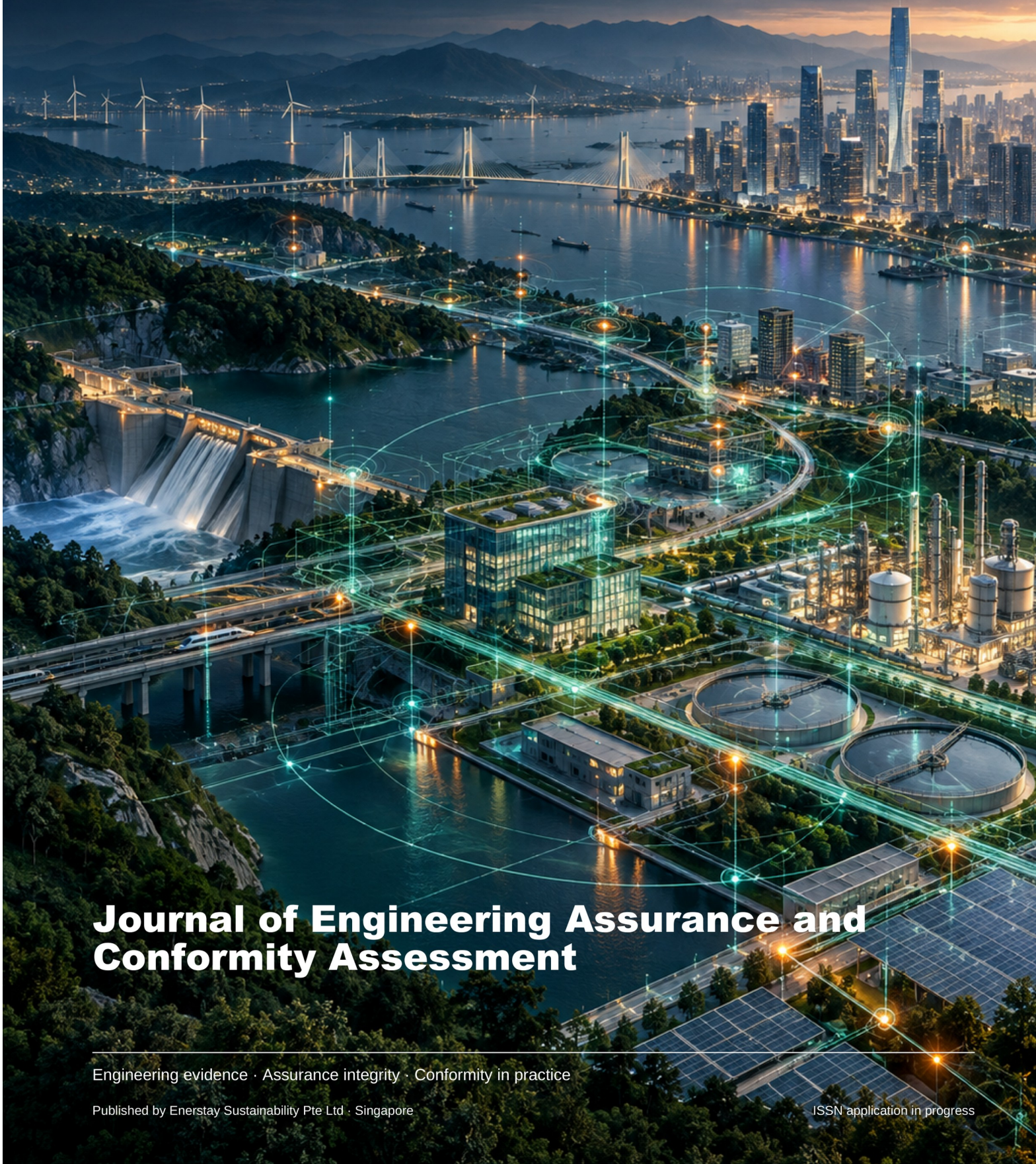
VOLUME 1 · ISSUE 1 PUBLISHED 1 NOVEMBER 2026

JEACA

engineering assurance
+ conformity assessment



Journal of Engineering Assurance and Conformity Assessment

Engineering evidence · Assurance integrity · Conformity in practice

Published by Enerstay Sustainability Pte Ltd · Singapore

ISSN application in progress

Publication and editorial information

First published online: 1 November 2026

JOURNAL TITLE	Journal of Engineering Assurance and Conformity Assessment (JEACA)
PUBLISHER	Enerstay Sustainability Pte. Ltd.
REGISTERED OFFICE	3791 Jalan Bukit Merah, #03-05, E-Centre @ Redhill, Singapore 159471
ISSUE DESIGNATION	VOLUME 1 · ISSUE 1 Published online 1 November 2026
FREQUENCY	Quarterly (4 issues per year)
WEBSITE	www.enerstaysustainability.com/jeaca/
ISSN	Application pending; no ISSN has been assigned or asserted
DOI	Not assigned; no article-level DOI is asserted

Editorial leadership

EDITOR-IN-CHIEF	Yuanzhe Li Enerstay Sustainability Pte. Ltd. 3791 Jalan Bukit Merah, #03-05, E-Centre @ Redhill, Singapore 159471 yuanzhe.li@enerstaysustainability.com · +65 8626 7896
EDITOR-IN-CHIEF	Micue Liu Hongci Enerstay Sustainability Pte. Ltd. 3791 Jalan Bukit Merah, #03-05, E-Centre @ Redhill, Singapore 159471 enquiries@enerstaysustainability.com · +65 8626 7896
ASSOCIATE EDITOR-IN-CHIEF	Haizhou Wang Enerstay Sustainability Pte. Ltd. 3791 Jalan Bukit Merah, #03-05, E-Centre @ Redhill, Singapore 159471 haizhou.wang@enerstaysustainability.com · +65 8626 7896
EDITOR	Yun Bai Enerstay Sustainability Pte. Ltd. 3791 Jalan Bukit Merah, #03-05, E-Centre @ Redhill, Singapore 159471 enquiries@enerstaysustainability.com · +65 8626 7896

Scope review and publication ethics

Aims and scope

JEACA publishes research, reviews, perspectives and applied methods at the intersection of engineering assurance, conformity assessment, environmental performance, digital measurement, carbon accounting, audit evidence and standards implementation.

Peer review

Submissions are assessed for relevance, methodological transparency, evidential sufficiency and reproducibility. This published issue does not assert review dates, reviewer identities or editorial decisions that are not recorded in the publisher files.

Research integrity

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Publication status

This is the first published online issue of JEACA. It was published on 1 November 2026 and contains five complete articles. No ISSN or DOI is asserted unless and until assigned.

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PUBLISHED 1 NOVEMBER 2026

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REVIEW ARTICLE

Microalgal Carbon Sequestration: A Systematic Review of Monitoring Approaches, Environmental Drivers, Biological Mechanisms, and Applied Systems

AUTHORS · Yuanzhe Li; WanLuoh Choo; Danqi Wang

PUBLISHED ONLINE · 1 NOVEMBER 2026

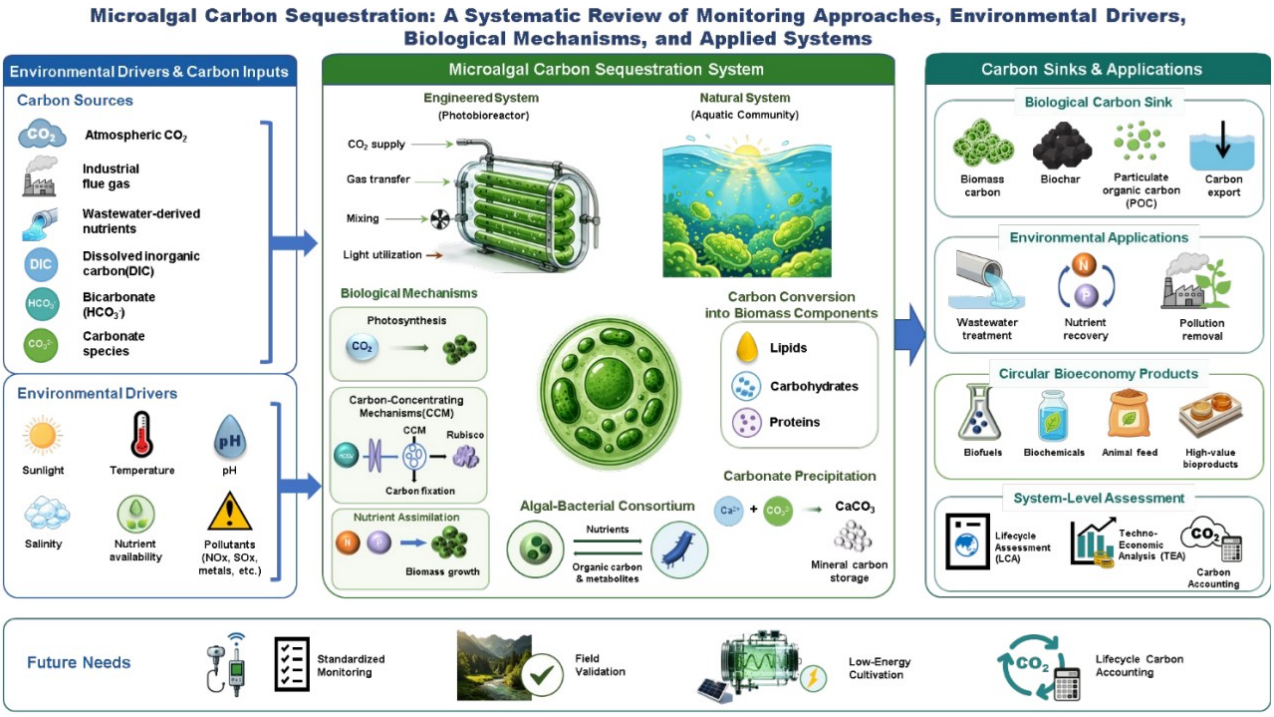
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ABSTRACT

Microalgae convert inorganic carbon into biomass and other carbon-containing products, while also supporting nutrient recovery and low-carbon bioproduct generation. This systematic review synthesizes evidence on microalgal carbon sequestration, with emphasis on monitoring indicators, environmental controls, biological mechanisms, cultivation systems, and applications in wastewater treatment, bioenergy, and circular resource use. Following screening, 109 studies were retained as the core evidence set because they explicitly addressed microalgae-related carbon sinks and reported methods or results for carbon monitoring, calculation, modelling, or performance assessment. Across these studies, carbon fixation and storage were shaped by species traits and cultivation context, particularly light, temperature, pH, nutrient supply, CO_2 availability, salinity, pollutant exposure, and reactor or ecosystem design. Common monitoring variables included biomass productivity, CO_2 removal, carbon-fixation rate, dissolved inorganic carbon, particulate organic carbon, photosynthetic efficiency, nutrient removal, and life-cycle or techno-economic metrics. Recent work has moved from single-species laboratory cultivation toward integrated systems, including photobioreactors, algal-bacterial consortia, wastewater-linked cultivation, flue-gas utilization, natural phytoplankton communities, and carbonatechemistry-based carbon management. The value of these systems depends on the fate of the fixed carbon—whether it remains in harvested biomass, enters products, or is released during downstream processing. When cultivation is paired with wastewater treatment and biomass valorization, fixed carbon can be routed into fuels, bioproducts, residual biomass, or other defined product streams. Durable carbon-sink contributions remain context-dependent and require full-boundary assessment of energy inputs, biomass fate, downstream processing, and long-term carbon storage. The evidence indicates that microalgal carbon-sink performance cannot be judged from CO_2 uptake alone. It also depends on whether fixed carbon is retained in biomass, transferred to durable products, mineralized, or rapidly returned to the atmosphere through downstream processing. Carbon-fixation rate, biomass fate, energy inputs, and product allocation should be reported within the same accounting boundary.

Keywords: *Microalgae; carbon sequestration; carbon fixation; carbon monitoring; photobioreactor; wastewater treatment; life-cycle assessment; circular bioeconomy*

Graphical Abstract



1. Introduction

1.1 The Global Challenge of Climate Change and Carbon Sinks

Rising greenhouse-gas concentrations remain a central driver of contemporary climate change. Atmospheric CO_2 continues to increase, with Mauna Loa observations exceeding 430 ppm, indicating that anthropogenic emissions remain well above levels compatible with near-term climate stabilization (NOAA GML, 2026; IPCC, 2023). Fossil-fuel combustion and land-use change, particularly deforestation, remain the dominant sources of this increase. Global carbon-budget estimates indicate that CO_2 emissions from fossil fuels and land-use change reached approximately 40.9 Gt CO_2 in 2023, underscoring the scale of mitigation required (Friedlingstein et al., 2023). International climate policy has therefore prioritized rapid emission reduction, while carbon-sink enhancement is discussed as a complementary strategy for residual emissions and delayed mitigation pathways. Carbon sinks take up more carbon than they release over a defined period, thereby slowing the accumulation of atmospheric CO_2.

Forests and grasslands remain central to this process because they store carbon in living biomass and soil organic matter (Yang et al., 2022; Sha et al., 2022). Interest has also expanded to biological carbon sinks outside conventional terrestrial systems. Microalgae are relevant in this context because many taxa grow rapidly, assimilate CO_2 photosynthetically, and generate biomass suitable for fuels, feed, nutraceuticals, or other bioproducts (Ahiahonu et al., 2023; Leong et al., 2021). Their cultivation can also be linked with wastewater treatment, allowing nitrogen and phosphorus removal to occur alongside CO_2 conversion into biomass (Abdelfattah et al., 2023; Sadvakasova et al., 2025). For this reason, microalgae should be evaluated as coupled carbon conversion and resource-recovery systems, rather than as stand-alone CO_2-capture organisms.

1.2 Microalgae: Emerging Frontiers in Carbon Sequestration

Microalgae contribute to biological carbon sequestration by converting dissolved or gaseous inorganic carbon into cellular and extracellular carbon pools. Through photosynthesis, these organisms assimilate CO_2 or bicarbonate and allocate fixed carbon to biomass, lipids, carbohydrates, pigments, extracellular products, or storage compounds. Under suitable cultivation conditions, microalgae can achieve high areal productivity and can be grown on non-arable land, saline water, wastewater, or engineered systems, reducing direct competition with conventional cropland (Ma et al., 2022; Daneshvar et al., 2022; Sadvakasova et al., 2025). Their relevance reflects both ecological tolerance and physiological plasticity.

Many taxa can grow in freshwater, seawater, brackish water, and wastewater, making them suitable for different carbon capture and resource-recovery systems (Abdelfattah et al., 2023; Ma et al., 2022). A key mechanism is the

CO_2-concentrating mechanism (CCM), which increases CO_2 availability around Rubisco and helps sustain carbon fixation in aquatic environments where dissolved CO_2 is often limited (Burlacot et al., 2022; Burlacot & Peltier, 2023). In Chlamydomonas reinhardtii, alternative photosynthetic electron pathways can help power the CCM by supporting proton gradients and bicarbonate-to-CO_2 conversion near the carboxylation site (Burlacot et al., 2022). Such variation in inorganic-carbon acquisition offers a basis for strain selection and process design. Light, salinity, nutrient supply, CO_2 concentration, gas transfer, and photobioreactor configuration all affect biomass production and carbon fixation (Banerjee et al., 2019; Ma et al., 2022; Sadvakasova et al., 2025). Flue-gas cultivation can provide a concentrated CO_2 source while linking carbon mitigation with biomass production (Ma et al., 2022; Mousavi et al., 2022). Wastewater integration adds nutrient removal and pollutant control to this framework (Abdelfattah et al., 2023; Zheng et al., 2023).

In this context, microalgae function as flexible biological platforms that connect CO_2 capture with wastewater treatment, flue-gas utilization, biofuel production, and circular-bioeconomy development (Leong et al., 2021; Ma et al., 2022; Sadvakasova et al., 2025).

1.3 Advancing Research in Microalgae Carbon Sinks

Microalgal carbon-sequestration research now spans algal physiology, environmental engineering, bioprocess design, and climate-mitigation assessment. A central objective is to improve CO_2-to-biomass conversion by coordinating light supply, nutrient availability, salinity, pH, CO_2 delivery, and gas-transfer conditions (Ma et al., 2022; Sadvakasova et al., 2025). Genetic and metabolic engineering are also being used to improve photosynthetic efficiency, increase CO_2 fixation, and redirect carbon flow toward lipids, carbohydrates, and other useful products (Priyadharsini et al., 2022; Sadvakasova et al., 2025). Reactor design is another determinant of performance. Photobioreactors can improve light distribution, mixing, gas–liquid mass transfer, and CO_2 exposure, but monitoring is needed to distinguish biological CO_2 fixation from gas transfer, carbonate chemistry, pH drift, and culture instability (Ma et al., 2022; Mousavi et al., 2022; Yahaya et al., 2025).

Microalgae are also relevant to blue-carbon research, although their role differs from mangroves, seagrasses, and tidal marshes because much of the fixed carbon may be rapidly recycled (Macreadie et al., 2021). In coastal systems, interactions among microalgae, seagrasses, corals, and microbial communities can alter local productivity and carbon cycling, but their contribution to long-term blue-carbon storage remains uncertain (Tew et al., 2021). Industrial integration offers another pathway: flue gas can provide a concentrated CO_2 source for algal growth, while the resulting biomass can be used for fuels, biochemicals, feed, or other products (Ma et al., 2022; Daneshvar et al., 2022; Choi et al., 2021).

This review has five objectives. First, it synthesizes current knowledge on microalgal carbon sequestration, with particular attention to CO_2 biofixation, biomass production, and carbon utilization in engineered cultivation systems (Ma et al., 2022;

Sadvakasova et al., 2025). Second, it evaluates how studies monitor and quantify microalgal carbon uptake, including direct measurements of biomass carbon and indirect measurements based on inlet–outlet CO_2 concentrations. Third, it examines how environmental conditions such as temperature, light, pH, salinity, nutrients, CO_2 concentration, and gas– liquid mass transfer shape algal growth and carbon fixation (Ma et al., 2022; Sadvakasova et al., 2025). Fourth, it reviews biological and genetic mechanisms, including CCMs, photosynthetic regulation, and species-specific stress responses (Priyadharsini et al., 2022; Burlacot & Peltier, 2023). Finally, it discusses practical applications in wastewater treatment, biofuel production, industrial carbon capture, and circular-bioeconomy systems, and identifies gaps in carbon accounting, reactor design, strain selection, scale-up, and life-cycle evaluation (Abdelfattah et al., 2023; Daneshvar et al., 2022; Sarwer et al., 2022; Ma et al., 2022; Choi et al., 2021).

2. Review Methodology

2.1 Protocol for Literature Search and Selection

The review followed a predefined search and screening protocol that specified the research questions, search strings, database source, screening steps, and inclusion and exclusion criteria (Figure 1).

2.1.1 Research Questions

The review was structured around three research questions:

RQ1: What is currently known about microalgal contributions to carbon sinks?

RQ2: How are microalgal carbon fixation, sequestration, and storage monitored or quantified in existing studies?

RQ3: What research trends, methodological gaps, and deployment challenges remain in this field?

2.1.2 Search Strategy

A structured search strategy was used to identify studies relevant to the three research questions. The strategy combined

predefined search strings, a publication-time window, and screening criteria. Two search strings were used:

- Search String 1: TS=("microalg*" AND ("carbon sink" OR "carbon capture" OR "carbon absorb*" OR

"carbon sequest*"))

- Search String 2: TS=("phytoplankton*" AND ("carbon sink" OR "carbon capture" OR "carbon absorb*"

OR "carbon sequest*") AND ("account*" OR "monitor*" OR "inventory*" OR "calculate*"))

The search strings were applied to the titles, keywords, and abstracts of publications. The search covered articles published up to 2026. In June 2026, the Web of Science search yielded 663 records. After article-type and abstract screening, 275 records were excluded, leaving 388 articles for detailed evaluation. Search String 2 narrowed the retrieval to studies addressing accounting, monitoring, calculation, or assessment of microalgal or phytoplankton-related carbon sinks (Table S1). The final synthesis was based on 109 core studies that met the relevance and evidence criteria.

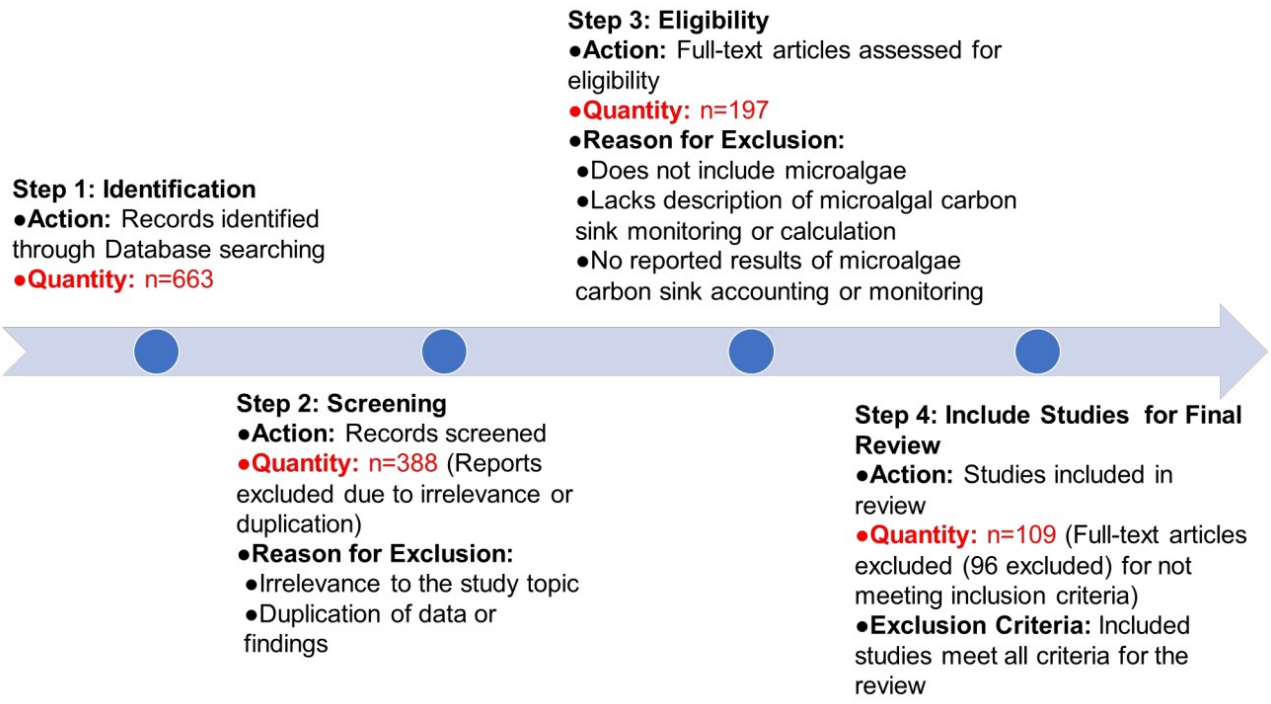


Figure 1. Protocol for literature search and selection.

2.2 Criteria for Inclusion and Exclusion

Inclusion and exclusion criteria were applied to keep the evidence base relevant and methodologically consistent. Articles were included when they involved microalgae or related algal systems, addressed carbon fixation, carbon sequestration, CO_2 capture, or carbon-sink processes, and reported methods or results related to monitoring, calculation, modelling, or performance assessment. Articles were excluded when they lacked microalgal relevance, carbon-sink relevance, methodological information, or reported accounting/monitoring results.

Screening was conducted in two stages: title/abstract screening followed by full-text eligibility assessment. Records were first screened for document type, microalgal relevance, and carbon-sink relevance, and the remaining articles were evaluated against the indicator- and evidence-based inclusion criteria. Full-text exclusions were mainly due to insufficient methodological detail, lack of carbon-related monitoring or calculation, or absence of microalgal relevance. The screening decisions and operational criteria are summarized in Table S2.

2.3 Tools and Techniques for Bibliometric Analysis

Bibliometric analysis was used to describe publication patterns, keyword structure, and thematic development in microalgal carbon-sink research. VOSviewer was used to construct keyword co-occurrence and bibliographic-coupling maps. The software was used to construct and visualize keyword co-occurrence and bibliographic-coupling networks, including clustered maps of related terms and sources (van Eck & Waltman, 2010). These analyses supported the narrative synthesis by identifying dominant themes, methodological patterns, and research gaps (Bai et al., 2022).

3. Overview of Research Landscape

3.1. Trends in Microalgae Carbon Sink Literature

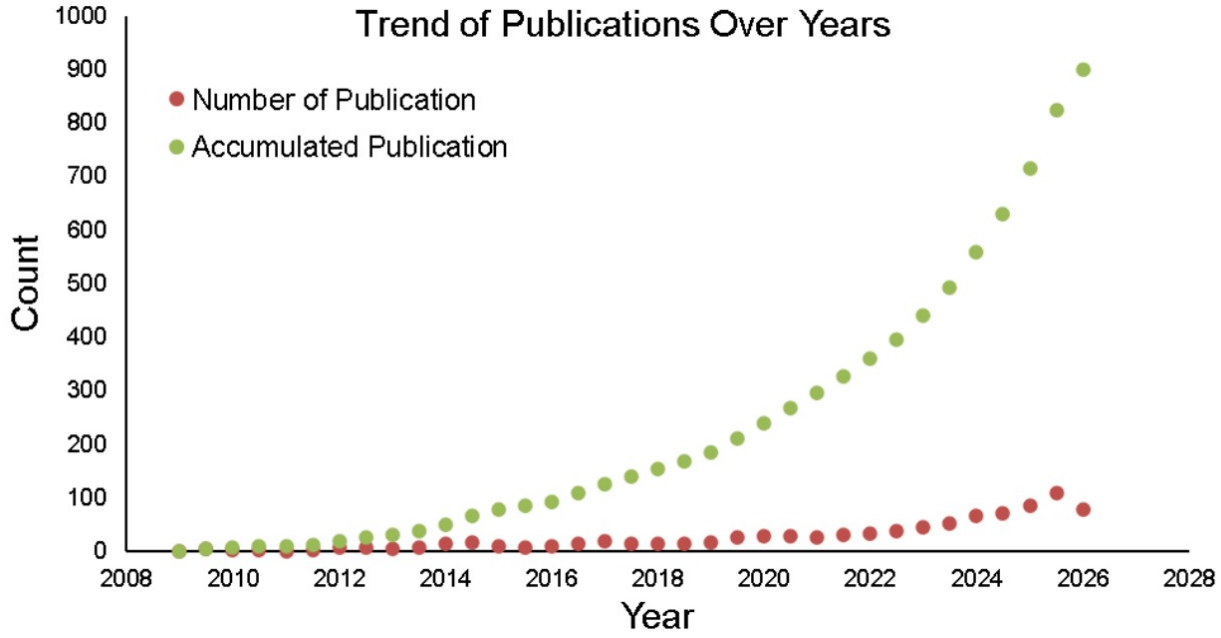


Figure 2. Publication trends in microalgae-related carbon fixation and carbon sequestration studies based on the updated search conducted

in June 2026.

Figure 2 shows how publications on microalgae-related carbon fixation and sequestration changed over time. The number of studies

increased over the reviewed period, especially among papers reporting monitoring or assessment indicators. This rise reflects growing interest in CO_2 fixation, biomass production, wastewater-linked cultivation, and carbon-accounting applications. It also suggests that the field

is moving beyond proof-of-concept cultivation toward monitoring, system integration, and deployment-oriented assessment. Studies on membrane and flat-panel photobioreactors illustrate this shift by linking CO_2 capture with biomass production and biofuel-related applications

(Senatore et al., 2021; Do et al., 2022).

Keyword co-occurrence analysis shows that the field is organized around several connected themes. Figure 3 summarizes these patterns using keyword co-occurrence and overlay visualization.

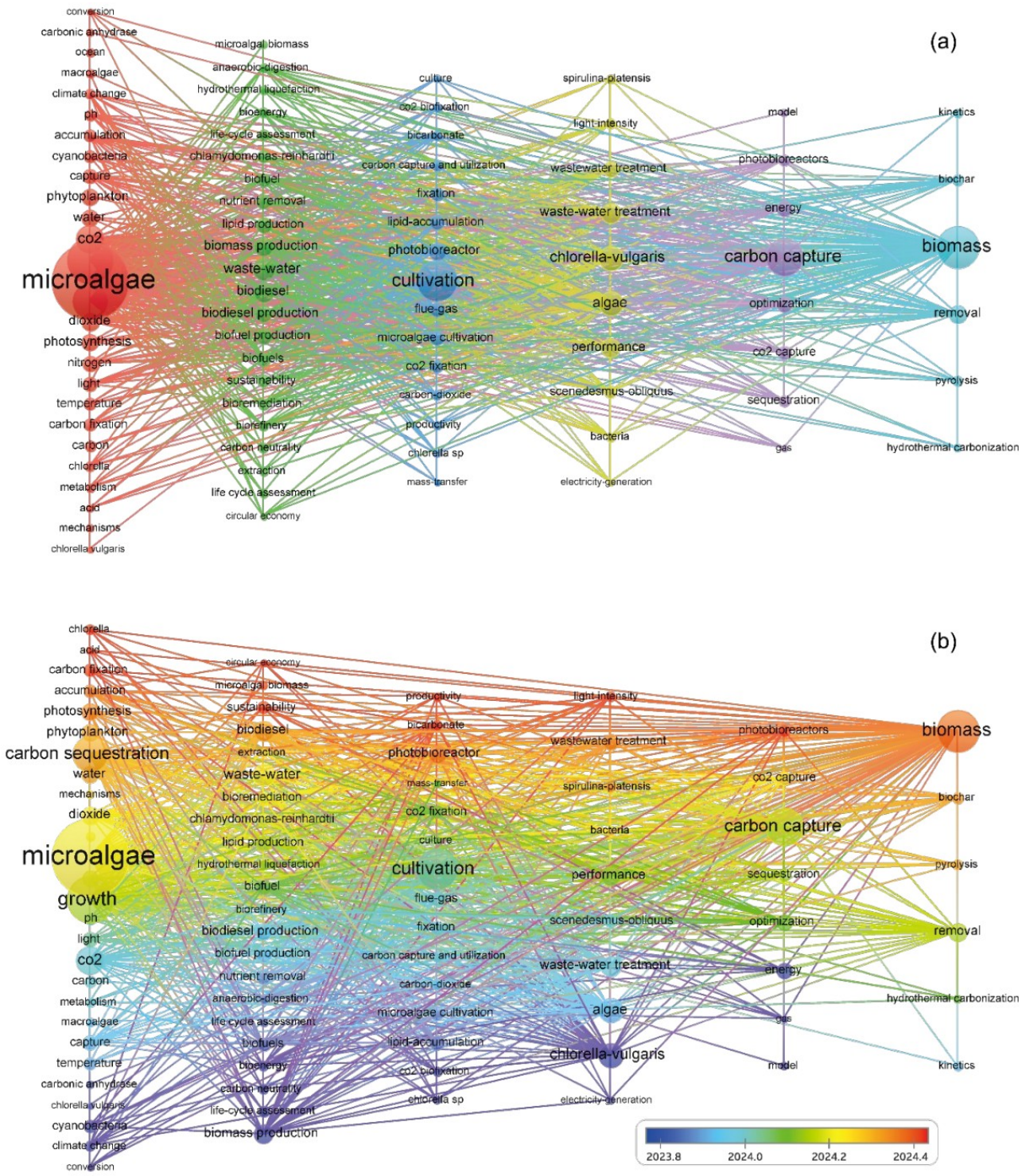


Figure 3. Keyword co-occurrence and overlay visualization maps generated using VOSviewer.

The keyword co-occurrence network in Figure 3(a), constructed using VOSviewer, shows the thematic structure of research on microalgae-related carbon sinks. The analysis included 88 keywords occurring at least 11 times and identified six clusters. Node size represents keyword frequency, whereas links indicate co-occurrence relationships. The most frequent terms were “microalgae”, “growth”, “biomass”, “cultivation”, “carbon capture”, and “carbon sequestration”, indicating that algal growth, biomass production, and carbon capture remain central themes. The clusters cover biological CO_2 fixation and photosynthesis, cultivation and photobioreactor optimization, biofuel and bioenergy production, wastewater treatment and nutrient removal, engineered carbon capture, and biomass-derived carbon management such as biochar and pyrolysis. Chlorella-related keywords also appear prominently, reflecting the role of this genus in studies of CO_2 fixation, biomass accumulation, and bioenergy production.

The overlay visualization in Figure 3(b) adds a temporal dimension to the keyword network. The color gradient represents the average publication year of each keyword, with warmer colors indicating more recent attention. Keywords such as “chlorella”, “photobioreactors”, “circular economy”, “microalgal biomass”, “carbon fixation”, “productivity”, and “lightintensity” point to recent emphasis on strain-level carbon fixation, Chlorella-based biomass accumulation, photobioreactor design, productivity enhancement, and circular-economy-oriented biomass use. This pattern is consistent with recent studies on Chlorella-mediated CO_2 fixation, photobioreactors for carbon dioxide control, and integrated systems combining carbon capture, biomass production, wastewater treatment, and bioenergy applications.

3.2. Geographical and Institutional Contributions

The geographical distribution of the reviewed studies shows a clear concentration of microalgae-related carbon-sink research in Asia, especially China. Chinese institutions contributed substantially to studies on photobioreactor development, wastewater-linked cultivation, algal biotechnology, and carbon-neutrality-oriented applications. Other Asian countries, including India, Malaysia, Japan, Thailand, and South Korea, also contributed to work on microalgae-based direct air capture, photobioreactor development, bioenergy conversion, and urban carbon mitigation strategies (Dayanath et al., 2026). North American and European studies were fewer in number but were strongly represented in biological and chemical carbon sequestration, photosynthetic enhancement, life cycle assessment, techno-economic evaluation, and process optimization (Gallagher et al., 2026; Naddeo et al., 2026; Sivaraj et al., 2026). Contributions from Africa and South America were more limited, but they indicate emerging interest in wastewater treatment, resource recovery, and sustainable bioenergy applications.

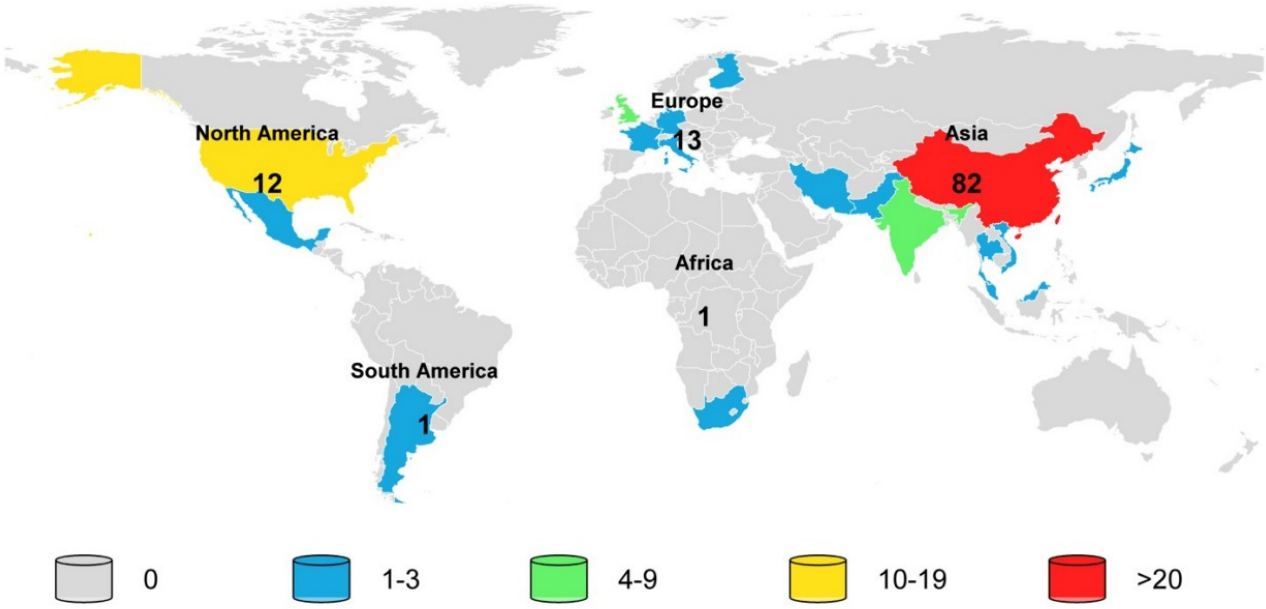


Figure 4. Spatial distribution of publications.

The journal-level bibliographic coupling map in Figure 5 shows how microalgae-related carbon-sink research is distributed across journals in biotechnology, environmental engineering, bioresource recovery, and energy systems.

Node size indicates the number of articles published in each source, whereas link strength reflects overlap in cited references. Bioresource Technology, Algal Research-Biomass, Biofuels and Bioproducts, Chemical Engineering Journal,

Journal of Environmental Management, and Journal of Environmental Chemical Engineering were among the most frequently represented sources. The coupling pattern indicates that these journals share a common knowledge base around algal cultivation, CO_2 capture, biomass production, wastewater treatment, bioenergy conversion, and environmental assessment.

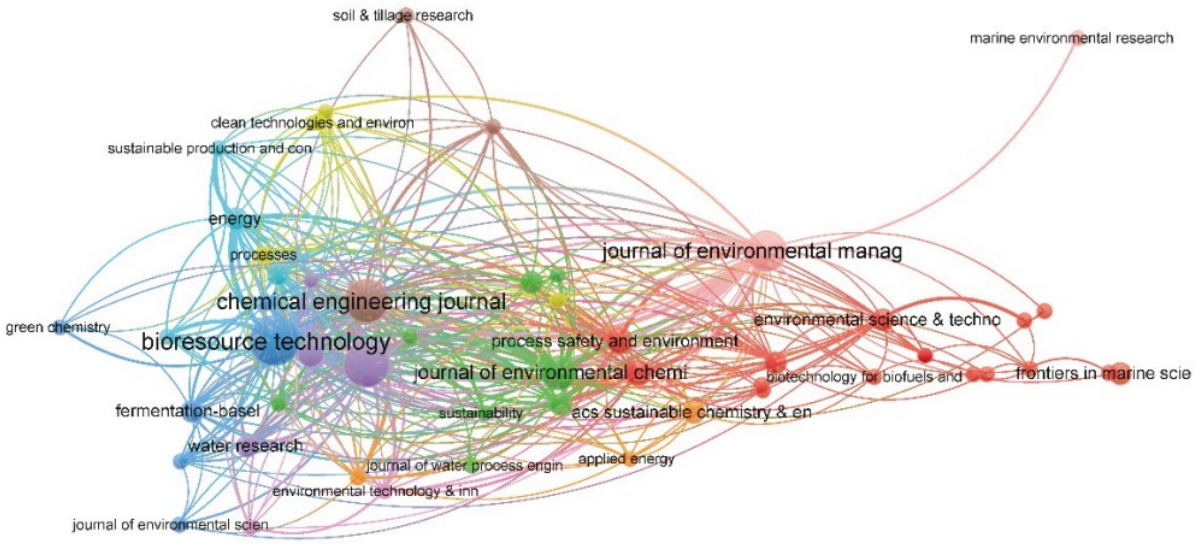


Figure 5. Journal-level bibliographic coupling map generated using VOSviewer.

3.3. Synthesis of Research Trends and Impact in Microalgal Carbon Sequestration

The reviewed literature shows a shift from isolated cultivation experiments toward integrated assessments of carbon fixation, system performance, and application relevance. The selected studies fall into overlapping themes: photobioreactor optimization, wastewater and flue-gas utilization, genetic or mechanistic enhancement, specialenvironment carbon sequestration, microalgal consortia, and modelling or assessment systems. The field is therefore no longer limited to biomass growth or CO_2 uptake; it now also considers cultivation design, environmental constraints, biological traits, and downstream carbon fate.

As summarized in Table 1, the strongest research emphasis was placed on controlled cultivation and photobioreactor systems, followed by practical applications such as wastewater treatment, flue-gas utilization, bioenergy production, and resource recovery. Genetic, physiological, and mechanistic studies formed another major group, reflecting interest in improving carbon fixation through strain selection, metabolic regulation, and stress tolerance. A smaller but distinct body of work addressed natural or special-environment carbon sequestration, including phytoplankton communities, marine systems, and carbonate-related processes. Together, these categories show that microalgal carbon-sink research is developing along both engineering and ecological pathways.

Table 1. Categories of the selected literature based on their study aims.

Aim Description Key Findings Paper Author (Year)

			Number (%)	
A1: Special Environment Carbon Sequestration	Evaluating microalgae-related carbon sequestration in unique aquatic, marine, pollution-affected, or other special environmental settings.	Studies show that environmental context, including marine/oceanic conditions, phytoplankton dynamics, nutrient status, and pollutant interactions, can strongly affect microalgal carbon sequestration capacity.	16 (14.7%)	Jia, JJ (2022); Lai, YS (2022); Nowicki, M (2022); Chen, HY (2023); Wu, JW (2023); Nguyen, HD (2024); Rao, YM (2024); Elghazy, E (2025); Sun, YZ (2025); Yan, JL (2025); Zhao, RR (2025); Chi, LB (2026); Fan, LW (2026); Hu, ZX (2026); Sun, ZH (2026); Yuan, DP (2026)
A2: Genetic/Mechanistic Enhancement	Clarifying and enhancing carbon sequestration through genetic, metabolic, enzymatic, nano-enabled, or mechanistic approaches.	Mechanistic studies indicate that photosynthetic regulation, carbon-concentrating mechanisms, enzyme activity, transcriptomic responses, and nano-enabled strategies can improve CO_2 fixation and biomass accumulation.	19 (17.4%)	Chen, B (2024); Singh, AK (2024); Wang, DT (2024); Wang, XY (2024); Wang, YC (2024); Xin, K (2024); Yuan, AL (2024); Hao, Z (2025); Hou, YQ (2025); Li, DA (2025); Liu, Y (2025); Ma, SY (2025); Wang, Z (2025); Yuan, RL (2025); Zhang, LH (2025); Fazal, T (2026); Li, XQ (2026); Sivaraj, S (2026); Wei, XR (2026)
A3: Bioyield Improvement Factors	Exploring environmental and operational conditions that enhance microalgal bioyield and carbon fixation.	Optimal or stage-specific combinations of light, CO_2 concentration, pH, aeration, temperature, nutrients, and cultivation conditions were identified as key drivers	20 (18.3%)	Huang, Y (2023); Wei, SJ (2023); Bhatti, SG (2024); Li, PC (2024); Liang, S (2024); Long, DL (2024); Wagner, H (2024);

		of biomass productivity and carbon fixation.		Dewangan, M (2025); Gupta, A (2025); He, JJ (2025); Kusi, PA (2025); Latagan, MJD (2025); Peng, HY (2025); Sobhi, M (2025); Tharasirivat, V (2025); Weerakkody, WS (2025); Yap, JX (2025); Gallagher, N (2026); Hou, YQ (2026); Zare, S (2026)
A4: Photobioreactor Effects	Investigating photobioreactor designs and reactor configurations for improved microalgae cultivation and carbon capture.	Novel photobioreactor designs, including direct-air-capture, light-guiding, membrane, and integrated bio-CCU systems, were shown to improve light utilization, mass transfer, carbon fixation, and process scalability.	21 (19.3%)	Chakravarty, S (2022); In-na, P (2022); Wang, RL (2022); Oliver, A (2023); Ren, CYF (2023); Vajdi, S (2023); Battula, SR (2024); Chen, MH (2024); Ruiz-Ruiz, P (2024); Du, X (2025); Fazal, T (2025); Feng, L (2025); Groff, MC (2025); Wu, YL (2025); Yahaya, E (2025); Yang, YF (2025); Dayanath, WMS (2026); Feng, L (2026); Naddeo, V (2026); Nazari, S (2026); Zhang, JM (2026)
A5: Practical Carbon Fixation Applications	Utilizing microalgae for practical environmental and industrial applications such as wastewater treatment, nutrient recovery, bioenergy, biochar, biorefinery, and circular bioeconomy.	Application-oriented studies demonstrate the dual or multiple benefits of microalgae for CO_2 mitigation, wastewater treatment, nutrient recovery, biofuel/bioenergy production, carbon emission reduction, and	19 (17.4%)	Cole, GM (2023); Ganeshan, P (2023); Wicker, RJ (2023); Ansari, FA (2024); Burrington, JD (2024); Chen, HY (2024); Gong, L (2024); Peng, HY (2024); Wen, CM (2024); Barla, RJ (2025);

		biomass valorization.		Gowd, SC (2025); Liu, YY (2025); Shuhaili, F (2025); Song, YJ (2025); Subramanian, KG (2025); Zha, JR (2025); Zhang, ZZ (2025); Zuorro, A (2025); Zhang, ZZ (2026)
A6: Special Microalgae Biological Properties	Examining unique biological characteristics and carbon fixation potential of selected microalgae species or strains.	Species- and strain-level studies reveal substantial variation in carbon fixation capacity, stress tolerance, photosynthetic performance, biomass accumulation, and suitability for bioenergy or carbon management applications.	6 (5.5%)	Zhang, ML (2023); Kikuchi, Y (2024); Kusi, PA (2024); Pekkoh, J (2024); Wang, JL (2024); Ma, JJ (2025)
A7: Microalgae Synergy	Exploring microalgal-bacterial, microbial, or integrated consortia systems for enhanced carbon sequestration and resource recovery.	Synergistic systems suggest that interactions among microalgae, bacteria, methanotrophs, or other microbial partners can enhance carbon capture, nitrogen recovery, wastewater treatment, and biomass generation.	2 (1.8%)	Wang, SQ (2025); Zhu, GY (2026)
A8: New Algae Cultivation Models	Developing novel cultivation, monitoring, modelling, or optimization methods to increase microalgal bio-production and carbon sequestration.	New cultivation and modelling approaches improve process control, monitoring, optimization, and scalability, supporting more precise and efficient microalgae-based carbon sequestration systems.	6 (5.5%)	Pruvost, J (2022); Kushwaha, OS (2023); De, DK (2024); Ma, SY (2024); Cho, BA (2025); Galang, MGK (2025)

4. Carbon Sink Dynamics and Microalgal Monitoring

4.1. Methodological Approaches and Indicators for Carbon Fixation Monitoring in Microalgae

This section examines how the reviewed studies quantified or monitored microalgal carbon fixation, sequestration, and related process performance. Table 2 groups the reported indicators into biological, photosynthetic, carbon-related, wastewater-related, nutrient-related, environmental, and system-level categories. Because individual

studies often reported several indicators, the categories were treated as non-exclusive. Table S3 provides a narrower keyword-based summary, while Table S4 lists the article-level eligibility rules used for this synthesis.

Table 2. Indicators and methods used to monitor microalgal carbon fixation and carbon-related performance.

Category	Indicator	Methodological Approach	Application	Pros	Cons	No. of Studies	Percentage
Biological Related	Dry weight/biomass yield	Gravimetric dry-weight measurement or harvested biomass quantification	Direct biomass and algal bio-yield assessment	Direct, comparable, and closely linked to carbon conversion	Requires harvesting/drying and can be labor-intensive	52	47.7%
	Cell density/algal density	Cell counting, optical density, or algal-density monitoring	Tracking microalgal growth and population dynamics	Rapid and suitable for repeated monitoring	May not directly represent dry biomass or carbon content	5	4.6%
	Biomass concentration/productivity	Spectrophotometry, dry-biomass calibration, or productivity calculation	Estimating growth rate, biomass accumulation, and culture performance	Rapid and suitable for cultivation optimization	Method-specific calibration is usually required	55	50.5%
	Lipid content/fatty acid profile	Solvent extraction, gravimetric lipid analysis, GC/GC-MS/FAME profiling	Assessing biofuel potential and carbon partitioning into lipids	Directly relevant to biodiesel and bioenergy applications	Extraction and derivatization are labor-intensive	36	33.0%
	Protein/carbohydrate/ biochemical productivity	Biochemical assays for proteins, carbohydrates, pigments, extracellular	Evaluating biomass quality and carbon allocation into value-	Links carbon fixation to biomass valorization	Requires multiple assays and may vary by extraction protocol	35	32.1%

		polymers, or proximate composition	added products				
Chlorophyll/ Photosynthesis Related	Chlorophyll content	Spectrophotometry, fluorometry, or pigment extraction	Proxy for active photosynthetic biomass and physiological status	Sensitive indicator of photosynthetic biomass	Varies with light, nutrient status, and stress	26	23.9%
	Photosynthetic efficiency /quantum yield	Chlorophyll fluorescence, F_v/F_m, electron transport, or photosystem indicators	Assessing photosynthetic capacity and carbon fixation potential	Non-destructive and mechanistically informative	Requires specialized equipment and careful interpretation	39	35.8%
Carbon Related	CO_2 removal/uptake/absorption	Gas analysis, inlet/outlet CO_2 monitoring, CO_2 absorption measurement, IR-based gas analysis	Quantifying direct CO_2 capture from air, flue gas, or culture systems	Directly represents carbon capture performance	Affected by mass transfer, carbonate chemistry, and system boundaries	66	60.6%
	Carbon fixation or storage rate	Carbon balance, biomass carbon conversion, CO_2-to-biomass calculation, or fixation-rate estimation	Estimating carbon converted into biomass or stable carbon pools	Links growth with carbon conversion or storage performance	Depends on biomass-carbon assumptions and system boundaries	57	52.3%
	Total carbon/TOC/TIC/DIC	TOC/TIC/DIC/ TC analysis, CHNS elemental analysis, or carbon mass balance	Quantifying carbon pools in biomass, water, wastewater, or treatment streams	Precise and useful for mass-balance calculation	Requires analytical instruments and sample preparation	33	30.3%
	Carbonate precipitation/alkalinity	Alkalinity, pH-carbonate chemistry, $CaCO_3$ precipitation,	Monitoring chemical carbon sequestration and	Captures non-biomass carbon storage pathways	Highly dependent on pH, ions, and carbonate equilibrium	16	14.7%

		XRD/FTIR/SEM mineral analysis	mineral carbon storage				
Oxygen/ Wastewater Related	Dissolved oxygen/oxygen evolution	Electrochemical DO probes, oxygen sensors, or oxygen evolution monitoring	Assessing photosynthetic oxygen production and metabolic activity	Fast and useful for real-time culture monitoring	Indirect indicator of carbon fixation; affected by mixing and aeration	8	7.3%
	COD/BOD/organic load	COD/BOD analysis and organic-load monitoring	Evaluating wastewater treatment and organic carbon removal	Relevant for coupled wastewater-carbon mitigation systems	Indirect measure of microalgal carbon fixation	8	7.3%
Nutrient/ Pollutant Related	Nitrogen and phosphorus removal	TN/TP, ammonium, nitrate, phosphate, nutrient assimilation assays	Assessing nutrient recovery and wastewater remediation during algal growth	Connects carbon fixation with resource recovery	Does not directly quantify carbon storage	31	28.4%
	Pollutant removal and stress/toxicity indicators	NO_x/SO_x removal, metal bioavailability, microplastic interaction, ROS/oxidative stress assays	Evaluating carbon fixation under pollutant or flue-gas stress conditions	Relevant to realistic environmental and industrial scenarios	Pollutant responses may confound carbon fixation signals	36	33.0%
Environmental/ Process Related	pH and carbonate-system conditions	pH meter, alkalinity measurement, culture-condition monitoring	Monitoring culture environment and carbonate equilibrium	Simple, rapid, and widely reported	Indirect indicator; should be combined with carbon or biomass metrics	38	34.9%
System/ Model Related	Life-cycle, techno-economic and emission metrics	LCA, techno-economic analysis, GWP, carbon-emission assessment, energy-balance	Evaluating system-level sustainability and deployment potential	Connects laboratory results with scale-up, economics, and policy relevance	Sensitive to boundary assumptions and scenario design	39	35.8%

		analysis					
	Modelling, monitoring and sensing indicators	Kinetic modelling, process modelling, machine learning, remote sensing, lidar, intelligent sensing	Supporting prediction, optimization, and large-scale monitoring of microalgal carbon sinks	Useful for scale-up, control, and spatial/temporal monitoring	Requires validation and depends on model assumptions	47	43.1%

Biological Indicators Biological indicators were central to most assessments of microalgal carbon-sink performance. In Table S3, biomass productivity or production appeared in 33 studies, chlorophyll or pigment indicators in 24 studies, growth rate in 20 studies, and biomass concentration or density in 16 studies. These variables describe algal growth, photosynthetic status, and the conversion of inorganic carbon into cellular biomass. Studies on Scenedesmus obliquus, Chlorella vulgaris, and Chlorella-based CO_2 absorption systems often paired biomass productivity with carbon-fixation metrics to evaluate performance under different light, CO_2, pH, and aeration conditions (Gallagher et al., 2026; Zare et al., 2026; Hou et al., 2026). Lipid-, protein-, and carbohydrate-related indicators were used to trace fixed carbon into biomass fractions with potential biofuel or bioproduct value (Liu et al., 2025a; Latagan et al., 2025; Hou et al., 2026).

Photosynthesis-related Indicators

Photosynthesis-related measurements were used to explain variation in carbon fixation under different cultivation conditions. Chlorophyll and pigment indicators were frequently reported, while photosynthetic efficiency, F /F , quantum

v m yield, electron transport, and photosystem-related variables were used to diagnose changes in light use and physiological stress. These indicators are useful because changes in photosynthetic activity can occur before measurable differences in final biomass yield or carbon accumulation. For instance, antioxidant nanozymes were shown to improve photosynthetic performance and CO_2 uptake under low-temperature and high-light stress, while stepwise light strategies enhanced carbon utilization and high-protein biomass production in a Chlorella-based CO_2 absorption–microalgae conversion system (Sivaraj et al., 2026; Hou et al., 2026). Other studies used chlorophyll content, photosynthetic electron transfer, or photosystem activity to explain the effects of bicarbonate supply, light restriction, UV conversion substrates, and high-CO_2 stress on algal carbon fixation (Li et al., 2024; Wang et al., 2024a; Ma et al., 2025a).

Carbon-specific indicators

Carbon-specific indicators provided the most direct evidence for carbon-sink assessment. These included CO_2 removal or uptake, carbon-fixation rate, carbon-sequestration capacity, dissolved inorganic carbon, particulate organic carbon, total organic carbon, biomass carbon, and carbon-balance metrics. They were usually calculated from gas-phase CO_2 removal, changes in dissolved inorganic carbon, biomass carbon conversion, or system-level carbon balances. Studies on direct air capture, membrane photobioreactors, and CO_2 absorption–microalgae conversion systems used these metrics to quantify CO_2 capture or conversion under defined operating conditions (Dayanath et al., 2026; Wang et al., 2022; Hou et al., 2025). In wastewater, marine, and ecosystem-scale studies, TOC, TIC, DIC, particulate organic carbon, elemental carbon, and carbon-flow metrics were important because carbon may be partitioned among biomass, dissolved inorganic carbon, organic matter, and particulate pools (Zhu et al., 2026; Sun et al., 2025; Yan et al., 2025).

Several studies also considered mineral or carbonate-associated carbon storage. Carbonate precipitation, alkalinity, $CaCO_3$ formation, and related carbonate-system indicators are especially relevant when carbon storage is not limited to biomass formation. For example, biological and chemical carbon sequestration by Scenedesmus obliquus was evaluated through both biomass growth and $CaCO_3$ precipitation, while Chlorella-based marine carbon sequestration studies quantified biomass productivity, pH regulation, alkalinity change, $CaCO_3$ precipitation, and total carbon capture (Gallagher et al., 2026; Fazal et al., 2025; Fazal et al., 2026). These studies show that microalgae may contribute to carbon sinks through both organic biomass accumulation and inorganic mineral-carbon pathways.

Oxygen- and wastewater-related indicators

Oxygen- and wastewater-related indicators were less frequent than biomass or carbon-specific metrics, but they were important in studies linking carbon fixation with wastewater treatment.

Dissolved oxygen and oxygen evolution were used mainly as proxies for photosynthetic activity and metabolic status. COD, BOD, TOC, nitrogen removal, phosphorus removal, and biomass recovery were reported in wastewater-linked systems to evaluate whether carbon mitigation occurred alongside organic-load reduction and nutrient recovery. Integrated anaerobic treatment and membrane photobioreactor systems, for example, used TOC, nitrogen removal, biomass recovery, and carbon emission assessment to evaluate simultaneous nutrient recovery and carbon mitigation (Zhu et al., 2026; Feng et al., 2025). These indicators do not directly quantify long-term carbon storage, but they help define the environmental and operational context in which microalgal carbon fixation occurs.

Nutrient- and pollutant-related indicators

Nutrient- and pollutant-related indicators were common in studies that coupled carbon fixation with wastewater treatment, flue-gas utilization, or stress-response assessment. Nitrogen and phosphorus variables included total nitrogen, ammonium, nitrate, phosphate, total phosphorus, and nutrient assimilation, and were frequently reported in wastewater, anaerobic-digestion effluent, aquaculture wastewater, and nutrient-recovery studies (Zhu et al., 2026; Feng et al., 2025; Song et al., 2025). Pollutant- and stress-related variables included NOx removal, SO_2 or sulfide stress, metal bioavailability, microplastic exposure, reactive oxygen species, and oxidative stress. These indicators are important because nutrient availability and pollutant stress can alter photosynthesis, growth, carbon allocation, and biomass recovery, thereby affecting the apparent carbon-sink performance of microalgal systems.

Environmental and process-related indicators

Environmental and process-related indicators, particularly pH and carbonate-system variables, were frequently used to interpret changes in carbon fixation. pH is central because it regulates the equilibrium among dissolved CO_2, bicarbonate, and carbonate ions, thereby affecting carbon availability, photosynthetic uptake, and carbonate precipitation. Several studies used pH-regulated bicarbonate cultivation, alkalinity monitoring, or carbonate chemistry to improve carbon supply, stabilize culture conditions, and promote carbon capture (Latagan et al., 2025; Du et al., 2025; Fazal et al., 2026).

However, pH should not be interpreted as an independent carbon-sequestration metric. Its value lies in explaining changes in CO_2 uptake, DIC/TIC dynamics, photosynthetic performance, and mineral carbon formation when used together with biomass and carbon-balance measurements.

System- and model-level indicators

System- and model-level indicators help connect culture-scale observations with deployment-oriented assessment. Lifecycle assessment, techno-economic analysis, global warming potential, carbon-emission assessment, carbon-footprint estimation, and energy consumption were used to evaluate whether

microalgal carbon-capture systems provide net environmental benefits after accounting for cultivation, gas delivery, harvesting, drying, reactor construction, and

downstream processing. Examples include microalgae-based direct air capture in developing megacities, renewable or light-guiding photobioreactors, CO_2 capture for coal-fired power plants, and wastewater-based microalgae biodiesel systems (Dayanath et al., 2026; Zhang et al., 2026a; Zhang et al., 2025a; Liu et al., 2025b). These studies show that high CO_2 fixation in culture does not necessarily translate into climate benefit unless system boundaries, energy demand, carbon fate, and economic feasibility are considered.

Modelling, monitoring, and sensing approaches further extend microalgal carbon-sink assessment beyond flask- or reactor-scale measurements. Kinetic models, process models, machine learning, intelligent prediction, ecosystem simulation, remote sensing, and lidar-based monitoring were used to optimize operating conditions, predict biomass productivity, estimate large-scale carbon-sequestration potential, and track carbon dynamics in aquatic environments.

For instance, machine learning was used to estimate global microalgae cultivation potential on marginal lands, intelligent predictive models were applied to optimize algal photobioreactors for greenhouse gas capture, and airborne dualwavelength lidar was used to retrieve phytoplankton absorption and productivity in the South China Sea (Chen et al., 2024a; Galang et al., 2025; Yuan et al., 2026). These methods are important for linking laboratory performance with spatially explicit and system-level carbon assessment.

4.2. Analysis of Microalgae Species Selection

Across the 109 studies, single-species experiments remained important but were not dominant. As shown in Table 3, 24 studies used a single species or strain, accounting for 22.02% of the dataset. These studies mainly focused on speciesspecific carbon fixation, biomass productivity, physiological tolerance, and responses to CO_2 concentration, light, pH, and nutrient conditions, with representative taxa including Chlorella, Scenedesmus, and Arthrospira species (Gallagher et al., 2026; Zare et al., 2026; Hou et al., 2026; Yuan et al., 2025). Multi-species, community-level, and system-oriented studies were also common, reflecting growing interest in comparative screening, natural phytoplankton communities, algal diversity, cyanobacterial assemblages, ecosystem-scale carbon sequestration, modelling, life-cycle assessment, and techno-economic analysis (Shuhaili et al., 2025; Latagan et al., 2025; Fazal et al., 2025; Jia et al., 2022; Sun et al., 2025; Yuan et al., 2026; Yan et al., 2025; Dayanath et al., 2026; Zhang et al., 2026a; Chen et al., 2024a).

Table 3. Distribution of microalgal species selection across the reviewed studies.

Species selection type	Number of Studies	Percentage
Single species/strain	24	22.02%
Multiple named species/strains	31	28.44%
Mixed species/consortium	9	8.26%
Natural community or broad taxonomic group	22	20.18%
Not species-specific or not reported	23	21.10%

The pie chart in Figure 6 presents the distribution of the number of microalgae species used across the studies. Singlespecies studies remain the largest category, with 24 studies accounting for 22.02% of the dataset, indicating that controlled single-strain systems are still widely used to evaluate species-specific carbon fixation, biomass production, and physiological responses. Studies involving two species account for 23 papers, or 21.10%, suggesting a growing interest in comparative species screening and paired-species systems. In contrast, studies using three, four, or five species are less common, representing 5 studies, 1 study, and 1 study, respectively. A further 12 studies, or 11.01%, involve six species or more, mainly reflecting research on algal diversity, phytoplankton communities, cyanobacterial assemblages, and

ecosystem-scale carbon sequestration. However, 27 studies, accounting for 24.77%, did not clearly specify the number of microalgae species used, often because they focused on modelling, life-cycle assessment, techno-economic analysis, or system-level carbon mitigation rather than species selection. The distribution in Figure 6 indicates that although singlespecies studies remain widely used, recent microalgal carbon-sink research now includes more two-species comparisons, community-level studies, and system-oriented analyses.

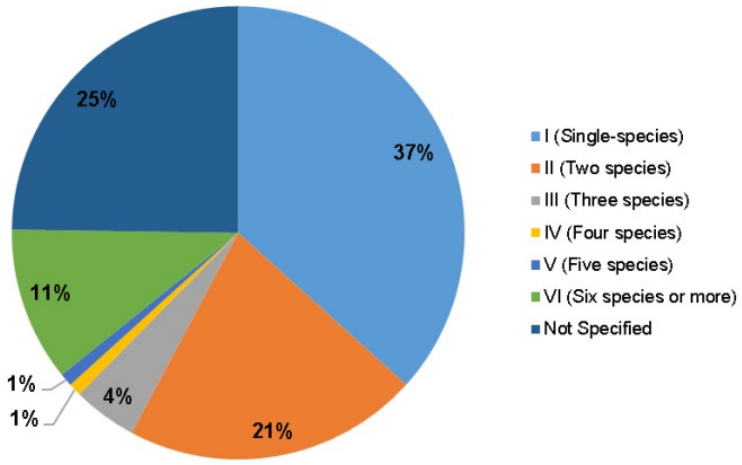


Figure 6. Microalgae species selection for study.

The distribution in Figure 6 indicates that species selection was uneven across the reviewed studies. Single-species or single-strain studies accounted for 24 records, or 22.02% of the dataset, while studies using multiple

named species or strains accounted for 31 records, or 28.44%. Natural or mixed microalgal communities represented 22 studies, and several studies focused on algal-bacterial consortia or other integrated biological systems. Controlled strain-based experiments remain important for evaluating physiological responses, carbon-fixation

mechanisms, biomass productivity, and responses to light, CO_2, pH, and nutrient conditions (Gallagher et al., 2026; Zare et al., 2026; Hou et al., 2026). At the same time, mixed communities and consortia are particularly relevant for wastewater-linked, field, and ecosystem-scale carbon processes.

4.3. Advances in Carbon Measurement Techniques

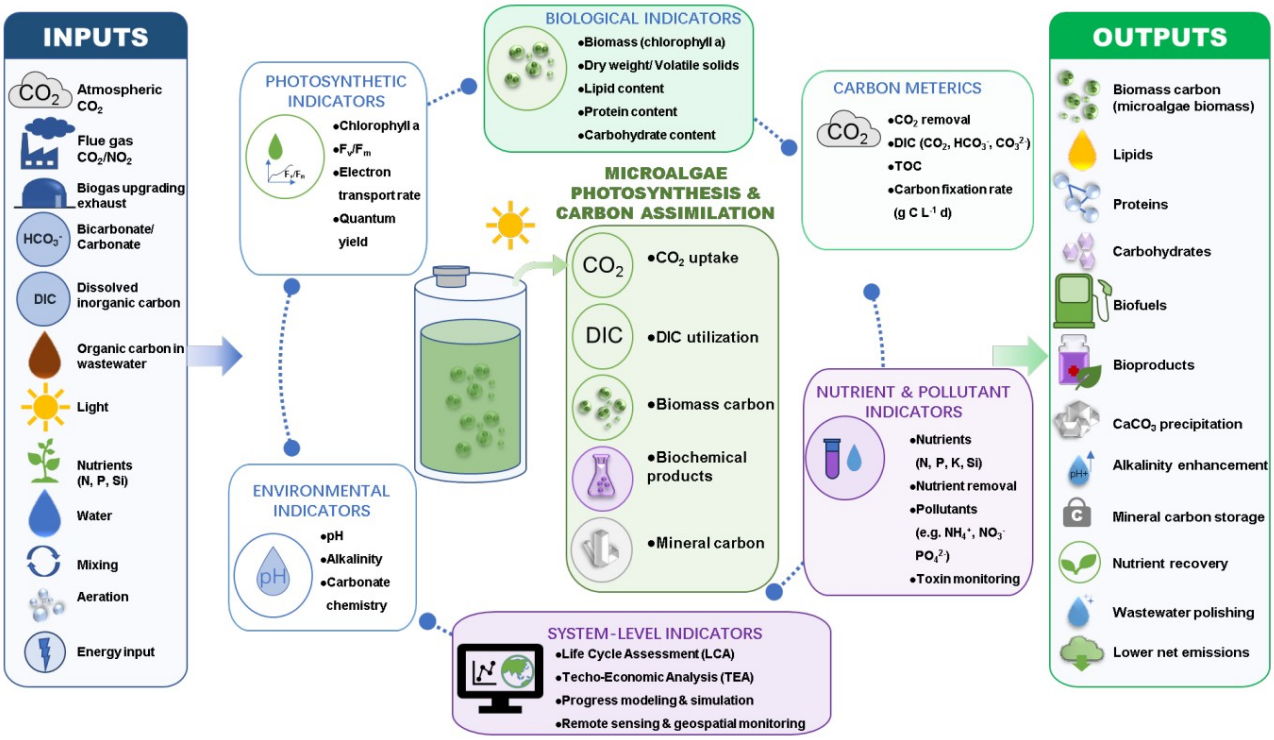


Figure 7. Framework of carbon inputs, monitoring indicators, and carbon sink outputs in microalgal carbon sequestration systems.

Figure 7 summarizes the main carbon inputs, monitoring indicators, and carbon-sink outputs considered in microalgal carbon sequestration

studies. The relevant inputs include atmospheric CO_2, flue gas, biogas-upgrading exhaust, bicarbonate, carbonate, dissolved inorganic

carbon, and wastewater-derived organic carbon. These carbon sources may be converted through photosynthesis and inorganic carbon utilization into biomass carbon, biochemical products, or mineral carbon forms such as carbonate precipitates. Because microalgal carbon fixation is shaped by species traits, carbon availability, light, pH, nutrient status, reactor configuration, and environmental stress, no single

indicator can adequately represent sequestration performance. Recent studies therefore combine biomass productivity, dry weight, chlorophyll content, photosynthetic efficiency, CO_2 removal, dissolved inorganic carbon, carbon-fixation rate, pH, alkalinity, nutrient removal, and pollutant response to describe carbon dynamics more comprehensively (Gallagher et al., 2026; Zare et al., 2026; Fazal et al., 2026; Zhu

et al., 2026).

Methodological choices depend on the scale and purpose of each study. Conventional measurements, including dry weight, biomass concentration, chlorophyll extraction, pH monitoring, and carbon balance, remain essential because they provide direct and interpretable evidence of algal growth and carbon conversion. However, these methods are often labor-intensive and are difficult to apply continuously or at large spatial scales. Automated monitoring, spectroscopic techniques, modelling, machine learning, life-cycle assessment, techno-economic analysis, and remote sensing therefore extend microalgal carbon-sink assessment beyond

short-term laboratory experiments. For example, lightguiding photobioreactor assessments combine biomass and carbon-fixation measurements with energy, environmental, and economic indicators, while intelligent modelling

and lidar-based approaches support prediction and monitoring across larger spatial or temporal scales (Zhang et al., 2026a; Galang et al., 2025; Yuan et al., 2026). This framework highlights that reliable assessment of microalgae-based carbon sinks requires biological performance, carbon accounting,

environmental conditions, and system-level feasibility to be considered together rather than through a single monitoring method.

5. Environmental Impact on Microalgal Carbon Sequestration

5.1. Ecological and Environmental Modulators of Carbon Sequestration

Table 4 summarizes the environmental factors that regulate microalgal growth and carbon-sequestration performance across the reviewed studies. In addition to light, temperature, pH, and nutrient availability, the literature emphasizes CO_2 concentration, inorganic carbon supply, alkalinity, salinity, aeration, mass transfer, and pollutant-related stress. Light intensity and photoperiod affect photosynthetic energy supply and carbon conversion into biomass, whereas temperature regulates enzyme activity, respiration, and stress tolerance (Hou et al., 2026;

Gallagher et al., 2026; Sivaraj et al., 2026). CO_2 concentration and bicarbonate availability are central to carbon fixation because they determine inorganic carbon supply, pH dynamics, and CO_2-to-biomass conversion efficiency (Zare et al., 2026; Latagan et al., 2025; Wang et

al., 2022). pH, alkalinity, and carbonate chemistry further influence carbon speciation and may promote mineral carbon storage through

carbonate precipitation (Fazal et al., 2025; Fazal et al., 2026; Lai et al., 2022). Nutrient availability, especially nitrogen and phosphorus, supports biomass accumulation and links carbon fixation with wastewater treatment and nutrient recovery (Zhu et al., 2026; Feng et al., 2025; Song et al., 2025). Salinity, seawater ion composition, aeration, mixing, and pollutants such as NOx, SO_2, sulfide, metals, and microplastics also affect microalgal physiology and carbon-sequestration performance under realistic cultivation or environmental conditions (Du et al., 2025; Shuhaili et al., 2025; Hu et al., 2026; Fan et al., 2026). These patterns indicate that microalgal carbon sequestration is controlled by

interacting environmental and operational factors rather than by a single variable.

Table 4. Environmental controls on microalgal growth, carbon fixation, and carbon-related outcomes.

Environmental Factor	Impact on Microalgal Growth	Effect on Carbon-related Outcomes	Specific Examples
Light intensity and photoperiod	Regulates photosynthetic energy supply, pigment acclimation, growth rate, and biomass productivity; insufficient light limits growth, whereas excessive light may induce photoinhibition.	Directly controls photosynthetic CO_2 assimilation and carbon conversion into biomass; optimized or stage-specific light regimes can improve carbon utilization efficiency.	Step-wise light strategies improved carbon utilization and high-protein biomass production in a *Chlorella*-based CO_2 absorption–microalgae conversion system; high light combined with elevated CO_2 enhanced *Scenedesmus obliquus* growth and carbon sequestration.
CO_2 concentration and inorganic carbon supply	Provides the main carbon substrate for photosynthesis and affects carbon-concentrating mechanisms, pH dynamics, and biomass accumulation.	Determines CO_2 uptake, DIC utilization, carbon fixation rate, and conversion of inorganic carbon into biomass or carbonate products.	Operational-factor mapping identified CO_2 availability as a major control of *Chlorella vulgaris* biofixation; bicarbonate- and carbonate-based cultivation improved carbon capture in several systems.
Temperature	Influences enzyme kinetics, membrane stability, respiration, photosynthetic repair, and stress	Modifies the balance between photosynthetic carbon fixation and respiratory carbon loss,	Temperature interacted with CO_2, light, aeration, and pH in *Chlorella vulgaris* cultivation; low-temperature stress

	tolerance.	thereby affecting net sequestration efficiency.	reduced microalgal performance but could be partly alleviated by antioxidant nanozymes.
pH, alkalinity and carbonate chemistry	Affects $CO_2/HCO_3^-/CO_3^{2-}$ speciation, enzyme activity, nutrient availability, membrane transport, and physiological stress.	Controls carbon availability, bicarbonate use, carbonate precipitation, alkalinity enhancement, and mineral carbon storage.	pH-regulated bicarbonate cultivation enhanced carbon capture and utilization; microalgae-driven carbonate precipitation contributed to biological and chemical carbon sequestration.
Nutrient availability	Nitrogen, phosphorus, and other nutrients regulate cell division, chlorophyll synthesis, protein production, and biomass quality.	Controls biomass accumulation and thus the amount of carbon converted into biomass; nutrient recovery also links carbon fixation with wastewater treatment.	Microalgal ammonium assimilation supported nitrogen recovery and carbon mitigation in a membrane photobioreactor; wastewater and aquaculture effluents supplied nutrients while supporting biomass production.
Salinity, seawater composition and ionic balance	Affects osmotic regulation, ion transport, aggregation, harvesting behavior, and strain-specific tolerance.	Influences bicarbonate uptake, carbon utilization, carbonate precipitation, and feasibility of marine or saline cultivation systems.	Optimizing calcium and magnesium in seawater medium promoted *Chlorella* growth and self-flocculation harvesting; halotolerant *Chlorella* strains differed in CO_2 capture dynamics.
Aeration, mixing and mass transfer	Regulates CO_2 delivery, oxygen stripping, hydrodynamic stress, cell suspension, and light exposure within cultures.	Improves gas–liquid transfer and carbon availability, although excessive aeration or inefficient hydrodynamics can increase energy demand and reduce net benefits.	Membrane carbonation, bubble-column photobioreactors, and off-site bicarbonation absorbers were used to improve CO_2 transfer and microalgal fixation efficiency.
Pollutants and environmental stressors	Flue-gas components, sulfide, NOx, metals, microplastics, and oxidative stress can inhibit or stimulate growth depending on concentration, exposure time, and strain tolerance.	Stress responses alter photosynthesis, carbon allocation, pollutant removal, and the stability of carbon capture systems under realistic operating conditions.	Simulated flue gas studies showed that moderate nitric oxide could enhance *Chlorella* carbon fixation and pollutant removal; microplastic–copper interactions changed phytoplankton photosynthetic carbon sequestration.

Table 5 summarizes how ecological interactions and environmental settings influence microalgal growth and carbon sequestration. The

reviewed studies indicate that microalgal performance is shaped by microbial partners, nutrient availability, wastewater composition, pollutants, and cultivation conditions. Microalgal–bacterial systems can enhance nutrient transformation, wastewater purification, biomass production, and carbon recovery, thereby coupling CO_2 fixation with resource recovery and pollutant removal (Zhu et al., 2026; Feng et al., 2025; Wang et al., 2025a). Methanotroph–microalgae symbioses further link methane conversion with microalgal carbon fixation, expanding

the role of algal systems beyond CO_2 capture alone (Li et al., 2026; Ruiz-Ruiz et al., 2024). Natural phytoplankton communities, cyanobacterial assemblages, and microphytobenthos contribute to ecosystem-scale carbon storage through primary production, particulate organic carbon formation, and biological carbon export (Jia et al., 2022; Sun et al., 2026; Yuan et al., 2026). In applied systems, wastewater nutrients may support algal growth and carbon fixation, whereas salinity, toxicity, flue-gas components, metals, and microplastics can alter photosynthesis and carbon-sequestration efficiency (He et al., 2025; Hu et al., 2026; Fan et al., 2026). This indicates that microalgal carbon

sequestration depends not only on species traits but also on ecological interactions and cultivation environments.

Table 5. Influence of ecological interactions on microalgal carbon fixation and carbon-related outcomes.

Type of Interaction	Description	Impact on Microalgal Growth	Effect on Carbon-related Outcomes	Study Details	Carbon-related Results
Microalgal–bacterial interactions	Interactions between microalgae and bacteria through nutrient exchange, organic carbon release, oxygen production, and microbial degradation processes.	Can improve nutrient availability, ammonium assimilation, wastewater purification, and culture stability, although outcomes depend on community composition and operating conditions.	May enhance CO_2 fixation by coupling photosynthesis with nutrient recovery, organic-matter removal, and microbial carbon transformation.	Integrated anaerobic treatment–membrane photobioreactor systems, bacteria–algae symbiosis, and continuous algal-bacterial biofilm systems.	Examples include effective nitrogen recovery, biomass carbon recovery, reduced net carbon emissions, and improved inorganic carbon fixation in algal-bacterial systems.
Methanotroph–microalgae symbiosis	Coupled systems in which methanotrophs convert methane-derived carbon while microalgae provide oxygen and assimilate inorganic carbon.	Biomass ratio and co-metabolic interactions influence growth performance, methane conversion, and community stability.	Links methane mitigation with microalgal carbon fixation, expanding carbon capture from CO_2-only systems to CH_4-related carbon pathways.	Methanotrophs–microalgae symbiotic systems designed to optimize biomass ratio, methane conversion, and carbon fixation mechanisms.	Optimized biomass ratios improved methane conversion and carbon fixation efficiency through co-metabolic regulation.
Microalgae with macrophytes or other photosynthetic organisms	Interactions between microalgae and duckweed, seaweed, bivalves, or other photosynthetic/associated organisms in wastewater, aquaculture, or aquatic ecosystems.	Can alter nutrient competition, light availability, biomass recovery, and water-quality improvement.	May increase total ecosystem or treatment-system carbon capture by combining algal biomass production with nutrient removal and organic carbon cycling.	Microalgae–duckweed aquaculture wastewater systems, red seaweed treatment systems, bivalve–microalgae co-cultivation, and oyster-aquaculture carbon studies.	Reported benefits include enhanced biomass recovery, nutrient removal, and increased phytoplankton-derived or algal biomass carbon pools.
Natural phytoplankton and aquatic communities	Community-level interactions among phytoplankton, cyanobacteria, diazotrophs, microphytobenthos,	Community composition, nutrient availability, water-column	Determines particulate organic carbon formation, carbon export,	Floodplain lakes, East China Sea shelf, South China Sea, tidal flats, marine ranching	Studies linked community composition and productivity to carbon sequestration,

	grazers, and aquatic environmental gradients.	structure, and disturbance events regulate primary productivity and algal biomass.	biological pump strength, and regional ecosystem carbon storage.	systems, and phytoplankton-derived particulate organic carbon studies.	particulate organic carbon formation, and ecosystem-scale carbon storage.
Nutrient and wastewater coupling	Coupling of microalgal growth with wastewater-derived nitrogen, phosphorus, ammonium, organic carbon, and other resources.	Wastewater nutrients can support biomass production, but nutrient imbalance, toxicity, salinity, or organic load may constrain growth.	Improves practical carbon mitigation by combining CO_2 fixation with nutrient recovery and wastewater polishing.	Municipal wastewater, anaerobic digestion effluent, piggery wastewater, aquaculture wastewater, and diluted swine wastewater systems.	Reported outcomes include simultaneous biomass production, nitrogen/phosphorus removal, carbon recovery into biomass, and lower net emissions.
Pollutant and stress-mediated interactions	Interactions between microalgae and flue-gas components, NOx, SO_2, sulfide, heavy metals, microplastics, organic pollutants, or oxidative stress.	Low or moderate stress may induce acclimation or pollutant transformation, whereas high stress can inhibit photosynthesis and biomass accumulation.	Pollutants can change carbon fixation by altering photosynthetic efficiency, carbon allocation, antioxidant responses, and pollutant-removal pathways.	Simulated flue gas, nitric oxide exposure, microplastic–copper complexes, sulfide stress, and organic pollutant removal in algal-bacterial biofilms.	Examples include enhanced CO_2 removal under moderate NO exposure and altered phytoplankton carbon sequestration under microplastic–metal interactions.
Reactor and cultivation-environment interactions	Interactions between microalgal physiology and engineered cultivation conditions, including photobioreactor design, aeration, mixing, light distribution, and mass transfer.	Reactor configuration controls light exposure, gas transfer, oxygen stripping, nutrient distribution, and hydrodynamic stress.	Improves carbon fixation when CO_2 delivery, light utilization, and energy demand are balanced; poor design can reduce net carbon benefits.	Membrane photobioreactors, light-guiding photobioreactors, bubble-column photobioreactors, attached cultivation, and biofilm systems.	Studies reported improved CO_2 transfer, biomass productivity, carbon fixation rate, and techno-environmental performance under optimized reactor conditions.

5.2. Role of Controlled Environments and Photobioreactors

Photobioreactors (PBRs) provide controlled cultivation environments for testing microalgal growth, CO_2 fixation, and biomass recovery under defined conditions. As summarized in Table 6, PBR performance depends not only on biological

productivity, but also on energy demand, light distribution, carbon delivery, mixing, aeration, harvesting, and process control. Energy efficiency is critical because lighting, aeration, pumping, harvesting, and downstream processing can offset part of the climate benefit of algal carbon capture. Recent PBR studies therefore combine

low-energy or renewableenergy-driven operation with life-cycle assessment, techno-economic analysis, and intelligent control systems to evaluate whether CO_2 capture can remain environmentally and economically feasible at larger scales (Dayanath et al., 2026; Zhang et al., 2026a; Naddeo et al., 2026; Zhang et al., 2025a).

Table 6. Design aspects of microalgal cultivation and their relevance to photobioreactor efficiency.

Design Aspect	Importance in Microalgal Cultivation	Strategies for Maximizing Efficiency
Energy efficiency	Determines whether CO_2 capture and biomass production can deliver a net environmental and economic benefit, especially when lighting, pumping, aeration, harvesting, and downstream processing are considered.	Use low-energy or renewable-energy-driven lighting, improve light-guiding structures, recover process energy where possible, and combine PBR operation with LCA, techno-economic analysis, and intelligent control of light, temperature, aeration, and CO_2 supply.
Scalability and modular deployment	Essential for moving from laboratory-scale carbon fixation experiments to pilot-, building-, industrial-, or city-scale deployment without losing productivity or increasing energy demand disproportionately.	Adopt modular PBR arrays, façade-integrated or building-integrated designs, attached cultivation, biofilm systems, and scalable monitoring frameworks that allow expansion while maintaining control of light, gas transfer, and culture stability.
Light exposure and distribution	Directly controls photosynthetic efficiency, photoinhibition risk, biomass productivity, and carbon conversion efficiency.	Use transparent materials, short optical paths, light-guiding devices, dynamic light transmission models, stage-specific light strategies, and spectral conversion approaches to improve light penetration and reduce dark zones.
Mixing, aeration and mass transfer	Controls CO_2 delivery, oxygen removal, nutrient distribution, cell suspension, hydrodynamic stress, and culture homogeneity.	Optimize gas flow, bubble-column operation, membrane carbonation, off-site bicarbonation, circulation patterns, and aeration intensity to improve gas–liquid transfer while limiting energy costs and shear stress.
Carbon delivery and pH control	Maintains inorganic carbon availability while avoiding pH inhibition, inefficient CO_2 loss, or carbonate imbalance.	Use bicarbonate supply, membrane carbonation, CO_2 absorption–microalgae conversion systems, pH-regulated operation, and carbonate/alkalinity management to stabilize carbon supply and improve fixation efficiency.
Monitoring, modelling and automation	Improves process control, reproducibility, prediction, and long-term operation under variable environmental or industrial conditions.	Integrate kinetic modelling, machine learning, intelligent sensing, predictive optimization, and real-time monitoring of biomass, CO_2 removal, pH, nutrients, light, and energy demand.

Scale-up of PBRs is constrained by the same variables that determine culture performance: light distribution, gas transfer, fouling, mixing energy, and temperature control. Modular reactor arrays, façade-integrated photobioreactors, attached cultivation, and biofilm-based systems have been explored to increase cultivation area while maintaining light exposure and mass-transfer efficiency (Vajdi et al., 2023; Chen et al., 2024b; Hou et al., 2025). Light exposure is another key factor, because uneven illumination, excessive light, or dark zones can reduce photosynthetic efficiency and biomass

productivity. Recent designs therefore use light-guiding structures, dynamic light-transmission models, and stage-specific light strategies to improve light penetration and carbon utilization (Ma et al., 2024; Hou et al., 2026; Wang et al., 2024a).

Mixing and aeration systems are also important because they regulate CO_2 delivery, oxygen removal, nutrient distribution, and culture homogeneity. Membrane carbonation, bubble-column operation, off-site bicarbonation absorbers, and optimized aeration strategies have been used to improve gas transfer while limiting energy costs and shear stress (Wang et al., 2022; Yahaya et al., 2025; Wu et al., 2025; Zare et al., 2026). Taken together, these studies show that recent PBR designs combine reactor geometry, carbon delivery, light management, and environmental assessment rather than optimizing a single component in isolation.

Table 7. Advantages of controlled environments for microalgal growth and carbon fixation.

Controlled-System Function	Relevance to Microalgal Growth and Carbon Fixation	Study Context
Optimized growth conditions	Controlled systems allow coordinated regulation of CO_2 supply, light, pH, nutrients, temperature and aeration, thereby improving biomass productivity and carbon fixation stability.	Operational-factor mapping and high-light/elevated-CO_2 cultivation studies evaluated how controlled culture conditions affect *Chlorella vulgaris* and *Scenedesmus obliquus* growth and carbon sequestration.
Consistent and efficient production	Controlled cultivation improves reproducibility of biomass accumulation, biochemical productivity and CO_2-to-biomass conversion under defined operating conditions.	Studies using dynamic light strategies, attached cultivation, bicarbonate supply and growth modelling assessed how controlled operation supports stable productivity and carbon utilization.
Direct air and flue-gas carbon capture	Engineered PBRs and carbon delivery systems can use atmospheric CO_2, flue gas, biogas-upgrading exhaust or bicarbonated media as carbon sources.	Direct air capture, membrane photobioreactor, off-site bicarbonation absorber and simulated flue-gas studies examined microalgal CO_2 capture under controlled gas-transfer conditions.
Enhanced carbon supply and pH control	pH-regulated operation, membrane carbonation and bicarbonate/carbonate management stabilize dissolved inorganic carbon availability and reduce carbon loss.	Membrane carbonation, bicarbonate cultivation and carbonate-enhanced systems were used to improve carbon transfer, pH stability and microalgal utilization of inorganic carbon.
Integration with wastewater treatment and nutrient recovery	Wastewater-integrated systems provide nitrogen, phosphorus and organic carbon while coupling biomass production with nutrient removal and carbon mitigation.	Membrane photobioreactor, piggery wastewater, aquaculture wastewater and bacteria–algae systems evaluated simultaneous nutrient recovery, wastewater polishing and carbon capture.
Process monitoring, modelling and scale-up assessment	Automated monitoring, modelling, machine learning, LCA and techno-economic analysis help evaluate long-term feasibility beyond short-term culture performance.	Predictive photobioreactor models, machine-learning assessments, light-guiding cultivation systems and multi-scenario LCA/TEA studies linked reactor operation with environmental and economic outcomes.

As summarized in Table 7, controlled cultivation environments offer several advantages for evaluating microalgal growth and carbon fixation. Photobioreactors and related engineered systems allow key variables such as CO_2 supply, light intensity, pH, temperature, nutrients, and aeration to be adjusted within defined ranges, making it easier to evaluate the responses of specific strains and cultivation modes. For example, operational-factor mapping of Chlorella vulgaris and

high-light/elevated-CO_2 cultivation of Scenedesmus obliquus showed that coordinated control of CO_2, light, temperature, pH, and aeration can improve biofixation performance and clarify the operating window for stable carbon conversion (Zare et al., 2026; Gallagher et al., 2026). Controlled systems also support more consistent biomass production. Dynamic light strategies, attached cultivation, bicarbonate supply, and growth modelling have been used to improve carbon utilization, biomass accumulation, and process reproducibility in microalgal CO_2 capture systems (Hou et al., 2026; Hou et al., 2025; Latagan et al., 2025; Cho et al., 2025).

Controlled environments are also important for direct air and flue-gas carbon capture. Recent studies have tested microalgae-based direct air capture systems, membrane photobioreactors,

off-site bicarbonation absorbers, and simulated flue-gas cultures to improve CO_2 transfer and evaluate carbon-capture performance under different gasdelivery modes (Dayanath et al., 2026; Wang et al., 2022; Wu et al., 2025; Hu et al., 2026). Beyond gas capture, pHregulated operation, membrane carbonation, bicarbonate cultivation, and carbonate management can stabilize dissolved inorganic carbon supply and reduce carbon loss, while also supporting carbonate precipitation in some systems (Lai et al., 2022; Du et al., 2025; Fazal et al., 2025; Fazal et al., 2026). Several studies further show that controlled systems can be combined with wastewater treatment and nutrient recovery, allowing microalgae to remove nitrogen and phosphorus while fixing carbon and producing

recoverable biomass (Zhu et al., 2026; Feng et al., 2025; He et al., 2025; Song et al., 2025). Together, these studies indicate that controlled cultivation systems are most useful when carbon delivery, environmental control, nutrient recovery, and process monitoring are considered together rather than when the systems are treated only as biomass production units.

5.3. Discussion on Environmental Factors and Adaptations

Environmental controls on microalgal carbon sequestration should be interpreted as interacting drivers rather than isolated variables. Light, temperature, pH, CO_2 availability, and nutrient supply jointly regulate photosynthesis, cellular C:N:P stoichiometry, biomass accumulation, and carbon allocation. Meta-analysis evidence shows that irradiance, temperature, nitrogen, and phosphorus availability affect phytoplankton elemental composition, which in turn influences carbon export and sequestration potential (Tanioka & Matsumoto, 2020). Recent synthesis work also emphasizes that warming and nutrient limitation interact in complex ways, altering phytoplankton productivity and ocean carbon–nutrient feedbacks (Hutchins & Tagliabue, 2024). This indicates that improving microalgal carbon fixation requires coordinated regulation of environmental conditions rather than optimization of a single factor.

Microalgae respond to environmental variation through changes in photosynthetic electron transport, carbonconcentrating mechanisms, pigment composition, nutrient-use efficiency, and metabolic carbon allocation. In Chlamydomonas reinhardtii, alternative photosynthetic electron pathways provide energy for the algal CO_2-concentrating mechanism, linking light reactions directly to inorganic carbon transport and fixation (Burlacot et al., 2022). Similarly, assembling metal-organic frameworks on Chlorella pyrenoidosa enhanced local CO_2 delivery and improved photosynthetic CO_2 fixation, showing that carbon availability near the cell surface can affect algal productivity (Li et al., 2023). These examples indicate that environmental adaptation is not only a survival response, but also a mechanism that helps maintain carbon fixation under variable light, carbon, and nutrient conditions.

Controlled cultivation systems, especially photobioreactors, provide useful platforms for

testing these interactions under defined conditions. By regulating light, CO_2 supply, pH, nutrients, temperature, mixing, and aeration, PBRs can identify operating ranges that improve biomass production and carbon fixation. However, culture-level CO_2 uptake does not

necessarily translate into durable climate benefit. Energy use, gas transfer, harvesting, drying, downstream processing, and biomass fate must also be considered when assessing net carbon mitigation. Recent reviews therefore argue that microalgae-based CO_2 sequestration should be evaluated together with biorefinery integration, techno-economic feasibility, and life-cycle performance (Ma et al., 2022). At the ecosystem scale, the same caution applies: lake phytoplankton can fix atmospheric CO_2 into biomass, but a substantial fraction of this organic carbon may later return to the atmosphere as CO_2 or CH_4, weakening long-term sequestration (Shi et al., 2024). Light-dependent methane production by the coccolithophore Emiliania huxleyi further shows that gross photosynthesis and net climate benefit are not always equivalent (Rao et al., 2024).

These examples show that microalgal carbon sequestration depends on interactions among environmental regulation, physiological plasticity, ecological processes, and system design. Future work should give greater attention to combined stressors, including warming, nutrient limitation, pH shifts, salinity, pollutants, and variable light regimes. It should also distinguish short-term CO_2 fixation from long-term carbon storage. Integrating physiology, reactor engineering, ecological monitoring, and life-cycle assessment will be necessary to determine when microalgae act primarily as rapid CO_2converting biomass producers and when they can contribute to more durable carbon sequestration.

6. Microalgal Biology and Carbon Sequestration Mechanisms

6.1. Species-Specific Carbon Fixation Strategies

Table 8 summarizes representative species- and taxon-specific strategies for microalgal carbon fixation. The reviewed studies do not focus

only on a few model species; instead, they cover freshwater, marine, saline, wastewater-linked, flue-gas-exposed, and community-level

systems. Chlorella remains one of the most frequently studied groups because of its tolerance to variable CO_2 supply, pH, light, nutrient availability, wastewater conditions, and reactor configurations. Studies on Chlorella vulgaris and related strains show that carbon-fixation

performance can be improved through operational-factor optimization, bicarbonate supply, flue-gas treatment, attached cultivation, and

carbonate-related carbon management (Zare et al., 2026; Hou et al., 2026; Hu et al., 2026; Du et al., 2025; Fazal et al., 2025).

Table 8. Carbon fixation strategies and application contexts of selected microalgal taxa.

Microalgal Taxon	Habitat or Cultivation Context	Adaptation Strategy	Carbon-Fixation or Carbon-Allocation Feature	Application Context
Chlorella vulgaris / *Chlorella* spp.	Freshwater, wastewater, flue-gas systems, bicarbonate / carbonate media, photobioreactors, and biofilm or attached cultivation systems.	Broad tolerance to variable CO_2 supply, pH, light and nutrient regimes; efficient inorganic carbon utilization; suitability for wastewater nutrients and controlled reactor operation.	Frequently reported as efficient under optimized CO_2, light, pH, bicarbonate or wastewater conditions; performance depends strongly on operational-factor combinations and	CO_2 biofixation, wastewater treatment, flue-gas mitigation, biomass valorization, biofuel/bioproduct production, and carbonate-related carbon management.

			carbon delivery mode.	
Scenedesmus obliquus / *Scenedesmus spp.*	Freshwater and engineered cultivation systems, including high-light / elevated-CO_2 experiments, wastewater-related systems, and comparative cultivation studies.	High tolerance to elevated CO_2 and light variation; capacity for biomass accumulation and, in some systems, coupled biological and chemical carbon sequestration.	High light and elevated CO_2 can enhance growth, biomass carbon storage and carbonate-related carbon sequestration, although responses vary with strain and culture chemistry.	CO_2 sequestration, biomass production, wastewater treatment, biodiesel-oriented cultivation, and screening of high-performing carbon-fixing strains.
Arthrospira platensis / *Spirulina*	Alkaline or high-pH cultivation systems and high-CO_2 or high-light stress environments.	Tolerance to alkaline conditions, high CO_2 and photoinhibition; production of protein- and pigment-rich biomass; improvement through mutagenesis or adaptive evolution.	Shows potential for stable CO_2 fixation and premium biomass production when CO_2 concentration, pH and light are controlled.	CO_2 biofixation, nutraceutical biomass, protein-rich biomass production, pigments, and circular bioeconomy applications.
Nannochloropsis oceanica	Marine microalgal cultivation and chemically adjusted carbon-supply systems.	Marine adaptation, lipid accumulation, and metabolic adjustment under nitrogen and carbon-source regulation.	Carbon fixation and lipid-rich biomass production can be supported under stable pH and adjusted nitrogen conditions, with carbon allocation linked to lipid and biomass production.	CO_2 fixation, lipid-rich biomass, biodiesel-related applications, and marine biorefinery systems.
Dunaliella tertiolecta / *Dunaliella* spp.	Saline or bicarbonate / carbonate media and modelled cultivation systems.	Tolerance to salinity and carbonate-rich media; flexible growth responses to light and temperature; suitability for process modelling.	Growth and carbon capture are strongly affected by light, temperature and bicarbonate / carbonate availability.	Algae-based carbon capture, saline cultivation, modelling-assisted process optimization, and strain screening.
Synechococcus elongatus / *Synechocystis* spp.	Cyanobacterial cultivation systems exposed to high CO_2, light stress or engineered metabolic conditions.	Adaptive evolution, genetic regulation, RuBisCO-related modification, and carbon-storage metabolism under high CO_2 or high-light conditions.	Adaptive or engineered strains can improve tolerance to high CO_2, redirect carbon flow and support biomass or biopolymer accumulation.	Mechanistic carbon fixation studies, engineered CO_2 fixation, PHB or biochemical production, and cyanobacterial chassis development.
Euglena gracilis	Photoautotrophic or flue-gas CO_2 fixation systems.	Nuclear mutagenesis and adaptive evolution to improve photoautotrophic growth and flue-gas CO_2 tolerance;	Improved strains can support enhanced growth under flue-gas CO_2 and modified	CO_2 fixation, paramylon production, stress-tolerant strain development, and bio-

		paramylon-related carbon storage.	carbon allocation.	based product generation.
Emiliania huxleyi / coccolithophores	Marine carbonate-forming systems and membrane carbonation or carbonate-chemistry studies.	Calcification, bicarbonate/carbonate use, and sensitivity to pH and CO_2 supply; links biological carbon fixation with inorganic carbonate processes.	Carbon transfer and pH control can be improved through membrane carbonation, but net carbon benefit depends on carbonate chemistry and associated gas fluxes.	Ocean carbon studies, carbonate chemistry, biological carbon pump research, and mineral carbon-related sequestration.
Natural phytoplankton / cyanobacterial communities	Lakes, estuaries, tidal flats, continental shelves, marine ranching systems, and open-ocean monitoring areas.	Community-level adjustment to nutrients, light, stratification, pollution, grazing and hydrological change; shifts in dominant taxa and carbon pathways.	Carbon sequestration is expressed through primary productivity, particulate organic carbon, carbon export, biological nitrogen fixation and ecosystem carbon flow.	Blue-carbon assessment, ecosystem carbon-budget evaluation, remote sensing, lidar monitoring, and natural carbon sink management.
Microalgal consortia / algal-bacterial systems	Wastewater treatment, anaerobic effluent treatment, biofilm systems, methane-conversion systems, and aquaculture wastewater platforms.	Resource sharing, microbial co-metabolism, nitrogen transformation, organic pollutant removal, oxygen exchange, and community stabilization.	Can couple CO_2 fixation with methane conversion, nutrient recovery, pollutant removal and biomass generation, but performance depends on community balance and operating conditions.	Wastewater treatment, nutrient recovery, methane mitigation, low-carbon treatment systems, and integrated circular bioeconomy platforms.

Chlorella vulgaris / Chlorella spp.: Broad Environmental Tolerance and Inorganic Carbon Utilization Chlorella remains one of the most frequently studied groups because of its tolerance

to variable CO_2 supply, pH, light, nutrient availability, wastewater composition, and flue-gas exposure. Studies on Chlorella vulgaris and related strains show that carbon fixation can be improved through operational-factor optimization, bicarbonate supply, flue-gas treatment, attached cultivation, and carbonate-related carbon management (Zare et al., 2026; Hou et al., 2026; Hu et al., 2026; Du et al., 2025; Fazal et al., 2025).

Scenedesmus obliquus / Scenedesmus spp.: Robust Growth under Elevated CO_2 and Light Variation Scenedesmus species are also widely represented, especially in studies evaluating elevated CO_2, high light, wastewaterlinked cultivation, and combined biological–chemical carbon sequestration. Their relevance lies in robust growth, biomass accumulation, and tolerance to controlled carbon and light conditions, although performance remains strain- and systemdependent (Gallagher et al., 2026; Ruiz-Ruiz et al., 2024; Subramanian et al., 2025).

Arthrospira platensis / Spirulina: Alkaline Tolerance and High-Value Biomass Production Cyanobacteria such as Arthrospira platensis and Spirulina provide another strategy based on alkaline tolerance, high- CO_2 adaptation, photosynthetic efficiency, and production of protein- or pigment-rich biomass. These traits make this group relevant not only for CO_2 biofixation but also for nutraceuticals, pigments, and circular bioeconomy applications (Yuan et al., 2025; Sobhi et al., 2025; Tharasirivat et al., 2025).

Nannochloropsis oceanica: Marine Adaptation and Lipid-Rich Biomass Nannochloropsis oceanica links marine CO_2 fixation with lipid-rich biomass production. Its carbon-fixation and biomassvalorization potential is closely related to nitrogen regulation, pH stability, lipid accumulation, and biomass conversion, making it relevant for biodiesel-oriented and marine biorefinery systems (Liu et al., 2025a).

Dunaliella tertiolecta / Dunaliella spp.: Salinity Tolerance and Carbonate-Rich Cultivation Dunaliella spp. are relevant to saline or bicarbonate-rich cultivation systems. Their growth responses to light, temperature, salinity, and carbonate availability make them useful for saline cultivation and modelling-assisted carbon capture studies (Cho et al., 2025; Sivaraj et al., 2026).

Synechococcus elongatus / Synechocystis spp.: Cyanobacterial Carbon Storage and Engineering Potential Synechococcus elongatus and Synechocystis spp. illustrate cyanobacterial strategies based on adaptive evolution, high- CO_2 tolerance, RuBisCO-related regulation, and redirection of carbon flow toward storage compounds or biochemical products. These taxa are useful for mechanistic studies of carbon fixation and for engineered CO_2-fixation platforms (Ma et al., 2025a; Tharasirivat et al., 2025; Wang et al., 2024a).

Euglena gracilis: Stress-Tolerant CO_2 Fixation and Paramylon Accumulation Euglena gracilis has been explored for flue-gas CO_2 tolerance and paramylon accumulation after mutagenesis or

adaptive evolution. This species is therefore relevant to stress-tolerant CO_2 fixation and paramylon-related bioproduct formation (Xin et al., 2024; Yuan et al., 2024).

Emiliania huxleyi / Coccolithophores: Calcification and Carbonate Chemistry Emiliania huxleyi and other coccolithophores connect biological carbon fixation with calcification and carbonate chemistry. Their contribution to carbon sequestration depends on pH control, carbon-transfer efficiency, carbonate formation, and associated gas fluxes, making them important for marine carbon cycling and mineral-carbon studies (Lai et al., 2022; Rao et al., 2024).

Natural Phytoplankton / Cyanobacterial Communities: Ecosystem-Scale Carbon Flow Natural phytoplankton and cyanobacterial communities extend the discussion from controlled cultivation to ecosystemscale carbon processes. Studies in lakes, estuaries, tidal flats, continental shelves, marine ranching systems, and openocean settings show that community composition, nutrient supply, light, stratification, and hydrology influence primary

productivity, particulate organic carbon formation, biological nitrogen fixation, and carbon export (Jia et al., 2022; Sun et al., 2026; Yuan et al., 2026; Yan et al., 2025).

Microalgal Consortia / Algal-Bacterial Systems: Coupled Carbon Fixation and Resource Recovery Microalgal consortia and algal-bacterial systems show that carbon fixation can be coupled with nutrient recovery, methane conversion, pollutant removal, and wastewater treatment. These systems rely on resource sharing, microbial cometabolism, nitrogen transformation, oxygen exchange, and community stabilization, making them relevant to integrated circular-bioeconomy and low-carbon treatment platforms (Zhu et al., 2026; Li et al., 2026; Feng et al., 2025; Song et al., 2025).

The studies summarized in Table 8 show that microalgal carbon fixation is strongly species- and context-dependent.

Future applications should therefore match taxa with the intended carbon source, cultivation environment, stress conditions, and downstream carbon fate rather than assuming that one species or strategy is universally optimal.

6.2. Genetic and Molecular Bases of Carbon Fixation and Allocation

Recent genetic and molecular studies link microalgal carbon fixation with strain improvement, transcriptomic regulation, genomic traits, adaptive evolution, and metabolic carbon allocation. As summarized in Table 9, enhanced carbon fixation is not controlled by a single gene or pathway. It depends on coordinated regulation of RuBisCO-related carbon assimilation, carbon-concentrating mechanisms, bicarbonate and CO_2 utilization, nitrogen assimilation, photosynthetic electron transport, stress tolerance, and allocation of fixed carbon into biomass or storage compounds.

Table 9. Genetic determinants and biotechnological strategies related to microalgal carbon fixation and carbon allocation.

Microalgal Taxon or System	Key Genetic Determinants or Regulatory Determinants	Carbon-Fixation or Allocation Pathway	Biotechnological or Cultivation Strategy	Application Context
Chlorella spp. / *Chlorella vulgaris*	RuBisCO-related carbon fixation genes, carbon-concentrating mechanism components, carbonic anhydrase activity, bHLH-mediated transcriptional regulation, and stress-responsive metabolic genes.	Calvin–Benson cycle, biophysical CCM, bicarbonate/CO_2 utilization, and mixotrophic carbon assimilation.	Operational-factor optimization, transcriptional reprogramming, mutagenesis breeding, attached cultivation, bicarbonate supply, and flue-gas adaptation.	CO_2 biofixation, carbon-waste remediation, wastewater treatment, flue-gas mitigation, biomass production, and biofuel/bioproduct generation.
Nannochloropsis oceanica	Assimilatory nitrate pathway genes, nitrogen metabolism regulators, lipid biosynthesis-related genes, and pH-responsive metabolic	Calvin–Benson cycle coupled with nitrogen assimilation and lipid carbon allocation.	Regulation of nitrogen source and pH-stabilized cultivation to redirect energy use and carbon allocation.	Marine CO_2 fixation, lipid-rich biomass production, biodiesel-related applications, and carbon-efficient biorefinery systems.

	regulation.			
Arthrospira platensis / Spirulina	Genes associated with high-CO_2 tolerance, photoinhibition resistance, photosynthetic regulation, and pigment/protein-rich biomass formation.	Cyanobacterial CO_2-concentrating mechanism and Calvin–Benson cycle.	Atmospheric and room-temperature plasma mutagenesis, adaptive selection, and high-CO_2/high-light tolerance improvement.	CO_2 biofixation, nutraceutical biomass, protein-rich biomass, pigment production, and stress-resilient cultivation.
Synechococcus elongatus / *Synechocystis* sp.	RuBisCO-related genes, phaAB, carbon storage genes, photosynthetic electron-transfer components, and high-CO_2 adaptive loci.	Cyanobacterial CCM, Calvin–Benson cycle, and carbon flux toward polyhydroxybutyrate or other storage compounds.	Adaptive laboratory evolution, RuBisCO/phaAB overexpression, and metabolic flow enhancement toward carbon storage products.	Engineered CO_2 fixation, PHB or biopolymer production, cyanobacterial chassis development, and mechanistic carbon-flow studies.
Euglena gracilis	Genes regulating photoautotrophic growth, flue-gas CO_2 tolerance, transcriptomic responses to CO_2 concentration, and paramylon accumulation.	Calvin–Benson cycle linked with paramylon carbon storage and photoautotrophic metabolism.	Nuclear mutagenesis, adaptive evolution, and transcriptome-guided analysis under different CO_2 concentrations.	Flue-gas CO_2 fixation, paramylon production, stress-tolerant strain development, and high-value biomass generation.
Mychonastes rotundus	Multi-omics-identified genes and regulatory networks associated with CO_2 fixation, photosynthesis, and carbon metabolism.	Calvin–Benson cycle and associated photosynthetic carbon assimilation pathways.	Multi-omics analysis to identify performance mechanisms and potential targets for strain improvement.	Mechanistic understanding of CO_2 fixation and screening of candidate strains for carbon capture.
Desmodesmus sp.	Genome-level determinants of acid tolerance, inorganic carbon fixation, stress response, and photosynthetic performance.	Calvin–Benson cycle and acid-tolerant carbon-fixing metabolism.	Isolation, whole-genome analysis, and screening of acid-tolerant carbon-fixing strains.	CO_2 fixation under acidic or flue-gas-related conditions and development of robust industrial strains.
Chlamydomonas pacifica / *Parachlorella kessleri*	Endogenous transcription factors, low-CO_2-inducible proteins, lipid accumulation genes, and CO_2-responsive regulatory components.	Calvin–Benson cycle, CCM-related CO_2 response, and carbon partitioning toward lipids.	Transcription-factor identification and overexpression; use of genetically modified strains with low-CO_2-responsive traits.	Enhanced biomass and biofuel productivity, lipid accumulation, and strain improvement for CO_2-responsive cultivation.
Algal-bacterial and methanotroph–microalgae systems	Community-level metabolic genes involved in methane oxidation, inorganic carbon fixation, ammonium assimilation, pollutant degradation, and co-metabolism.	Microalgal photosynthetic carbon fixation coupled with bacterial / methanotrophic carbon and nitrogen metabolism.	Optimization of biomass ratio, biofilm carriers, co-culture design, and microbial community regulation.	Methane conversion, wastewater treatment, nitrogen recovery, pollutant removal, and integrated carbon mitigation.

In Chlorella spp., carbon fixation is often associated with broad tolerance to variable CO_2 supply, pH, light, nutrient availability, and flue-gas conditions. Recent studies have examined operational-factor optimization, transcriptional

reprogramming, mutagenesis breeding, attached cultivation, and flue-gas adaptation as routes to improve CO_2 biofixation and biomass production (Zare et al., 2026; Yang et al., 2025; Li et al., 2025; Hou et al., 2026; Hu et al., 2026). For Nannochloropsis oceanica, regulation of the assimilatory nitrate pathway and nitrogen metabolism can influence energy use and carbon allocation, linking carbon fixation with lipid-rich biomass production under stable pH conditions (Liu et al., 2025a).

Cyanobacteria and related taxa show another set of genetic and biotechnological strategies. In Arthrospira platensis, plasma mutagenesis and adaptive selection have been used to improve tolerance to photoinhibition and high CO_2, supporting CO_2 biofixation and protein- or pigment-rich biomass production (Yuan et al., 2025). In Synechococcus elongatus and Synechocystis sp., adaptive evolution, RuBisCO-related regulation, and phaAB/RuBisCO overexpression can redirect carbon flow toward biomass or storage compounds such as PHB (Ma et al., 2025a; Tharasirivat et al., 2025).

These studies indicate that carbon-fixation improvement is closely connected to photosynthetic regulation and downstream carbon-allocation pathways.

Other taxa provide more specialized examples. Euglena gracilis has been improved through nuclear mutagenesis and adaptive evolution for photoautotrophic growth under flue-gas CO_2, while transcriptome analysis has linked CO_2 concentration with paramylon accumulation (Xin et al., 2024; Yuan et al., 2024). Multi-omics analysis of Mychonastes rotundus has also been used to identify mechanisms underlying CO_2 fixation performance (Zhang et al., 2025b). In Desmodesmus sp., whole-genome analysis helped characterize acid-tolerant carbon-fixing traits, which are relevant for robust cultivation under acidic or flue-gas-related conditions (Wang et al., 2024b). Genetically modified Parachlorella kessleri and transcription-factor overexpression in Chlamydomonas pacifica further show how targeted regulatory manipulation can support biomass accumulation, lipid production, and biofuel productivity (Singh et al., 2024; Gupta et al., 2025).

At the community level, algal-bacterial and methanotroph–microalgae systems extend biological regulation beyond a single strain. In these systems, carbon fixation is coupled with methane conversion, ammonium assimilation, pollutant degradation, and microbial co-metabolism. Optimizing biomass ratio, biofilm carriers, and community composition can therefore improve both carbon mitigation and wastewater treatment performance (Li et al., 2026; Wang et al., 2025a; Zhu et al., 2026; Feng et al., 2025). The evidence summarized in Table 9 suggests that future strain and community engineering should move from single-gene improvement toward integrated regulation of carbon uptake, photosynthetic energy supply, stress tolerance, and carbon allocation. This approach may better support microalgal systems designed for carbon fixation, bioproduct formation, resource recovery, and circular-bioeconomy applications.

6.3. Discussion on Microalgal Biology Contributions to Carbon Sinks

Microalgae contribute to carbon sinks through photosynthetic carbon fixation, biomass formation, carbon export, and interactions with aquatic food webs. Their contribution is not determined by photosynthetic rate alone; it also depends on nutrient availability, light, temperature, community composition, and the fate of fixed carbon after algal growth. These controls matter because phytoplankton elemental composition and biological carbon export can change substantially with nitrogen, phosphorus, irradiance, and temperature conditions (Tanioka & Matsumoto, 2020; Henson et al., 2022).

Microalgal Photosynthesis as a Carbon Capture Mechanism Photosynthesis is the main pathway by which microalgae convert inorganic carbon into organic matter. In aquatic systems, this process is influenced by light supply, dissolved inorganic carbon, nutrient status, and temperature. These factors regulate biomass accumulation and the chemical composition of algal cells, including C:N:P ratios and carbon allocation. A meta-analysis of marine phytoplankton showed that nutrient supply, irradiance, and temperature can shift algal elemental stoichiometry, with direct implications for carbon export and sequestration efficiency (Tanioka & Matsumoto, 2020).

Ecological Role of Microalgae in Natural Carbon Sinks In natural waters, microalgae and phytoplankton form the base of aquatic food webs and contribute to the biological carbon pump. Fixed carbon can be transferred to grazers, released as dissolved organic matter, recycled by microbes, or exported as sinking particulate organic carbon. The strength of this pathway varies with community structure, environmental conditions, and the balance between recycling and export. Recent work has emphasized that many processes controlling ocean biological carbon export remain poorly represented in climate models, making the future response of carbon export uncertain (Henson et al., 2022). Similarly, model-based analysis suggests that phytoplankton diversity may shift under climate change, which could alter productivity, trophic transfer, and carbon cycling (Henson et al., 2021).

Carbon-Concentrating Mechanisms and Physiological Adaptation Microalgal carbon fixation is also shaped by physiological adaptation. Under changing light, temperature, and nutrient conditions, microalgae can adjust photosynthetic machinery, nutrient uptake, carbon-concentrating mechanisms, and metabolic pathways.

Experimental work in the Southern Ocean showed that warming and iron availability changed diatom growth, nutrient uptake, and transcriptional responses, with different diatom groups showing distinct physiological strategies (Jabre et al., 2021). More broadly, recent synthesis work argues that warming and nutrient limitation interact in taxon- and nutrient-specific ways, affecting phytoplankton growth-temperature optima, elemental-use efficiency, and ocean carbon–nutrient feedbacks (Hutchins and Tagliabue, 2024).

Microalgal Bioreactors for Controlled Carbon Sequestration Compared with natural systems, bioreactors provide greater control over light, CO_2 supply, pH, nutrients, mixing, and temperature. This control helps separate the effects of individual drivers and supports strain screening or process optimization. However, high growth in culture does not automatically imply durable climate mitigation. The final value of a microalgal system depends on whether fixed carbon is retained in biomass, long-lived products, sediments, or mineral forms, and whether the energy demand of cultivation and processing remains low. This distinction between carbon

fixation and durable carbon storage is consistent with recent studies showing that biological carbon export can respond adaptively to warming, but with strong regional and ecological dependence (Lomas et al., 2022; Henson et al., 2022).

Microalgal biology supports carbon sinks through three linked processes: photosynthetic CO_2 fixation, physiological adaptation to changing environments, and ecological transfer or storage of fixed carbon. These processes should be evaluated together rather than separately. Future work should connect strain-level traits with community ecology, carbon export, biomass fate, and system-level carbon accounting. This integrated approach will help distinguish systems that primarily convert CO_2 into short-lived biomass from those that can contribute to more stable and measurable carbon sequestration.

7. Microalgal Applications in Sustainability and Industry

7.1 Field Studies Review on Marine Microalgae Carbon Sequestration

Field-oriented studies complement laboratory and photobioreactor-based research because marine microalgal carbon sequestration is strongly shaped by environmental variability. In natural waters, carbon fixation by microalgae and phytoplankton changes with light, nutrient supply, hydrodynamics, community composition, carbonate chemistry, and pollution stress. Recent field-related studies have therefore shifted attention from biomass accumulation alone to broader ecosystem processes, including particulate organic carbon formation, biological carbon export, and regional carbonbudget assessment (Jia et al., 2022; Sun et al., 2025; Yan et al., 2025).

Unlike controlled cultivation studies, field studies usually rely on indicators that are indirect but ecologically meaningful.

These include chlorophyll-based variables, phytoplankton biomass, primary productivity, particulate organic carbon, dissolved inorganic carbon, alkalinity, community composition, and carbon export potential. Work in lakes, coastal waters, tidal flats, continental shelves, marine ranching areas, and the South China Sea shows that phytoplankton carbon sequestration depends not only on photosynthetic uptake, but also on grazing, microbial recycling, sedimentation, and

hydrodynamic transport (Jia et al., 2022; Sun et al., 2026; Yuan et al., 2026; Yan et al., 2025).

Monitoring technologies now play a larger role in evaluating marine microalgal carbon sinks. Traditional field sampling remains useful, but it is often too sparse to capture short-term blooms or rapid changes in carbon fixation. Remote sensing, airborne lidar, ecosystem modelling, and carbon-flow analysis provide broader spatial and temporal coverage and can help estimate phytoplankton productivity, absorption, and carbon-storage capacity or carbon-storage tendency in marine or aquaculture-influenced systems (Yuan et al., 2026; Sun et al., 2025). These approaches support future monitoring frameworks that combine field observation, high-frequency measurements, modelling, and standardized carbon accounting.

Field evidence also shows that high algal productivity does not necessarily indicate durable carbon sequestration. Fixed carbon may remain in biomass, enter food webs, form particulate organic carbon, sink to sediments, or return to the water and atmosphere through respiration and microbial decomposition. Environmental stressors, including nutrient enrichment, pollutants, microplastics, metals, and flue-gas-related compounds, may further alter photosynthetic performance and carbon allocation (Hu et al., 2026; Fan et al., 2026). Field assessment should therefore distinguish short-term CO_2 uptake from longer-term carbon storage and net climate benefit.

Field evidence therefore supports treating marine microalgal carbon sequestration as an ecosystem process rather than as a direct extrapolation of laboratory fixation rates. Controlled experiments remain necessary for identifying physiological mechanisms and optimal cultivation conditions, but field studies are needed to test how these

mechanisms operate under variable light, nutrients, hydrodynamics, community interactions, and pollution pressures. Future work should combine field sampling, remote sensing, high-frequency monitoring, laboratory validation, carbon-flow modelling, and life-cycle carbon accounting to better estimate the contribution of marine microalgae to regional and longer-term carbon sinks (Table S5).

7.2 Microalgae in Wastewater Treatment and Biofuel Production

Microalgae provide a useful link between wastewater treatment and biomass-based resource recovery. In nutrient-rich effluents, they assimilate nitrogen and phosphorus while using inorganic carbon for photosynthesis. The resulting biomass can be directed to biodiesel, biogas, biochar, fertilizer, feedstock, or other bioproducts. This distinguishes microalgal treatment from conventional wastewater polishing: nutrients are not only removed from water but also partly recovered as biological material (Zhu et al., 2026; Feng et al., 2025; He et al., 2025).

Table 10. Microalgal taxa used in wastewater treatment and biofuel production.

Microalgal Taxon or Consortium	Wastewater-Treatment Relevance	Lipid or Biofuel Relevance	Heavy-Metal or Pollutant Removal Relevance	Other Application Contexts
Chlorella vulgaris / *Chlorella* spp.	Frequently used in municipal, aquaculture and industrial wastewater because of stable growth and efficient uptake of nitrogen and phosphorus; also contributes to COD and organic-load reduction when coupled with bacteria.	Moderate to high, depending on strain and culture conditions; lipid accumulation is affected by nitrogen level, CO_2 supply, light and harvesting stage.	Can contribute to metal adsorption and organic pollutant removal, especially in mixed wastewater systems; efficiency depends on pH, biomass dose and wastewater chemistry.	CO_2 biofixation, wastewater polishing, biomass production, fertilizer or feedstock recovery, biodiesel and other bioproducts.
Scenedesmus obliquus / *Scenedesmus* spp.	Suitable for nutrient-rich wastewater and runoff-related systems; commonly reported for nitrogen removal, biomass accumulation and tolerance to variable cultivation conditions.	Moderate lipid potential with relatively robust biomass productivity; often considered for biodiesel-oriented cultivation when nutrient and light conditions are optimized.	Shows moderate adsorption or tolerance to wastewater-associated pollutants and metals; performance is strain- and system-dependent.	CO_2 fixation, wastewater treatment, biodiesel feedstock, biomass valorization and comparative strain screening.
Arthrospira platensis / *Spirulina*	Less frequently used as a primary wastewater-treatment species than *Chlorella* or *Scenedesmus*; more often cultivated in controlled	Generally lower lipid-oriented value; biomass is more valuable for protein, pigments and nutraceutical products than for biodiesel.	Can adsorb or accumulate heavy metals under suitable conditions; useful biomass for bioremediation studies involving Pb, Cd or other	CO_2 utilization, protein-rich biomass, pigments, nutraceuticals and stress-tolerant cultivation.

	alkaline systems.		metals.	
Nannochloropsis oceanica / *Nannochloropsis* spp.	Not usually the first choice for nutrient removal in conventional wastewater; more often used in marine or controlled cultivation systems.	High relevance for lipid-rich biomass and biodiesel-related applications; lipid accumulation is closely linked to nitrogen metabolism and culture pH.	Limited application as a heavy-metal adsorbent compared with green algae or cyanobacterial biomass.	Biodiesel, omega-3 fatty acids, marine bioproducts and carbon-efficient biorefinery systems.
Dunaliella tertiolecta / *Dunaliella* spp.	Potentially useful in saline or alkaline wastewater, but less commonly used in municipal wastewater treatment than *Chlorella* or *Scenedesmus*.	Carbon may be directed toward carbohydrates or stress metabolites; lipid value is generally lower than *Nannochloropsis* in biofuel-oriented systems.	Not commonly reported as a main heavy-metal removal species in the selected literature.	Saline cultivation, bioethanol-oriented carbohydrate production, beta-carotene production and modelling-assisted carbon capture.
Mixed microalgal consortia / algal-bacterial systems	Often effective for nutrient recovery, organic matter removal and wastewater polishing because algae and bacteria share oxygen, CO_2, nitrogen and organic substrates.	Biofuel potential depends on community composition and harvesting strategy; mixed biomass may support biodiesel, biogas, biochar or other downstream routes.	Can improve removal of organic pollutants, nutrients and some metals through combined uptake, adsorption, biodegradation and precipitation processes.	Wastewater treatment, methane conversion, nutrient recovery, pollutant removal, biomass recovery and circular bioeconomy systems.

Among the taxa used in wastewater-related studies, Chlorella and Scenedesmus remain the most common choices. Their relevance reflects stable growth, tolerance to variable nutrient loads, and the capacity to accumulate biomass under relatively simple cultivation conditions. Chlorella vulgaris and related strains have been applied in municipal, aquaculture, and industrial wastewater systems, often with simultaneous CO_2 fixation and nutrient removal (Zare et al., 2026; Hou et al., 2025; Zhu et al., 2026). Scenedesmus species are similarly useful in nutrient-rich effluents and runoff-related systems, especially where robust growth and biomass recovery are required (Gallagher et al., 2026; Ruiz-Ruiz et al., 2024; Subramanian et al., 2025).

The biofuel value of these taxa depends strongly on culture conditions. Nitrogen limitation, CO_2 supply, light intensity, pH, and harvesting time can alter carbon allocation among proteins, carbohydrates, and lipids. For this reason, high nutrient removal and high lipid accumulation do not always occur simultaneously. In practice, cultivation systems often need to balance wastewater treatment efficiency against the desired biomass composition for biodiesel or other downstream uses (Liu et al., 2025b; Cho et al., 2025).

Other taxa have more specialized roles. Nannochloropsis is more relevant to lipid-rich biomass and marine bioproducts than to

conventional wastewater treatment (Liu et al., 2025a). Dunaliella species are better suited to saline or alkaline systems and may be useful where carbohydrate production, beta-carotene recovery, or saline wastewater treatment is targeted (Cho et al., 2025; Sivaraj et al., 2026). Arthrospira/Spirulina is generally less commonly used as a primary nutrient-removal group, but its protein- and pigment-rich biomass gives it value in nutraceutical and bioproduct

applications. It may also contribute to heavy-metal adsorption under suitable conditions (Yuan et al., 2025; Sobhi et al., 2025; Tharasirivat et al., 2025).

Mixed microalgal consortia and algal-bacterial systems are well suited to wastewater treatment because real wastewater rarely involves a single isolated process. In these systems, microalgae provide oxygen and assimilate nutrients, while bacteria degrade organic matter and recycle carbon and nitrogen. This cooperation can improve nutrient recovery, organic pollutant removal, and treatment stability. It also broadens the potential use of harvested biomass, including biofuel production, methane conversion, biochar production, and circular resource recovery (Zhu et al., 2026; Li et al., 2026; Feng et al., 2025; Song et al., 2025; Wang et al., 2025a).

Despite these advantages, large-scale application still faces practical constraints. Culture contamination, variable wastewater composition, harvesting costs, energy demand, and downstream processing can reduce overall environmental benefit. Microalgae should therefore not be framed only as biodiesel feedstocks or only as wastewater treatment tools. A more realistic view is to treat them as components of integrated resource-recovery platforms, where carbon fixation, nutrient removal, pollutant control, biomass valorization, and life-cycle performance are evaluated together (Zhu et al., 2026; Feng et al., 2025; Song et al., 2025).

7.3 Microalgae in Sustainable Development and Carbon-Management Strategies

Microalgae can support low-carbon industrial development when they are integrated into resource-recovery systems rather than treated as stand-alone carbon sinks. In flue-gas or wastewater-linked cultivation, CO_2, nutrients, and residual resources can be converted into biomass for fuels, feed, fertilizers, biochar, or other products. This fits the logic of a circular bioeconomy, but the climate benefit depends on system boundaries, energy demand, and the final use or fate of biomass. Recent techno-economic and life-cycle studies show that algal fuels and products can reduce greenhouse-gas emissions under some scenarios, but performance varies strongly with biomass composition, coproduct allocation, energy inputs, and downstream processing (Wiatrowski et al., 2025; Zhang et al., 2025a).

Carbon-neutrality claims for microalgal systems require transparent carbon accounting. ISO 14068-1:2023 emphasizes quantification, emission reduction, and transparent management of residual emissions, with direct emission reduction and

value-chain removals prioritized before offsetting. This distinction is important because high CO_2 uptake in a photobioreactor does not automatically demonstrate carbon neutrality. Mixing, lighting, harvesting, drying, extraction, and transport can offset part of the carbon benefit if they are energy-intensive. Direct flue-gas cultivation may avoid CO_2 separation and pressurization, but pilot-scale studies still show that cost and net emissions depend on carbon source, reactor design, algal species, and conversion pathway (Singh et al., 2025).

Microalgae are therefore most useful when they function as part of integrated resource-recovery platforms.

Photobioreactors, wastewater-linked systems, and algal-bacterial consortia can combine CO_2 use, nutrient recovery, pollutant removal, and biomass valorization. However, scale-up remains difficult because cultivation stability, harvesting cost, product recovery, and life-cycle performance are still major constraints. Recent reviews of microalgal biorefineries highlight the need for low-energy harvesting, better strain selection, standardized LCA/TEA methods, and stronger links between cultivation design and product value (Geng et al., 2025).

On this basis, microalgae should be viewed as enabling components of low-carbon industrial systems, not as a universal route to carbon neutrality. Their contribution depends on whether fixed carbon is retained in valuable biomass, durable products, biochar, carbonate minerals, or other relatively stable forms. Future work should focus on improving carbon-use efficiency, reducing energy and water inputs, integrating wastewater or flue-gas streams, and applying life-cycle assessment before making carbon-neutrality or carbon-removal claims.

8. Limitations and Future Directions

8.1. Limitations of the Study

Although microalgae have clear potential for carbon fixation, resource recovery, and low-carbon applications, their performance remains difficult to compare across studies. Reported carbon-fixation rates vary with species, strain, cultivation mode, CO_2 source, light, temperature, pH, nutrient conditions, reactor configuration, and carbon-accounting assumptions. Studies also use different indicators, including biomass productivity, CO_2 removal, lipid yield, dissolved inorganic carbon, carbon-fixation rate, and life-cycle greenhouse-gas balance. Because these metrics represent different parts of the carbon pathway, inconsistent reporting makes cross-study comparison and quantitative synthesis difficult.

Another limitation is the gap between laboratory performance and field or industrial operation. Laboratory cultures are usually maintained under stable light, nutrient, temperature, and gas-supply conditions, whereas real systems face contamination, seasonal shifts, variable wastewater quality, unstable CO_2 delivery, and hydrodynamic constraints.

Photobioreactors can improve environmental control, but scale-up introduces additional problems, including light attenuation, oxygen accumulation, gas-transfer limits, biofouling, culture instability, and higher operational costs (Kim et al., 2022; Peter et al., 2022).

Energy and material inputs remain a central concern. Mixing, aeration, harvesting, drying, and lipid extraction can consume substantial energy, which may reduce the net climate benefit of algal biofuels or carbon-capture systems.

Recent life-cycle studies show that the sustainability of microalgae-derived biodiesel is sensitive to cultivation assumptions, energy sources, nutrient inputs, and downstream processing choices (Bradley et al., 2023). Spatially resolved techno-economic and life-cycle modelling also shows that algal biofuel performance depends strongly on location, productivity, electricity mix, hydrogen supply, and nutrient markets (Quiroz et al., 2023).

Economic feasibility is another persistent barrier. Large-scale systems require costly cultivation infrastructure, efficient harvesting, reliable conversion technologies, and stable markets for fuels or coproducts. Biofuel production alone is unlikely to support broad commercial deployment in many scenarios. Integrated biorefinery models, in which fuels are produced alongside proteins, pigments, fertilizers, or other coproducts, may improve economic feasibility, but they also introduce trade-offs among product recovery, energy demand, process complexity, and environmental performance (Pérez-Almada et al., 2023).

Regulatory and ecological issues also remain unresolved. Non-native or genetically modified strains may improve CO_2 uptake, stress tolerance, or lipid production, but their use outside closed systems requires careful containment, ecological risk assessment, and monitoring. In addition, carbon-neutrality or carbon-removal claims should not be based only on

culture-level CO_2 uptake. Such claims require full system boundaries, including cultivation, harvesting, processing, transport, product use, and end-of-life carbon fate.

The central limitation is that microalgal carbon sequestration is highly context-dependent. Future studies should report cultivation conditions more consistently, use comparable carbon-accounting metrics, include energy and material inputs, and clearly distinguish short-term CO_2 conversion from durable carbon storage. Without these improvements, it will remain difficult to determine when microalgal systems provide measurable climate mitigation rather than simply producing short-lived biomass.

8.2. Future Directions

Future research priorities are summarized in Table S5 and should begin with more consistent measurement and comparison of microalgal carbon sequestration. Current studies still differ in their use of biomass productivity, CO_2 removal, carbon-fixation rate, lipid yield, nutrient recovery, and life-cycle emissions. A more useful benchmarking framework should report species, culture mode, carbon source, light, pH, nutrients, energy input, biomass composition, and final carbon fate in a consistent way. Such reporting would make laboratory, pilot-scale, field, and modelling studies easier to compare.

Strain and consortium resilience also require greater attention. Many strains perform well under stable laboratory conditions but lose productivity when CO_2 supply, light, temperature, salinity, or wastewater composition fluctuates.

Recent reviews argue that scalable microalgal carbon capture will require strains and consortia that maintain carbon conversion under dynamic CO_2 supply and variable environmental conditions, rather than only under optimized culture settings (Diankristanti and Ng, 2025; Sadvakasova et al., 2025).

Future cultivation systems should also reduce energy and water demand. Renewable electricity, waste heat, low-energy harvesting, wet biomass conversion, and direct use of wastewater or flue gas could improve net performance. Reactor design should be developed together with strain selection, carbon delivery, harvesting strategy, and downstream processing. Recent work on microalgal biorefineries emphasizes that commercialization depends not only on lipid yield, but also on process integration, coproduct recovery, harvesting efficiency, and life-cycle performance (Sharma et al., 2026).

Economic analysis should therefore move beyond single-product biofuel production. For deployment-oriented economic analysis, microalgal systems should be evaluated as multi-output processes that combine carbon capture, nutrient recovery, wastewater treatment, fuels, proteins, pigments, and other bioproducts. For marine microalgae in particular, system-level integration across cultivation, harvesting, conversion, and product valorization is likely to be more important than improving any single step in isolation (Feng et al., 2026a).

Finally, pilot-scale projects should be designed to test real operating conditions rather than only ideal productivity. These projects should evaluate culture stability, contamination risk, energy balance, product recovery, life-cycle carbon accounting, biomass fate, and ecological safety. Progress will require collaboration among biology, engineering, environmental assessment, economics, and policy. If these issues are addressed together, microalgae will be better

positioned as components of integrated carbon-management and circular-bioeconomy systems rather than as standalone biofuel or CO_2 capture technologies.

9. Conclusion

This systematic review shows that microalgae can contribute to carbon management by converting inorganic carbon into biomass and other carbon-containing products while supporting nutrient recovery, pollutant removal, and biomass valorization. Their carbon-sink contribution, however, depends on species traits, cultivation conditions, carbon source, system design, energy input, and the final fate of fixed carbon. The reviewed studies indicate a shift from single-species laboratory cultivation toward integrated systems, including algal-bacterial consortia, wastewater-linked cultivation, flue gas utilization, natural phytoplankton communities, carbonate-chemistry management, and optimized photobioreactors.

These systems can connect CO_2 fixation with resource recovery, biomass production, biofuels, biochar, fertilizers, and other bioproducts, making microalgae relevant to circular-bioeconomy and low-carbon industrial strategies. Several challenges still limit interpretation and deployment. Carbon-fixation rates, CO_2 removal efficiencies, biomass productivity, and life-cycle metrics are not consistently reported across studies, which limits cross-study comparison. Laboratory performance also cannot be directly extrapolated to field or industrial systems because real applications involve variable light, CO_2 supply, wastewater composition, contamination risk, energy demand, harvesting costs, and downstream processing. Future work should prioritize standardized carbon accounting, robust strain and consortium selection, field validation, low-energy cultivation and harvesting, life-cycle assessment, and clearer evaluation of biomass fate.

Microalgae should be viewed as flexible components of integrated carbon-management systems rather than as a standalone solution for carbon neutrality.

Funding: This research was funded by Enerstay Sustainability Pte Ltd (Singapore) Grant Call (Call 1/2023) _GHG (Project ID SCU-001), Singapore.

Data Availability Statement: The data that support the findings of this study are available from the corresponding author upon reasonable request.

Conflicts of Interest: All authors declare no Competing Financial or Non-Financial Interests.

Author Contributions

[Author contribution statement reserved for final publication.]

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Appendix 1. Table S1. The refinement standard of Search String 2 in the

literature search strategy is a deliberate methodological choice.

Key elements	Rationale of refinement
Screening relevance	Search String 2 was refined to retrieve studies that explicitly address microalgae-related carbon sink accounting, monitoring, calculation, measurement, modelling or assessment. This matched the screening objective of identifying articles with usable evidence on carbon sink quantification rather than papers that only mentioned microalgae and carbon in general terms.
Increased precision	The added methodological terms narrowed the results toward studies reporting carbon fixation, CO_2 removal, dissolved inorganic carbon, particulate organic carbon, life-cycle assessment, techno-economic analysis, remote sensing, photobioreactor performance or related monitoring indicators. This improved the relevance of the final evidence base and reduced records with only marginal links to carbon sinks.
Methodological rigor	Because the review focuses on how microalgal carbon sinks are evaluated, Search String 2 emphasized studies with measurable or traceable methods. This helped retain papers that reported experimental monitoring, carbon balance, model-based estimation, life-cycle accounting, field observation or system-level performance assessment.
Evidence-based exclusion	The screening file shows that many excluded records lacked a description of microalgal carbon sink monitoring or calculation, or did not report results related to accounting or monitoring. The refined string therefore acted as an evidence filter, helping separate method-relevant studies from broader background or application papers.
Focused analysis	The refinement limited the review to studies that could support analysis of carbon sink mechanisms, indicators, performance metrics and monitoring practices. This avoided an overly broad synthesis and allowed the review to focus on species selection, environmental drivers, photobioreactor design, wastewater-linked systems, field monitoring and life-cycle assessment.
Policy and application relevance	Terms related to monitoring, accounting and assessment improved the practical relevance of the search. Studies with these elements are more useful for carbon management, technology evaluation, life-cycle accounting, carbon-neutrality claims and future monitoring/reporting/verification frameworks.
Quality control	A targeted search string reduced the likelihood of including articles that only mentioned microalgae, biofuels or CO_2 without providing substantive information on carbon sink quantification. This strengthened consistency between the search strategy, screening criteria and the final review questions.
Transparency and reproducibility	The refinement makes the literature selection process easier to justify and reproduce. It clarifies why papers were prioritized when they contained explicit evidence on monitoring, calculation, accounting or performance assessment, and why papers without such evidence were excluded during screening.

Table S2. Inclusion and exclusion criteria for article screening.

Screening stage	Inclusion criteria	Exclusion criteria	Rationale	Operational decision
Initial relevance screening	The article explicitly involved microalgae, phytoplankton, cyanobacteria or closely related algal systems.	Articles were excluded if the study did not include microalgae or a comparable algal biological target.	This criterion ensured that the review remained focused on microalgal systems rather than broader plant, soil, forest, macroalgal or purely physicochemical	Retain only records with a clear microalgal or phytoplankton target in the title, abstract, keywords or full metadata.

			carbon-capture studies.	
Topic alignment	The article addressed carbon fixation, carbon sequestration, CO_2 capture, carbon sinks, carbon mitigation, carbon accounting or related carbon-cycle processes.	Articles were excluded if they focused on microalgae but did not address carbon-related outcomes or carbon sink relevance.	This criterion linked the selected literature directly to the research questions on microalgal contributions to carbon sinks and carbon management.	Retain studies with explicit carbon-related wording or measurable carbon outcomes.
Methodological relevance	The article described a method, indicator or framework for monitoring, calculating, measuring, modelling or assessing microalgal carbon sink performance.	Articles were excluded if they lacked a description of microalgal carbon sink monitoring or calculation.	The review aimed to evaluate how microalgal carbon sinks are quantified and monitored, not only whether carbon sequestration was mentioned.	Prioritize records reporting carbon fixation rates, CO_2 removal, biomass carbon, dissolved inorganic carbon, particulate organic carbon, LCA/TEA, remote sensing, photobioreactor performance or ecosystem carbon indicators.
Evidence availability	The article reported results relevant to microalgal carbon sink accounting, monitoring, calculation or performance assessment.	Articles were excluded if they discussed possible carbon sink roles but did not report accounting, monitoring or performance results.	This criterion reduced the inclusion of conceptual or tangential papers and strengthened the evidence base for synthesis.	Retain studies that provided experimental, field, modelling, life-cycle, techno-economic or monitoring results.
Evidence strength for core set	The article contained direct title- or abstract-level evidence of methods, performance metrics, carbon-specific outcomes or strong relevance to microalgal carbon sink assessment.	Articles were deprioritized at the stricter core step if they lacked direct method/performance wording or a strong carbon-specific metric.	This step separated highly relevant evidence from broader but less directly usable studies.	Include records meeting the core evidence rule in the final core dataset; retain less direct papers only in the broader candidate set.
Study type eligibility	Eligible studies included laboratory experiments, field studies, photobioreactor studies, modelling studies, life-cycle or techno-economic assessments, omics/mechanistic	Non-research items, records without sufficient methodological information, and papers outside the carbon-sink scope were excluded.	A broad but methodologically grounded study-type rule allowed the review to capture both mechanistic and application-oriented evidence.	Classify retained studies by evidence type and category after inclusion.

	studies, and applied wastewater or flue-gas studies.			
Category assignment	Included articles were assigned to thematic categories such as special environment carbon sequestration, genetic/mechanistic enhancement, photobioreactor effects, practical applications, microalgae synergy, and new cultivation/model systems.	Records that could not be mapped to a relevant review category after screening were not retained in the core synthesis.	Category assignment supported structured synthesis across mechanisms, monitoring, cultivation systems and applications.	Use the final category labels to organize tables, figures and narrative synthesis.
Quality-control and reproducibility check	The article had sufficient bibliographic and methodological metadata to justify its inclusion and support later verification.	Records with insufficient information to determine relevance, method or reported results were excluded or downgraded during screening.	Transparent criteria helped maintain reproducibility and reduce subjective selection bias.	Document inclusion/exclusion reasons and retain screening metadata for auditability.

Table S3. Related study perspectives.

Category	Item	No	Percentage
Biological related	Biomass productivity / production	33	30.28%
	Chlorophyll / pigment indicators	24	22.02%
	Growth rate / specific growth rate	20	18.35%
	Biomass concentration / biomass density	16	14.68%
	Photosynthetic efficiency / Fv/Fm	15	13.76%
	Lipid productivity / lipid production	11	10.09%
	Lipid content / lipid accumulation	10	9.17%
	Carbohydrate content / carbohydrate production	3	2.75%
	Biomass composition / biochemical composition	1	0.92%
	Protein content / protein production	1	0.92%
Carbon element related	Carbon sequestration rate / capacity	20	18.35%
	Dissolved inorganic carbon / DIC	20	18.35%
	Carbon fixation rate / capacity	14	12.84%
	Carbon sink / carbon storage	12	11.01%
	Carbon content / total carbon	8	7.34%
	Carbon fixation efficiency	8	7.34%
	Alkalinity / carbonate precipitation	6	5.50%
	Carbon balance / mass balance	5	4.59%
	Carbon capture rate / capacity	5	4.59%
	Total organic carbon / TOC	4	3.67%

	Carbon conversion / utilization efficiency	3	2.75%
	Dissolved organic carbon / DOC	2	1.83%
	Particulate organic carbon / POC	2	1.83%
	Carbon uptake rate	1	0.92%
Carbon dioxide related	Direct air capture / flue gas CO_2	20	18.35%
	CO_2 concentration / dissolved CO_2	16	14.68%
	CO_2 tolerance / high-CO_2 adaptation	9	8.26%
	CO_2 fixation rate	8	7.34%
	CO_2 removal efficiency / percentage	7	6.42%
	CO_2 absorption capacity / rate	4	3.67%
	CO_2 consumption / utilization efficiency	3	2.75%
	CO_2 fixation efficiency	3	2.75%
	CO_2 removal rate	3	2.75%
	CO_2 uptake rate	3	2.75%
	CO_2 capture efficiency / rate	1	0.92%
	CO_2 sequestration rate / capacity	1	0.92%
Water quality and nutrient related	pH	23	21.10%
	Pollutant / heavy metal removal	15	13.76%
	Nutrient removal / consumption	8	7.34%
	Dissolved oxygen / oxygen budget	7	6.42%
	Nitrogen removal / recovery	6	5.50%
	Phosphorus removal / recovery	6	5.50%
	COD removal	5	4.59%
System and product related	Remote sensing / modelling indicators	30	27.52%
	Biofuel / biodiesel properties	18	16.51%
	Life-cycle assessment / LCA	17	15.60%
	Techno-economic analysis / TEA	14	12.84%
	Harvesting / flocculation / recovery	7	6.42%

able S4. Inclusion and exclusion criteria applied for articles consideration.

Category		Remarks
Inclusion Criteria	-	Articles were included when they directly addressed at least one research question related to microalgal carbon sinks, carbon fixation or sequestration, carbon accounting, monitoring methods, or research trends and knowledge gaps.
	-	Articles had to involve microalgae, phytoplankton, cyanobacteria, microalgal consortia, algal-bacterial systems, or closely related algal biological systems.
	-	Articles were retained when they reported or assessed carbon fixation, CO_2 capture, carbon sequestration, carbon mitigation, carbon storage, carbon conversion, or carbon-related ecosystem/process indicators.
	-	Articles were prioritized when they described methods for monitoring, calculating, modelling, measuring, or assessing microalgal carbon sink performance, including experimental, field, reactor, omics, LCA/TEA, or modelling approaches.
	-	Articles had to provide usable results related to carbon fixation, CO_2 removal,

	biomass carbon, dissolved or particulate carbon, carbon balance, life-cycle performance, techno-economic performance, or other relevant indicators. - Peer-reviewed research articles and review-like papers with sufficient bibliographic and methodological information were considered. Records with adequate title, abstract, keywords, and/or full-text information were prioritized for detailed screening. - The screening focused on records that could be evaluated reliably based on available English metadata and full-text information. The updated corpus emphasized recent publications, with the final core evidence set concentrated between 2022 and 2026.
Exclusion Criteria	- Articles were excluded when they did not include microalgae, phytoplankton, cyanobacteria, or comparable algal biological systems. - Articles were excluded when they focused on microalgal biology, biofuels, wastewater treatment, or other applications without a clear link to carbon sinks, carbon fixation, CO_2 capture, or carbon accounting/monitoring. - Records were excluded when they lacked a description of microalgal carbon sink monitoring, calculation, measurement, modelling, or assessment. This was the most common broad-stage exclusion reason in the screening file. - Records were excluded when they discussed potential carbon sink relevance but did not report results for carbon accounting, monitoring, calculation, modelling, or performance assessment. - At the stricter core-screening step, records were deprioritized when they lacked direct title- or abstract-level method/performance wording or strong carbon-specific metrics, even if they were retained in the broader candidate set. - Records with inadequate bibliographic, methodological, or result-level information to support reliable classification were excluded or downgraded during screening. - Purely physicochemical carbon-capture studies, macroalgal studies, terrestrial plant or soil carbon studies, and papers not relevant to microalgal carbon sink assessment were excluded.

Table S5. Future Research Directions for Enhancing

Microalgal Contributions to Carbon Sinks

Research Area	Objective	Strategies	Expected Outcomes
Field monitoring of natural microalgal carbon sinks	Quantify carbon fixation, carbon export and storage by marine and freshwater phytoplankton under natural environmental variability.	Combine field sampling with high-frequency monitoring, chlorophyll-based indices, particulate organic carbon, DIC/alkalinity measurements, remote sensing, lidar and ecosystem carbon-flow modelling.	More reliable estimates of microalgal contributions to regional carbon budgets and clearer distinction between short-term CO_2 uptake and longer-term carbon storage.
Environmental-driver and adaptation studies	Clarify how light, temperature, pH, nutrients, salinity, CO_2 supply and pollutants interact to regulate	Design multifactor experiments and field–laboratory validation studies that link physiological responses, photosynthetic efficiency, carbon allocation and community composition.	Improved understanding of the environmental windows that support high carbon fixation and stable biomass or particulate carbon

	microalgal growth and carbon sequestration.		formation.
Optimized cultivation and photobioreactor systems	Improve carbon fixation in controlled systems while reducing energy input and operational costs.	Develop scalable photobioreactors with better light distribution, CO_2 delivery, pH control, mixing, aeration, attached cultivation, membrane carbonation and intelligent process control.	Higher productivity, improved CO_2-use efficiency, better process stability and greater feasibility for pilot- or industrial-scale deployment.
Carbon delivery and carbonate chemistry management	Increase inorganic carbon availability and reduce CO_2 loss during microalgal cultivation.	Optimize bicarbonate/carbonate supply, alkalinity regulation, membrane carbonation, off-site bicarbonation and pH-stat operation under different culture conditions.	More efficient CO_2-to-biomass conversion, improved DIC utilization and stronger links between biological carbon fixation and mineral carbon storage.
Genetic, metabolic and strain engineering	Enhance intrinsic carbon fixation capacity, stress tolerance and carbon allocation in selected microalgae.	Use genomics, transcriptomics, adaptive evolution, mutagenesis, CRISPR-based editing and synthetic biology to target CCMs, RuBisCO-related regulation, photosynthetic electron transport and storage pathways.	Robust strains with higher CO_2 tolerance, improved photosynthetic efficiency, enhanced biomass accumulation and greater potential for bioproduct formation.
Microalgal consortia and wastewater-linked systems	Couple carbon fixation with nutrient recovery, pollutant removal and microbial co-metabolism.	Develop algal-bacterial, methanotroph–microalgae, biofilm and wastewater-integrated systems; optimize community composition, biomass ratio, nitrogen recovery and organic pollutant degradation.	Multifunctional systems that combine CO_2 mitigation, wastewater treatment, methane conversion, biomass recovery and circular resource use.
Life-cycle, techno-economic and carbon-accounting frameworks	Assess whether microalgae-based systems provide net climate benefits after considering full system boundaries.	Conduct LCA, TEA, energy-balance analysis and carbon accounting from cultivation to harvesting, processing, product use and end-of-life carbon fate.	More realistic evaluation of net carbon removal, cost-effectiveness, emission reduction potential and technological readiness.
Policy, market and carbon-credit integration	Support practical adoption of microalgae-based carbon management technologies.	Develop carbon-credit methodologies, policy incentives, standards for monitoring/reporting/verification, and market pathways for algal biomass, biofuels, biochar, fertilizers and high-value products.	Greater investment confidence, clearer commercialization routes and stronger integration of microalgal systems into carbon-management and circular-bioeconomy strategies.
Data standardization and cross-disciplinary collaboration	Improve comparability among field, laboratory, modelling and industrial-scale studies.	Standardize reporting of carbon fixation rates, biomass carbon content, system boundaries, environmental conditions and uncertainty; promote collaboration among ecology, biotechnology, engineering and economics.	More comparable datasets, improved model validation and faster translation from physiological mechanisms to deployable carbon sink

systems.

REVIEW ARTICLE

Trusted Digital Infrastructure for Building Energy MRV and Carbon-Market Integration: A Design-Oriented Review and Reference Architecture for Singapore

AUTHORS · Yuanzhe Li

AFFILIATIONS · Carbon Neutrality Institute, China University of Mining and Technology, Xuzhou 221116, China; School of Civil and Environmental Engineering, University of Auckland, Auckland 1010, New Zealand

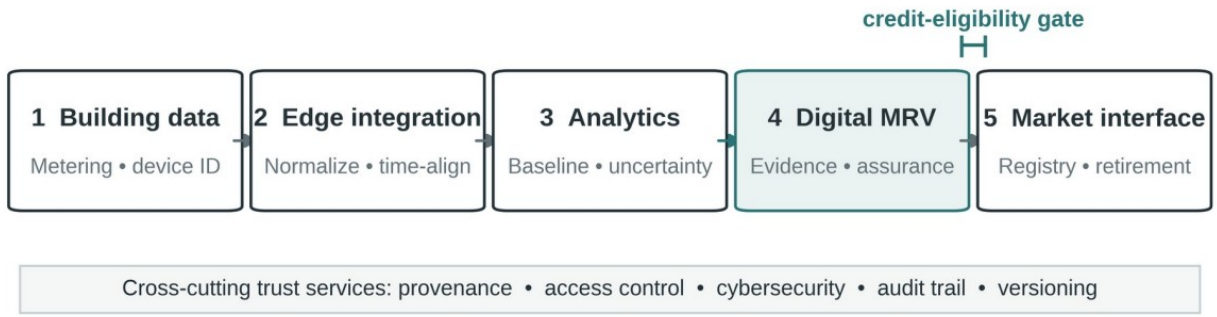
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ABSTRACT

Building-energy platforms are increasingly capable of collecting high-frequency operational data, yet the path from monitored efficiency gains to audit-ready decarbonization evidence remains fragmented. This paper presents a designoriented review and reference architecture that integrates building Internet of Things (IoT), edge-cloud computing, analytics, digital measurement, reporting and verification (dMRV), and carbon-market interfaces. Singapore is used as a regulatory case because it combines ambitious green-building targets, an economy-wide carbon tax, and an international carbon-credit framework. The proposed architecture separates six functions - physical sensing, edge integration, data and compute services, analytics, MRV and assurance, and external registry or market interfaces under a cross-cutting governance plane for identity, provenance, cybersecurity, model versioning, and auditability. A key design principle is that verified energy savings are evidence inputs rather than carbon credits: methodology eligibility, additionality, independent assurance, registry issuance, and, where relevant, Singapore's international-credit eligibility rules remain distinct downstream gates. The review identifies trade-offs between edge and cloud processing, centralized databases and distributed ledgers, and automation and verifier independence, and translates them into a five-phase implementation roadmap. The contribution is a computer-science-centered blueprint for building-energy systems that must be interoperable, verifiable, and regulator-ready rather than merely data-rich.

Keywords: *Building energy informatics; Digital MRV; Edge-cloud computing; Data provenance; Carbon markets; Singapore*

Graphical abstract



Conceptual overview. Building data are transformed into auditable MRV evidence before any separate carbon-credit eligibility and registry

workflow is invoked

1 Introduction

Buildings are a difficult but information-rich decarbonization domain. Their energy demand is determined by interacting technical and behavioral processes, including heating, ventilation and air-conditioning (HVAC), lighting, plug loads, weather, occupancy, operating schedules and maintenance condition. Reviews of building energy management consistently show that digital monitoring and control can reveal substantial efficiency opportunities, but they also show that measured performance is highly context-dependent [1-3]. For computer science, the resulting challenge is not simply to collect more data. It is to maintain a trustworthy chain from heterogeneous physical measurements to analytical models, operational decisions and externally auditable claims.

Singapore provides a useful case in which to examine this chain. The Singapore Green Building

Masterplan sets the '80-80-80 by 2030' targets: 80% of buildings by floor area are to be greened, 80% of new developments by gross floor area are to meet Super Low Energy standards from 2030, and best-in-class green buildings are targeted to achieve an 80% energy-efficiency improvement relative to 2005 [4]. BCA reported that, as of December 2025, approximately 66% of buildings had been greened and best-in-class improvement had reached 72% [4]. At the same time, the carbon tax rose to S$45 per tCO2e for emissions year 2026, while eligible International Carbon Credits (ICCs) may be surrendered for up to 5% of taxable emissions under the national ICC framework [5,6]. These policies create demand

for high-quality digital evidence, but they do not imply that a measured building energy saving automatically becomes a tradable or tax-eligible carbon credit.

The distinction is important. A conventional energy-management system optimizes operations against engineering or business objectives. A dMRV system must additionally preserve source identity, calibration state, baseline assumptions, data-quality decisions, model versions, exceptions, reviewer actions and the evidentiary basis for each reported result. A carbon-credit workflow adds further gates: the activity must be eligible under an applicable crediting methodology; additionality and other integrity criteria must be satisfied; validation and verification must occur with appropriate independence; and issuance, transfer and retirement must be recorded in a registry.

Singapore's carbontax use adds another eligibility layer because only credits accepted under the ICC framework can be used for that purpose [5,6]. Conflating these layers creates a high-risk software design in which a technically valid efficiency result can be mistaken for a financially or regulatorily valid instrument.

The original policy intuition behind an integrated monitoring-to-market platform is therefore retained in this review, but it is recast as a computer-science problem: how should such a platform be decomposed so that each state transition is observable, reproducible and governable? Recent work on building dMRV, blockchain-enabled MRV, verifiable building data, and digital carbon-management systems provides important building blocks [15-20]. However, much of this literature treats data acquisition, verification, registry interaction and market exchange as separate subsystems. The central gap is architectural: there is limited guidance on how these components can be composed while preserving evidence lineage and institutional separation.

This review addresses three questions. RQ1 asks which software and data architecture can connect building telemetry to dMRV without locking the system to a single vendor, cloud or registry. RQ2 asks which trust controls are required so that analytical outputs can be reproduced and independently assured. RQ3 asks how a digital platform can interface with carbon-credit and market systems without implying that verified savings are themselves credits. The resulting contribution is a layered reference architecture, a state-oriented dMRV workflow, a set of architecture invariants and failure controls, and a five-phase deployment roadmap for Singapore. The objective is not to prescribe a single national software product. It is to define a modular trust architecture that can be implemented by multiple providers while remaining interoperable and auditable.

2 Review scope and design principles

This article is a design-oriented narrative review rather than a systematic review or meta-analysis. It synthesizes three bodies of evidence that are directly relevant to the architecture problem: building energy monitoring and management [1-3,23-25], digital MRV and carbon-data systems [15-20], and computer-science standards or design patterns for IoT interoperability, data provenance, secure access and edge-cloud processing [7,8,12-14,21,22]. Singapore policy and regulatory sources are treated as boundary conditions for the case analysis [4-6]. International energy-management and measurement-and-verification standards are used as functional references rather than as software specifications [9-11]. The purpose of this synthesis is to derive design requirements and identify where technical and institutional controls must meet. AI-assisted tools were used to support language refinement, structural reorganization, and preparation of draft schematic elements; the author reviewed and verified the final manuscript content and remains fully accountable for it.

A design-oriented review is appropriate because the target problem is socio-technical. Accuracy at the meter does not

guarantee validity at the claim layer; similarly, a cryptographically tamper-evident database does not establish methodological additionality. The architecture must therefore encode technical quality, process state and organizational responsibility separately. Four principles guide the design. First, interoperability should be achieved through canonical data models, explicit schemas and versioned APIs rather than through a mandatory technology stack. Second, every material transformation should be reproducible from identified inputs, configuration and code or model versions. Third, assurance independence should be represented as an access-control and workflow property, not left to informal practice. Fourth, credit eligibility and registry state should be modeled as downstream states that cannot be inferred from an MRV result alone.

The distinction between evidence and state is central. Evidence is an object that supports a claim: a meter reading, calibration certificate, baseline model snapshot, anomaly disposition, calculation result or verifier approval. State describes where an asset, assessment or credit sits in a controlled process: baseline locked, quality-control accepted, assurance pending, verified, registered, issued or retired. An implementation should allow state changes only when specified evidence predicates are satisfied. This is closer to a workflow engine or state machine than to a dashboard. It also creates a natural interface between engineering data systems and external assurance or registry services.

Table 1 summarizes the requirements derived from the literature and the Singapore case. The requirements deliberately separate protocol interoperability from semantic interoperability, and cyber protection from evidentiary provenance. This avoids a common architectural

shortcut in which encryption is treated as equivalent to trust. Encryption can protect confidentiality and integrity in transit, but it cannot explain whether

the source meter was correctly mapped, whether a baseline model was changed after the reporting period, or whether an independent

reviewer accepted an exception. Those questions require provenance, versioning and process controls.

Table 1 Design requirements derived from the literature and the Singapore case

Design requirement	Computer-science interpretation	Failure if absent	Architectural response
Interoperability	Protocol and semantic compatibility across buildings, vendors and services	Vendor lock-in; inconsistent units, timestamps or meanings	Protocol adapters, canonical schema, schema registry and conformance tests
Provenance	Trace every material result to source, transformation, model and actor	Claims cannot be reconstructed or audited	Persistent IDs, evidence manifests, versioned transformations and append-only history
Reproducibility	Recompute a result using pinned inputs and a declared model or code version	Baseline drift; irreproducible savings estimates	Frozen assurance versions, input snapshots and parameter records
Security and privacy	Protect devices, data, credentials and administrative actions	Tampering, leakage or unauthorized model changes	Least privilege, device identity, encryption, logging and incident-to-evidence linkage
Assurance separation	Represent independent review as workflow and access-control constraints	Self-approval or conflicts of responsibility	Role separation, verifier workspace, authenticated state changes and exception records
Market-state integrity	Keep MRV results distinct from issued and retired units	Double counting or premature labeling of credits	Eligibility gate, authoritative registry IDs, reconciliation and explicit retirement state

All schematic figures in this article were redrawn as vector-oriented diagrams using Python/Matplotlib. They intentionally avoid decorative icons, photorealistic assets and technology-specific logos. This figure style reflects the architecture itself: boxes denote functional responsibilities, arrows denote allowed information or state transitions, and the governance plane represents controls that apply across layers.

3 Reference architecture for trusted building-energy MRV

Figure 1 presents a six-layer reference architecture with a separate governance plane. The lower layers acquire and normalize data; the middle layers provide storage, computation and analytics; the upper layers package evidence for assurance and expose controlled interfaces to external systems. The architecture is deliberately service-oriented. It does not require a single national database or a single cloud operator. Multiple implementations can be conformant if they preserve the same interface contracts, evidence semantics and governance invariants.

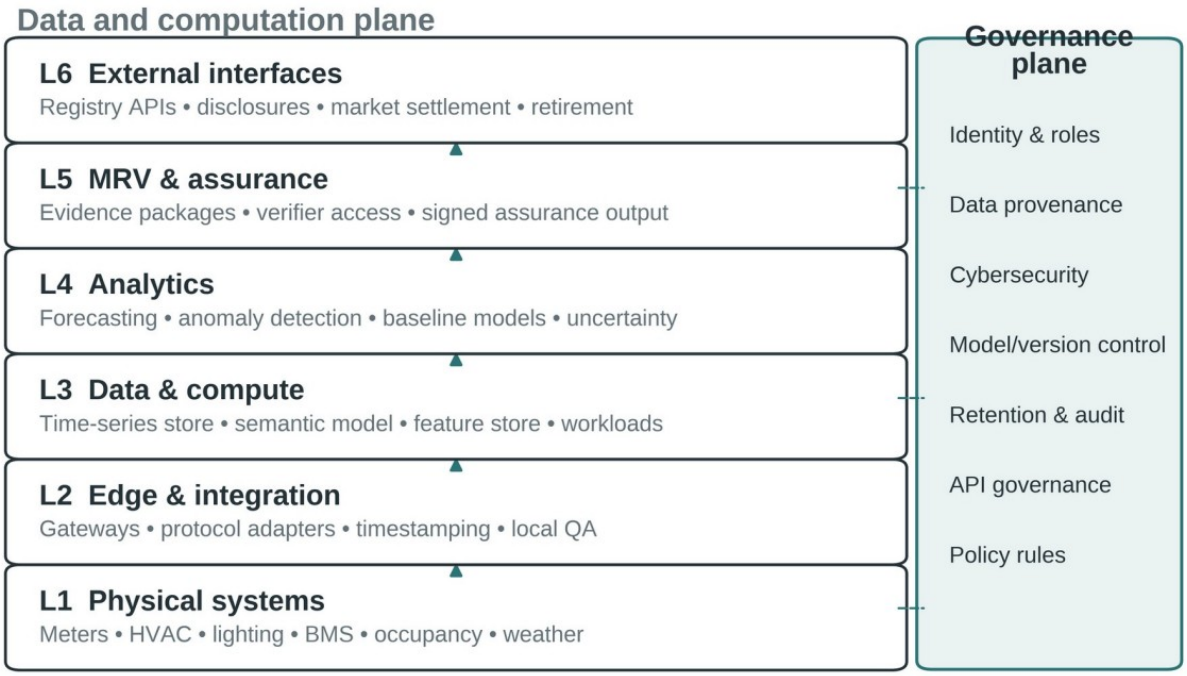


Fig. 1 Layered reference architecture for trusted building-energy data, digital MRV, assurance, and external interfaces

Layer L1 contains physical systems: revenue-grade or sub-metering devices, building management systems, HVAC controllers, lighting systems,

occupancy sensors and environmental measurements. Each source should have a persistent identifier, declared unit, sampling policy and calibration or maintenance reference. The aim is to prevent the 'anonymous time-series' problem, in which values arrive at a platform without enough metadata to establish what was measured and under what conditions. For legacy buildings, this layer may include protocol gateways and virtual meters derived from existing BMS points, but derived values should remain explicitly distinguished from directly measured values.

Layer L2 provides edge integration. Its responsibilities include protocol adaptation, timestamping, local buffering, schema validation, quality checks and, where justified, limited analytics. MQTT and related publish-subscribe approaches are useful for low-overhead telemetry, while national IoT guidance emphasizes secure and interoperable deployments [7,12]. Edge processing reduces bandwidth and can maintain basic functions during cloud interruption, but it creates governance obligations: local algorithms must be versioned, configuration changes logged, and data discarded at the edge only under documented retention rules. A system that performs undocumented filtering before upload may be operationally efficient but evidentially weak.

Layer L3 is the data and compute substrate. A typical implementation will combine a time-series store, metadata catalog, semantic layer, object storage for documentary evidence, and a compute environment for analytics. The semantic layer is especially important when data arrive from buildings of different vintages and vendors. Terms such as 'chiller power', 'air-side energy' or 'occupancy' should map to canonical concepts with explicit units and context. Schema registries and versioned transformations can make such mappings reviewable. Cloud services are natural candidates for elastic compute and cross-site aggregation, but workloads that involve commercially sensitive data or local latency constraints can remain at the edge. The architecture therefore treats edge and cloud as a deployment continuum rather than a binary choice [21,22].

Layer L4 contains analytics. It includes forecasting, anomaly detection, baseline modeling, normalization for weather or occupancy and uncertainty estimation. AI and machine learning can improve pattern recognition, but model performance alone is insufficient for MRV. A baseline used to support an external claim must be reproducible. Let the predicted baseline demand at time t be represented as

E_hat(b,t) = f(x_t ; theta_v) (1) where x_t contains approved explanatory variables and theta_v denotes the parameter set associated with a specific model version v. The version identifier, training window, feature definition and preprocessing pipeline should be fixed when the baseline is locked. A later model update may improve forecasting accuracy, but it should not silently rewrite an already assured reporting period.

Layer L5 implements MRV and assurance. It packages input data, quality decisions, model outputs, uncertainty information and reviewer actions into evidence bundles. The same layer exposes controlled workspaces for internal analysts and independent verifiers. Access separation matters: the user who changes a baseline model should not be able to approve the resulting assurance state using the same credentials. Digital signatures or equivalent authenticated approvals can be used for formal state changes, while sampling, exception handling and professional judgment remain human responsibilities. Automation should reduce the volume of routine checking, not collapse independence.

Layer L6 provides external interfaces. These may include registry APIs, disclosure systems, ESG reporting platforms, green-building portals, project databases or market settlement services. The architecture isolates these interfaces because external rules change more quickly than core telemetry infrastructure. A registry connector should be replaceable without redesigning the data plane. Likewise, a market interface should consume only eligible issued units and should not write directly into raw MRV records.

The governance plane crosses all six layers. It includes identity and roles, provenance, cybersecurity, model and version control, retention, API governance and policy rules. NIST zero-trust principles are relevant because trust should be granted per request and per resource rather than inherited from network location [14]. W3C PROV provides a useful conceptual vocabulary for entities, activities and agents in an evidence chain [13]. In practice, a lighter domain-specific schema may be preferable, but it should preserve the same questions: what object was produced, by which process, using which inputs, under whose responsibility? These controls are architecture-wide because a break in any one layer can invalidate an otherwise correct result.

4 Data acquisition, interoperability and analytics

4.1 Data acquisition and temporal integrity

Building telemetry is rarely clean. Meters can drift, clocks can be misaligned, controllers can reset, points can be renamed during commissioning, and vendors can change data-export formats. For operational dashboards these problems are often patched manually; for dMRV they must be explicit. The ingestion service should therefore distinguish at least four timestamps: device event time, gateway receipt time, platform ingestion time and correction time. Late or corrected values should not overwrite history without preserving the previous version. A quality flag should travel with the observation so that downstream algorithms can distinguish measured, estimated, substituted and rejected values.

Calibration and maintenance metadata should be first-class records rather than attachments buried in an audit folder. The useful unit of provenance is not simply 'meter M produced value V', but 'meter

M, under calibration state C and mapping version S, produced V at time T'. This design allows a verifier to query all results affected by an expired calibration certificate or mapping change. It also allows automated quality rules to identify gaps before they propagate into a reporting calculation.

4.2 Semantic interoperability

Protocol interoperability is necessary but not sufficient. Two platforms can both exchange JSON over HTTPS while assigning different meaning to the same field. A canonical building-energy model should therefore separate physical quantity, unit, asset, location, sampling interval, aggregation rule and quality state. Mappings from vendor schemas to the canonical model should be versioned and testable. The model does not need to be globally universal; it needs stable identifiers and explicit extension points so that sites can add domain-specific terms without breaking common services.

This semantic layer also supports comparison across buildings. Benchmarking is only meaningful when floor area, operating hours, weather normalization and building-use categories are consistently represented. In Singapore, where modern commercial towers coexist with older and strata-titled assets, semantic mediation is a practical precondition for scaling [4]. A national or sectoral platform should therefore publish data contracts and conformance tests rather than require every building to adopt the same BMS vendor.

4.3 Edge-cloud allocation

The choice between edge and cloud processing should be made per workload. Device authentication, buffering, basic range checks and latency-sensitive control belong naturally near the building. Portfolio benchmarking, model training and cross-site anomaly detection benefit from elastic cloud resources. Privacy-sensitive occupancy features may be transformed at the edge so that only aggregated variables leave the site. This split is consistent with broader edge/fog computing practice, where application placement balances latency, bandwidth, reliability and resource constraints [21,22]. The architecture should record where a transformation ran because reproducibility depends on both code and execution context.

4.4 Baselines, adjustment and uncertainty

Energy savings are counterfactual: the post-intervention consumption is observable, but the consumption that would have occurred without the intervention is not. Standards such as ISO 50015 and the IPMVP therefore emphasize baseline definition, relevant variables and adjustment [10,11]. In software terms, a baseline is a versioned model with an approved scope and validity interval. A simple savings expression is

S_t = E_hat(b,t) - E_a,t (2) where E_a,t is actual consumption and E_hat(b,t) is the adjusted baseline. If energy savings are converted to an emissions estimate, an emissions factor EF_t can be applied under a documented accounting rule:

ER = Sum_t (S_t x EF_t) (3) The system should retain the factors, source dates and unit conversions used in Eq. (3). It should also retain uncertainty or sensitivity information when the result is material. A single point estimate without evidence of data quality or model uncertainty may be sufficient for an operations dashboard but is weak evidence for external assurance.

4.5 AI governance

AI can support anomaly detection, missing-data estimation, predictive maintenance and adaptive baseline models. The risk is that convenience encourages continuous model changes. For claim-bearing models, the architecture should support a separation between 'development' and 'assurance' versions. A development model can evolve continuously; an assurance model is frozen for a defined period and released through controlled change management. Feature sets, hyperparameters, training data ranges and validation metrics should be recorded. Explainability should be proportional to the decision: a model used to alert a facility manager can tolerate more opacity than one whose output determines a financial claim.

4.6 Privacy and cybersecurity

Fine-grained energy data can reveal occupancy, production schedules or commercially sensitive operating patterns. Singapore's IoT guidance emphasizes security and interoperability, and the architecture should treat both as design requirements [7,8]. Controls include device identity, mutual authentication where appropriate, encrypted transport, least-privilege access, secret rotation, network segmentation and auditable administrative actions. Security events should be linked to affected evidence objects. For example, a compromised gateway should trigger review of all data ingested through that gateway during the exposure window rather than merely produce a generic cyber incident ticket.

5 Digital MRV and independent assurance

Figure 2 models dMRV as a controlled state sequence rather than a final calculation. State A defines scope and boundary: assets, meters, reporting period and organizational responsibility. State B validates and locks the baseline model. State C applies data-quality controls. State

D quantifies savings or emissions reductions. State E exposes the result and supporting evidence for independent assurance. State F is the

verified output. An exception path returns the workflow to measurement, modeling or documentation when an evidence predicate fails.

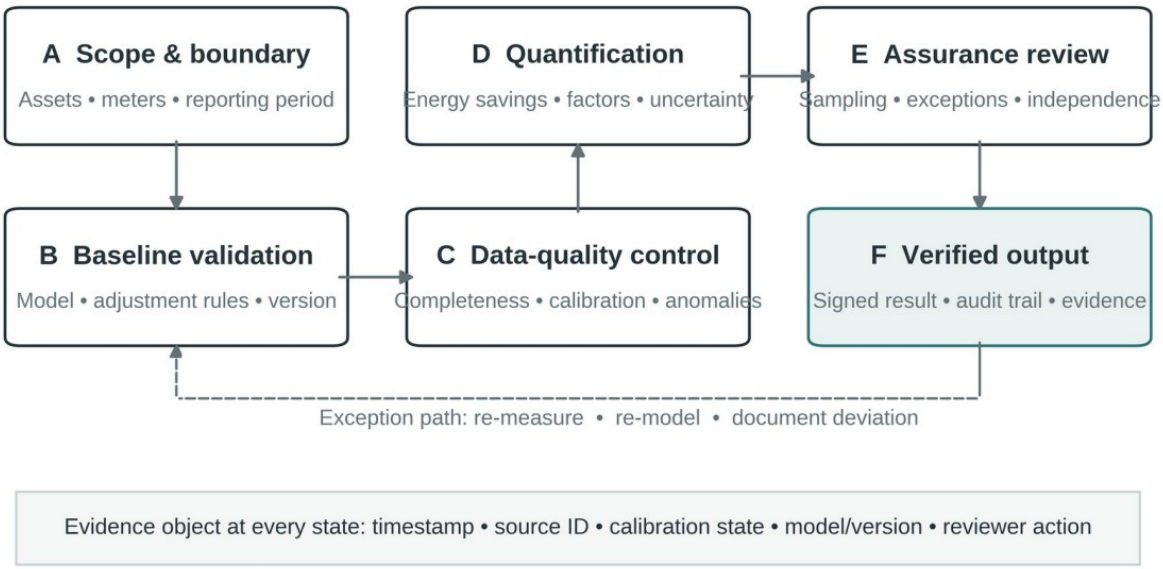


Fig. 2 State-oriented digital MRV workflow in which every transition is supported by a versioned evidence object and exceptions return to

controlled rework

This sequence is intentionally more conservative than a fully automated 'measurement-to-credit' pipeline. Digital MRV research shows that automation can reduce transaction costs and improve transparency, especially when evidence is structured and machine-readable [15-18]. Yet assurance is not simply another algorithmic stage. Professional verification involves materiality, risk assessment, sampling, evaluation of controls and judgment about exceptions. The software should make those activities easier to perform and more reproducible, while maintaining the organizational separation that gives assurance its value.

5.1 Evidence objects and provenance

A practical dMRV platform can represent each evidence item as a structured object O_k with fields such as asset_id,

source_id, event_time, value, unit, quality_state, calibration_ref, transformation_id, model_version and reviewer_action.

A cryptographic digest may be calculated as

h_k = H(O_k || h_(k-1)) (4) to create a tamper-evident chain where appropriate. Equation (4) does not require a blockchain. The same property can be implemented with append-only logs, write-once storage, transparency logs or database mechanisms with controlled administration. The design requirement is detectable modification and traceable history, not a specific ledger technology.

Blockchain-based building MRV has been proposed precisely because distributed ledgers can strengthen tamper evidence and shared state across parties [15,16]. More recent reviews, however, emphasize that blockchain adds complexity and does not solve poor input data, weak methodology or governance by itself [20]. For a national-scale platform, the default should therefore be technology-neutral: use a conventional database when one accountable operator and robust audit controls are sufficient; use a distributed ledger when multiple parties need shared state without a single trusted writer; and

avoid publishing commercially sensitive telemetry on public chains.

5.2 Architecture invariants

Five invariants should hold regardless of implementation. I1, provenance completeness: every material claim is traceable to identified source evidence. I2, model reproducibility: a baseline or quantification result can be recomputed from pinned inputs, code or model version and parameters. I3, role separation: users who create or modify claimbearing evidence cannot unilaterally set the independent assurance state. I4, non-destructive

correction: corrected data retain a link to the previous value and reason for change. I5, eligibility separation: the verified state does not imply credit issuance or carbon-tax eligibility. These invariants provide stronger test criteria than generic statements that a platform is 'secure' or 'transparent'.

5.3 Verifier workspace and sampling

A verifier interface should support risk-based sampling rather than force review of every observation. Useful functions

include stratification by building, meter, data-quality flag, model residual, intervention type and materiality contribution. The interface should allow the verifier to inspect the exact input snapshot used for a reported number, reproduce the calculation, raise an exception and record its resolution. Where automated anomaly scores are used to prioritize samples, the score itself should be versioned and should not hide items from the reviewer.

5.4 Signed outputs and evidence packages

The output of dMRV should be a compact, portable evidence package, not merely a PDF certificate. A package can include a machine-readable manifest, result values, uncertainty summary, model identifiers, evidence hashes, exception log and an authenticated assurance statement. Human-readable reports remain necessary, but the machinereadable layer enables registry or disclosure systems to validate identifiers and state without re-keying information. This is the point at which a software architecture can materially reduce transaction costs: it standardizes the evidence hand-off while preserving independent review.

6 Carbon-credit and market integration

The carbon-market interface is where technical overclaiming is most likely. A verified reduction is an evidence result; a carbon credit is a serialized unit issued under a defined program and methodology after required procedures are completed. Figure 3 therefore places an explicit eligibility gate between verified results and registration. The gate may test activity eligibility, additionality, baseline rules, monitoring requirements, double-counting controls, host-country requirements and program-specific conditions. If criteria are not met, the workflow returns a reject or revise state rather than issuing a unit.

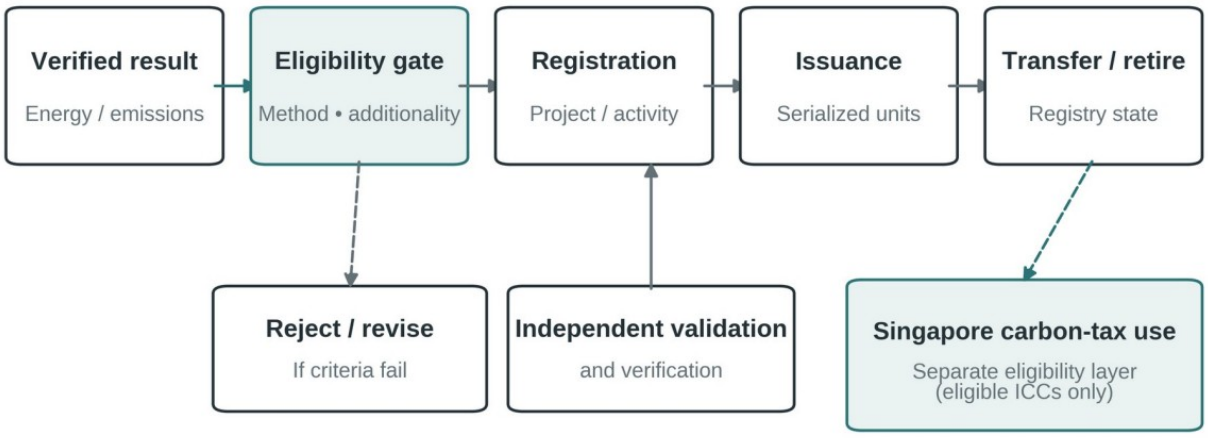


Fig. 3 Separation of verified performance results, carbon-credit eligibility, registry issuance, and Singapore carbon-tax eligibility

6.1 Methodology and additionality gate

Energy-efficiency projects can be difficult to credit when efficiency improvements are already legally required, financially attractive or supported by overlapping incentives. Additionality is therefore not a field that can be inferred from meter data. The platform can assist by assembling evidence on policy requirements, investment decisions, implementation dates and baseline conditions, but the methodology determines how those facts are evaluated. The software should store the methodology identifier and version used for each project so that eligibility can be re-evaluated if rules change.

6.2 Validation, registration and issuance

Independent validation or verification should be represented as a separate actor and state. Once a project satisfies the selected program's requirements, a registry may create a project record and, after successful verification, issue serialized units. The registry is the authoritative source for issuance and retirement state. A local building platform may cache registry data for performance and analytics, but it should not create a competing 'shadow' balance of credits. External identifiers and reconciliation jobs are therefore core integration functions.

6.3 Transfer, settlement and retirement

Market functions should operate only on issued units whose current registry status is known. Smart contracts can automate conditional settlement or transfer instructions, but they should not replace the authoritative registry unless the crediting program explicitly uses that ledger. This distinction prevents a marketplace from appearing to create environmental units simply because it can tokenize data. Reviews of blockchain carbon-market systems show that distributed ledgers can improve traceability, but benefits depend on governance, interoperability and off-chain data integrity [16,20].

Retirement is a terminal environmental state. Once a unit is retired for a claim, the platform should prevent subsequent transfer and should record the claim context where the registry permits. This can be modeled as a state-machine invariant: issued -> transferred is repeatable subject to ownership rules; issued or transferred -> retired is final. Reversal mechanisms, where permitted by a program, should be explicit exceptional states rather than silent edits.

6.4 Singapore's carbon-tax boundary

Singapore's ICC framework creates an additional layer that is separate from ordinary voluntary-market trading. From 1 January 2024, taxable facilities may use eligible ICCs to offset up to 5% of taxable emissions; the carbon tax is S$45 per tCO2e for emissions year 2026 [5]. The eligibility list is program-, host-country- and methodology-specific and is reviewed over time [6]. Consequently, a platform should not label all registry-issued credits as 'carbon-tax eligible'.

Instead, it should query or ingest the current eligibility rules and maintain an eligibility assertion with an effective date and source. A unit can therefore be validly issued and tradable but not eligible for Singapore carbon-tax surrender.

6.5 Market interface design

A market-facing API should expose only the minimum information required for discovery and settlement: unit type,

project identifier, vintage, methodology, registry, status, selected integrity attributes and provenance links. Raw building telemetry, personally identifiable data and proprietary baseline details should remain behind controlled access. Buyers or auditors who need deeper evidence can request permission through the assurance layer. This separation supports transparency without turning operational building data into public market data.

The result is a layered trust model. Building measurements establish the physical evidence; dMRV establishes a verified performance result; the crediting program establishes eligibility and issuance; the registry establishes unit state; and, where relevant, Singapore's ICC rules establish tax-use eligibility. Each layer can invalidate a downstream claim, and none should be collapsed into a single 'verified' flag.

7 Five-phase implementation roadmap for Singapore

The reference architecture can be implemented incrementally. Figure 4 reframes the original five-phase strategy as a sequence of capability releases with explicit exit criteria. This reduces the risk of building a visually impressive national platform before the evidence model and governance controls are mature.

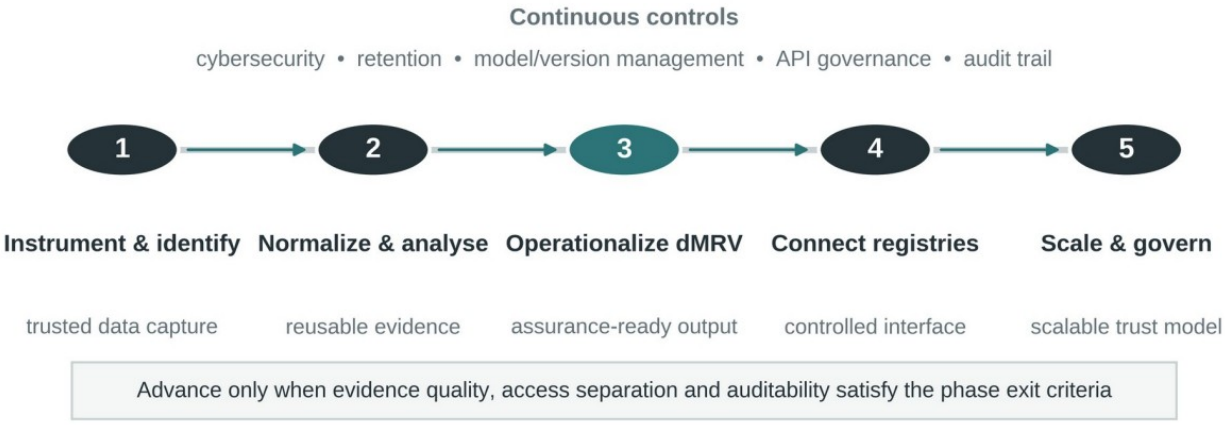


Fig. 4 Five-phase implementation roadmap with cross-cutting controls and evidence-based exit criteria

Phase 1 - Instrument and identify. The first phase creates a trusted data capture layer. Participating buildings map meters and BMS points to persistent asset identifiers, establish time synchronization, record protocol and calibration information, and define ownership of each data source. Pilot sites should deliberately include heterogeneous building types and legacy systems because interoperability problems are more informative than demonstrations on a single modern building. Exit criteria include stable source identity, measurable data completeness and reproducible mapping from physical asset to digital point.

Phase 2 - Normalize and analyse. The second phase introduces the canonical semantic model, data-quality rules, baseline services and cross-building benchmarking. The system should publish API schemas and conformance tests so that multiple vendors can contribute data. This phase is also the appropriate point for edge-cloud workload policies and privacy controls. Exit criteria include versioned transformations, defined quality states, reproducible baseline calculations and an auditable record of model changes.

Phase 3 - Operationalize dMRV. The third phase adds evidence manifests, exception management, sampling support, verifier workspaces and signed assurance outputs. A limited set of building-efficiency use cases should be selected to test the workflow end to end. The objective is not to maximize the number of automated calculations but to demonstrate that an independent reviewer can reproduce a material result from the evidence package. Exit criteria include role separation, traceable exceptions, repeatable quantification and successful independent assurance.

Phase 4 - Connect registries and credit workflows. Only after the MRV layer is stable should the platform integrate methodology eligibility, project registration, issuance status, transfer and retirement. Registry adapters should be modular and reconciled against authoritative external state. Singapore ICC eligibility should be implemented as

a separate rules service with effective dates rather than hard-coded into project records. Exit criteria include prevention of duplicate unit state, successful reconciliation, explicit eligibility provenance and correct handling of rejected or retired units.

Phase 5 - Scale and govern. The final phase addresses national-scale operations, API governance, model monitoring, cybersecurity exercises, retention policy, verifier capacity and regional interoperability. At this stage the platform is less a technology project than a governed digital infrastructure. Change-management procedures should cover schema evolution, algorithm upgrades, new registry connectors and rule changes. A public technical specification can support competition among implementation providers while preserving compatibility.

Government support remains relevant, but incentives should be tied to evidence quality rather than platform enrollment alone. Grants or retrofit schemes can require conformant data interfaces, while Green Mark or related programs can reuse validated performance data where appropriate. Such reuse should reduce reporting burden without conflating certification schemes. BCA, NEA, IMDA and market actors have different mandates; the architecture should allow them to consume the evidence needed for their functions rather than force all governance into a single institution [4-8].

For ASEAN replication, the portable asset is the architecture contract, not Singapore's policy logic. Neighboring jurisdictions can reuse the telemetry, provenance, assurance and registry-interface patterns while replacing tax rules, eligible methodologies and data-sovereignty controls. This modularity is more realistic than exporting a single platform instance because legal and market rules vary substantially across countries.

8 Discussion

8.1 Computer-science contribution

The main contribution of this review is to shift the unit of design from a 'carbon dashboard' to an evidence-bearing distributed information system. The architecture combines concepts from IoT integration, edge-cloud computing, provenance, access control, workflow engines and external API governance. The resulting system is judged not only by prediction accuracy or throughput but also by reproducibility and authority: can an external party determine what data and model created a result, who approved it, and which system has the authority to change its state? These properties align the building-decarbonization problem with trusted computing and software engineering rather than treating it solely as an energy-policy problem [13,14,18,19].

8.2 Architectural trade-offs

Several design choices have no universally correct answer. Centralized databases simplify administration and can provide strong audit controls, while distributed ledgers can reduce dependence on a single writer when multiple organizations share state [15,16,20]. Edge processing reduces latency and exposure of raw data, while cloud processing simplifies fleet-wide analytics and model management [21,22]. Highly adaptive AI models may improve forecasting, while locked models are easier to assure. The reference architecture does not eliminate these trade-offs; it makes them explicit and places governance requirements around them.

A practical example is baseline recalibration. An operator may want the baseline model to adapt as occupancy

changes. From an operational perspective this is sensible; from an assurance perspective continuous adaptation can make the counterfactual unstable. One solution is dual-track modeling: an adaptive operational model runs continuously, while an assurance model is versioned and frozen for each reporting interval. Differences between the two can be monitored as an indicator of model drift. This pattern preserves operational value without sacrificing reproducibility.

8.3 Failure modes and controls

Table 2 summarizes failure modes that arise when the platform is scaled. Several are conventional data-engineering issues, such as missing telemetry and API mismatch. Others are socio-technical, such as baseline gaming or conflicts between project development and verification.

The controls therefore combine technical mechanisms and governance. This is deliberate. No cryptographic mechanism can determine

whether a verifier is organizationally independent; no policy statement can recover a missing time-series. Reliable dMRV requires both.

Table 2 Representative failure modes and controls for a scalable dMRV platform

Failure mode	Detection signal	Control	Residual risk / owner
Sensor drift or expired calibration	Bias against peer meters; calibration expiry alert	Calibration metadata, drift tests, quarantine affected interval	Physical metering owner
Clock or interval misalignment	Unexpected load-shape shifts; duplicate or missing intervals	Multi-timestamp ingestion, clock health checks, deterministic resampling	Data-platform operator
Missing or substituted data	Completeness ratio; quality-state distribution	Documented substitution rules and materiality threshold	MRV analyst and verifier
Baseline or model drift	Residual deterioration; model-version change	Separate development and assurance models; locked reporting versions	Model owner
Undocumented data transformation	Hash or manifest mismatch; irreproducible result	Versioned transformation pipeline and immutable execution log	Platform operator
Assurance conflict	Same role creates evidence and sets final assurance state	Segregated roles, access policies and reviewer authentication	Assurance organization
Credit-eligibility conflation	MRV result displayed as issued unit without registry ID	Distinct schemas and state machine; methodology/eligibility gate	Crediting workflow owner
Registry/API inconsistency	Local balance differs from authoritative registry	Scheduled reconciliation, idempotent APIs and error queue	Registry integration operator

The most consequential failure mode is semantic overclaiming: a user interface shows a quantity as a 'carbon credit' before methodology, assurance and registry states are satisfied. This is not merely a wording problem because downstream systems may trade or report the quantity. The architecture

prevents it by separating data types and state transitions. An MRV result should have a schema distinct from a registry unit. Conversions between them should be performed only by a crediting workflow that records the applicable methodology and external identifier.

8.4 Economic and organizational implications

Better digital evidence can reduce repeated manual data extraction and improve audit targeting, but a national platform does not remove the costs of metering, calibration, modeling, assurance or methodology compliance. For small building owners, aggregation may still be necessary. An aggregator can operate a shared data and MRV service while preserving project-level source identifiers and evidence. This design allows transaction costs to be shared without making all projects indistinguishable.

Verifier capacity is another constraint. Scaling to many buildings could overwhelm a purely manual assurance model. The appropriate response is selective automation: automatic completeness checks, calibration expiry alerts, residual analysis and risk-based sampling can focus human attention on material exceptions. The platform should measure verifier workload and false-positive rates so that automation is evaluated as an assurance support tool rather than assumed to be beneficial.

8.5 Regional portability

The architecture is portable because the data and trust layers are more stable than policy rules. Building telemetry, provenance and controlled assurance are needed regardless of whether a jurisdiction uses a carbon tax, emissions trading scheme or voluntary market. The registry and policy adapters can therefore be jurisdiction-specific. Regional interoperability would benefit from shared identifiers, machine-readable methodology metadata and common evidence manifests, while allowing each country to retain authority over eligibility and claims.

The broader implication is that climate infrastructure should be designed as composable digital public infrastructure. Open interface specifications and conformance tests can allow commercial providers to compete on analytics and user experience without fragmenting evidence semantics. Public governance can then focus on the rules that determine trust rather than on operating every software component.

9 Limitations and future research

This review has four limitations. First, it is a design-oriented synthesis rather than a systematic review; the literature was selected for relevance to architecture, dMRV and the Singapore case, so it should not be interpreted as an exhaustive bibliometric account. Second, the proposed architecture has not been benchmarked through a productionscale prototype. Performance characteristics such as telemetry throughput, storage cost, verifier workload and registry latency therefore remain to be measured. Third, carbon-credit methodologies and national eligibility rules evolve. The separation of eligibility into an external rules layer reduces this risk but does not eliminate the need for continuous legal and methodological review. Fourth, the architecture focuses on operational energy and associated emissions. Wholelife carbon, embodied emissions, renewable-energy certificates and other environmental attributes would require additional accounting boundaries and evidence types [24,25].

Future research should implement the reference architecture in at least two heterogeneous building portfolios and evaluate it along both computer-science and assurance dimensions. Technical metrics should include ingestion reliability, schema-conformance error rate, provenance completeness, model reproducibility, query latency and API reconciliation. Assurance metrics should include exception density, percentage of evidence automatically accepted, reviewer time per material item and frequency of post-verification correction. Security evaluation should include compromised-device and unauthorized-model-change scenarios.

A second research direction is formal verification of state transitions. The architecture invariants can be encoded in a

workflow specification so that illegal transitions, such as verified -> issued without an eligibility record or retired -> transferred, are rejected by construction. A third direction is privacy-preserving portfolio analytics, including federated or edge-assisted learning when raw building data cannot be centralized. A fourth is machine-readable methodology logic. If methodology requirements can be represented as versioned rules with human override and explanatory evidence, registry integration can become more reliable without pretending that additionality or materiality can be fully automated.

Finally, human factors merit more attention. Facility managers, data engineers, verifiers, regulators and market operators interact with the same evidence at different levels of abstraction. Interface design should expose enough provenance for trust without overwhelming routine users. Usability studies could therefore complement system benchmarks and help determine how much technical detail should be displayed at each role.

10 Conclusion

Digital building decarbonization requires more than sensors, dashboards and a carbon marketplace. It requires a software architecture that preserves the meaning and authority of data as they move from physical measurement to analytical result, independent assurance and, where applicable, carbon-credit issuance. This review proposes a sixlayer reference architecture with a cross-cutting governance plane, a state-oriented dMRV workflow and explicit separation between verified energy savings and carbon-credit states.

For Singapore, this separation is especially important. The country's Green Building Masterplan, carbon tax and ICC framework create strong incentives for high-quality digital evidence, but they operate through distinct regulatory mechanisms [4-6]. A trustworthy platform should therefore integrate them through modular interfaces rather than collapse them into one automated conversion. The same principle improves regional portability: telemetry, provenance and assurance services can be shared across jurisdictions, while methodology and policy rules remain replaceable adapters.

The proposed five-phase roadmap begins with source identity and interoperability, then adds semantic normalization and analytics, operationalizes digital MRV, connects external registries, and finally scales governance. The architecture is intentionally technology-neutral. Cloud, edge, AI and distributed ledgers are implementation choices whose value depends on the problem they solve. The invariant requirements are stronger: reproducibility, provenance completeness, role separation, non-destructive correction and explicit eligibility state. Designing around these properties can turn building-energy data from a passive monitoring resource into trustworthy digital infrastructure for decarbonization.

Statements and Declarations

Funding: This research was funded by Enerstay Sustainability Academy (Singapore), Grant Call 2/2025 - CUMT (Project ID SDG-2025).

Competing interests: The author declares no competing financial or non-financial interests that are directly or indirectly related to this work.

Ethics approval: Not applicable.

Consent to participate: Not applicable.

Consent for publication: Not applicable.

Data availability: No new datasets were generated or analyzed for this review. The article synthesizes published literature, technical standards and publicly available Singapore policy and regulatory sources.

Code availability: No software prototype or analysis code was evaluated in this review. The schematic figures were prepared programmatically using Python/Matplotlib.

Author Contributions: [reserved for final publication]

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RESEARCH ARTICLE

An Audit-Ready Framework for Carbon Management in Hotel Food and Beverage Systems Using Hybrid Carbon Accounting and Evidence Chain Governance

AUTHORS · Yuanzhe Li; Jiahui Wu; Zeean Tay; Haizhou Wang; Zhiyong Wu

AFFILIATIONS · Carbon Neutrality Institute, China University of Mining and Technology, Xuzhou 221116, China; Enerstay Sustainability Academy, Enerstay Sustainability Pte. Ltd., Singapore 118136

PUBLISHED ONLINE · 1 NOVEMBER 2026 DOI · NOT ASSIGNED ACCESS · FREE TO READ; ALL RIGHTS RESERVED

ABSTRACT

Hotel food and beverage (F&B) operations are frequently embedded within aggregated hotel carbon inventories and are therefore under-managed as a distinct, carbon-intensive subsystem. Evidence from food-system LCA syntheses and hospitality-sector decarbonization initiatives indicates that dominant hotel emission drivers often sit outside traditional building-energy boundaries, particularly where restaurants, banquets, bars, and central kitchens are material revenue streams 1–3. In these settings, F&B emissions are shaped by upstream agricultural production, processing and packaging, cold-chain logistics and refrigerants, menu structure and portioning, kitchen engineering, and waste generation. We propose the Food & Beverage Carbon-Efficiency System (F&B–CES 2.0), an audit-ready framework that integrates hybrid activity-based and spend-based accounting, an ISO-aligned four-layer evidence chain, and a tiered Copper–Silver– Gold rating mechanism linking performance thresholds to data integrity and governance maturity. The system supports decision-making in menu engineering, procurement governance, capital planning, and product-level lowcarbon dish/banquet claims.

Keywords: *Hotel food & beverage decarbonization; Hybrid activity-based and spend-based carbon accounting (AB+SB); Audit- ready evidence chain; Menu engineering and procurement governance; Food waste avoided emissions and circularity*

1. Introduction

Food and beverage (F&B) operations have historically been treated as a supporting function within hotel management, and carbon measurement practices in hospitality have largely followed this framing. In many hotel inventories, F&B- related impacts are either embedded within aggregated operational energy, water, and waste accounts or treated as generic procurement emissions, rather than governed as a distinct product system with its own lifecycle boundary, data architecture, and decision levers [1-2]. This practice is increasingly misaligned with both scientific and market signals [3- 4]. From the science side, the lifecycle evidence base indicates that upstream food production and land-use drivers dominate the climate impacts of many foods; therefore, the same kitchen and the same service format can exhibit markedly different footprints depending on ingredient class and sourcing model [1,2,5]. A concrete hotel example is a banquet menu that switches a signature centerpiece from ruminant-based items to poultry

or plant-forward alternatives: on-site cooking energy may change little, but upstream emissions can shift substantially because the dominant driver is the ingredient profile rather than the cooking station. From the market and governance side, expectations are moving toward auditable, product-level claims and assured disclosures, including climate disclosure regimes and third-party assurance, which increases the risk of reporting aggregated numbers that cannot be defended at the dish or banquet level [4-6]. Operationally, hotel kitchens and cold chains behave more like micro-industrial facilities than like generic “building subsystems.” Their carbon intensity depends on duty cycles, thermal leakage, process sequencing, and equipment selection, while food waste dynamics are shaped by forecasting, portioning, buffet design, and service style [7-9]. These mechanisms were made unusually visible by COVID-19 disruptions: many hotels reduced or redesigned buffets, simplified menus, increased room service and takeaway packaging, and operated with staffing constraints.

In some properties, reduced buffet overproduction lowered food waste, while increased single-use packaging shifted burdens upstream and into waste management. Post-pandemic recovery has not simply returned to 2019 patterns, and corporate clients increasingly request verifiable sustainability information for events [10-11].

This paper addresses the gap by presenting F&B–CES 2.0, a systems-based carbon-efficiency framework that moves beyond footprint quantification toward integrated management, verification, and performance rating. The contribution is threefold. First, we define a hybrid AB+SB accounting architecture within an ISO 14067-compatible lifecycle boundary that is tailored to hotel F&B realities. Second, we propose a verification logic that operationalizes ISO 14064-3 style evidence requirements through a four-layer evidence chain, enabling audit-ready data governance rather than ad hoc calculations. Third, we translate the framework into an implementable tiered rating (Copper–Silver–Gold) that aligns performance thresholds with data integrity, traceability, and

operational maturity. Throughout, we emphasize decision-relevant functional units (per cover, per kg dish, and per banquet/event) so that measurement can be translated into menu engineering, procurement governance, CapEx planning, and defensible product-level claims.

2. Limitations of Current Approaches

Existing approaches to hotel carbon measurement have delivered important progress in transparency, but they remain poorly suited to governing F&B as a carbon-intensive product system. The limitation is structural: most methods are designed for property-level comparability or corporate reporting completeness, not for identifying and controlling the physical and organizational mechanisms that generate emissions in kitchens, cold chains, and

menus. When this mismatch is ignored, carbon numbers are either too aggregated to guide action or too fragile to verify [12].

2.1 Boundary leakage and upstream invisibility (AB without governed SB)

Activity-based accounting (AB) provides high-confidence measurement for on-site energy and for waste quantities when weighing is in place. However, AB alone cannot capture the upstream dynamics that dominate most food footprints. In day-to-day hotel practice, this means that a carbon inventory can look “stable” while the true drivers are shifting: a chef changes a recipe to include more cheese for perceived quality; a supplier changes processing or packaging; a hotel substitutes imported ingredients during supply disruptions; or a bar program increases premium imported spirits [13-15]. These changes can materially alter lifecycle emissions even when kitchen kWh and waste mass remain similar.

Spend-based (SB) estimation grounded in MRIO databases can provide boundary completeness for heterogeneous purchasing categories when primary data are unavailable. Yet in hospitality it is often used informally, without explicit rules for category eligibility, mapping logic, and overlap control with AB categories. This creates boundary leakage, double counting, and high-uncertainty blind spots, especially for outsourced production and imported beverages. The core issue is not that SB is “inaccurate” per se, but that without governance it cannot function as an auditable completeness layer [16].

2.2 Operational reality is a coupled socio-technical system

Hotel kitchens behave as engineered production systems with duty cycles, thermal leakage, process sequencing effects, and behavioral variability that is not captured when kitchen loads are merged into whole-building electricity [17]. As a result, common interventions such as equipment replacement or cold-room retrofits are difficult to justify, prioritize, or verify. Food waste illustrates the same problem from another angle. Waste is frequently tracked for cost reasons, but rarely as a verifiable mitigation pathway because waste categories are not linked to causal levers. A breakfast buffet can generate persistent overproduction because attendance is uncertain; a banquet can generate plate waste because portion size must satisfy service expectations; and an all-day dining menu can generate preparation loss because of low turnover for niche items. Without stable categorization and process evidence, reductions cannot be translated into avoided emissions credibly [18].

2.3 Weak linkage between carbon metrics and service economics

Most hotel decision-making is constrained by service commitments, brand positioning, and revenue expectations, particularly in MICE operations. Carbon metrics that are not normalized to service units (covers, events) and economic units (revenue) are hard to compare across properties and easy to misinterpret. This is a key reason why hotels can report whole-property indicators while remaining unable to decide which menus, procurement categories, and process changes would most credibly reduce emissions [19-20].

All above listed limitations prevent hotels from closing the management loop: measurement does not translate into diagnosis, diagnosis does not translate into prioritized interventions, and interventions cannot be verified and

institutionalized. F&B–CES 2.0 is designed as a response to this failure mode by integrating boundary design, evidence design, and performance translation.

3. The F&B–CES 2.0 Framework

F&B–CES 2.0 is structured as a management system rather than as a single accounting method. It integrates three management lines—Accounting, Audit & Verification, and Rating—with six interacting systems that correspond to the dominant causal mechanisms of F&B emissions in hotels (Fig. 1). The accounting line answers “what is the footprint and where does it originate” by enforcing explicit lifecycle boundaries and consistent functional units. The audit and verification line answers “how do we prove it” by specifying a traceable evidence architecture that links raw data, operating procedures, and model outputs to governance controls [13,21]. The rating line answers “how well are we performing” by turning carbon efficiency, circularity maturity, and data

integrity into a comparable tiered outcome. For orientation, Fig. 1 summarizes the overall architecture of F&B–CES 2.0, Fig. 2 summarizes the lifecycle boundary and AB+SB coverage logic, and Fig. 3 summarizes the evidence-chain design. The decision-oriented metric set used throughout the framework is summarized in Table 4, while the condensed tiering logic is shown in Table 1.

These cross-references make it straightforward to compare F&B–CES 2.0 against existing national and regional hotel assessment frameworks in a structured way.

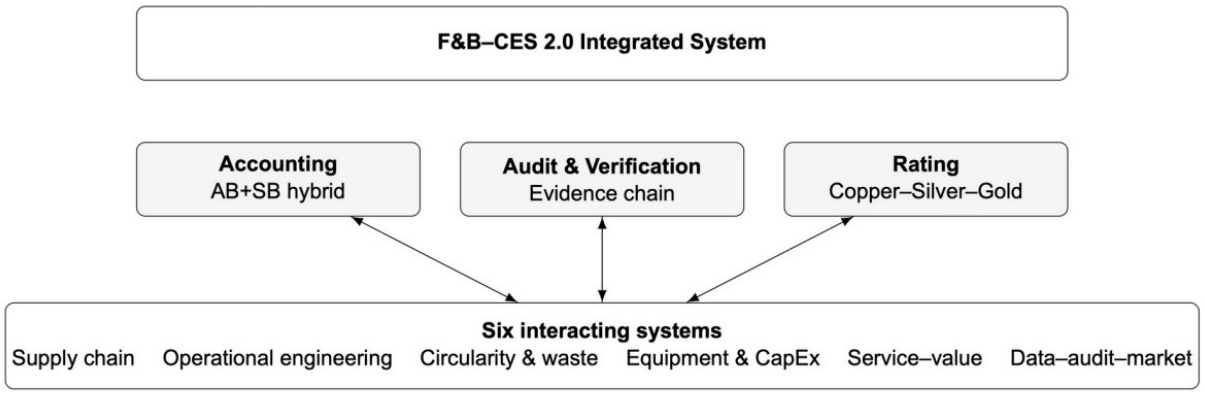


Figure 1. System-level structure of F&B–CES 2.0.

The six systems represent the operational domains that must be jointly managed for F&B decarbonization. Supply Chain Carbon Architecture addresses the reality that upstream ingredients and

beverages typically dominate lifecycle impacts, requiring multi-tier supplier segmentation, traceability, and menu-structure diagnostics [24-25]. Back-of-House Operational Engineering recognizes that kitchens and cold chains are engineered energy systems, and that process mapping and duty-cycle management are required to interpret meters and to prioritize interventions. Circularity and Food Waste provides the logic to treat avoidable food waste as a measurable mitigation pathway, aligning waste prevention, segregation, and recovery with avoided-emissions accounting under explicit rules [22-23,26]. Equipment and CapEx Decarbonization extends beyond operational energy into lifecycle impacts of capital goods and refrigerants, allowing hotels to rank investments by carbon payback and marginal abatement logic. Service–Value Coupling links carbon intensity to service design and revenue structure, enabling benchmarking across hotel segments and avoiding perverse incentives that penalize higher-service formats without recognizing value creation.

Finally, Data–Audit–Certification– Market Integration connects measurement to assurance and communication, enabling low-carbon dish/banquet claims, internal carbon pricing, and procurement governance that can withstand external scrutiny [9,11]. Table 1 provides a condensed view of the tiering logic. The full 48-indicator checklist can be maintained as a separate audit annex aligned with the evidence chain.

Table 1. Condensed indicator-tiering structure for F&B–CES 2.0.

System	Selected indicators (examples)	Tier logic
Supply chain	High-carbon SKU share; ruminant ratio; traceable Gold: low high-carbon ingredients; MSEI; SB coverage	Gold: low high-carbon share and high traceability; Silver: mid; Copper: high and low traceability
Operational engineering	kWh/cover; refrigeration kWh/cover; equipment utilization; thermal leakage	Gold: low energy intensity and high utilization; Silver: mid; Copper: high losses
Circularity & waste	Prep loss; plate waste; segregation accuracy; recovery rate; FWAE	Gold: low waste and high recovery; Silver: mid;
		Copper: high waste
Equipment & CapEx	Annualized embodied carbon; carbon payback; low-GWP refrigerant share	Gold: short payback and low-GWP; Silver: mid; Copper: long payback
Service–value	RCE; banquet kg CO_2e/cover; SII	Gold: low RCE at given service level; Silver: mid; Copper: high
Data–audit– market	AB+SB coverage; traceability; certified low-carbon dishes/banquets	Gold: high coverage and certification; Silver: partial; Copper: limited

4. Accounting System: AB+SB Hybrid within Lifecycle Boundaries

4.1 Lifecycle Boundary and functional units

The accounting boundary follows a product-oriented lifecycle logic compatible with ISO 14067 and ISO 14040/14044, separating upstream supply-chain impacts, on-site operations, and capital goods while preserving traceability to decision levers (Fig. 2) [9,14-15]. Upstream processes include agricultural production, primary processing, packaging, certification and compliance attributes (where material), and logistics, with explicit handling of cold-chain transportation when relevant. Operational processes include kitchen energy and fuel use, refrigeration electricity and refrigerant leakage where data allow, water use and dishwashing, laundry impacts attributable to F&B service, and end-of-life handling for food waste and packaging consistent with the chosen waste accounting rules. Capital goods include kitchen equipment, cold-chain infrastructure, and refurbishments, which are annualized to avoid distorting year-to-year operational comparisons and to reflect that CapEx decisions are a material lever in decarbonization planning [16-17,27-28].

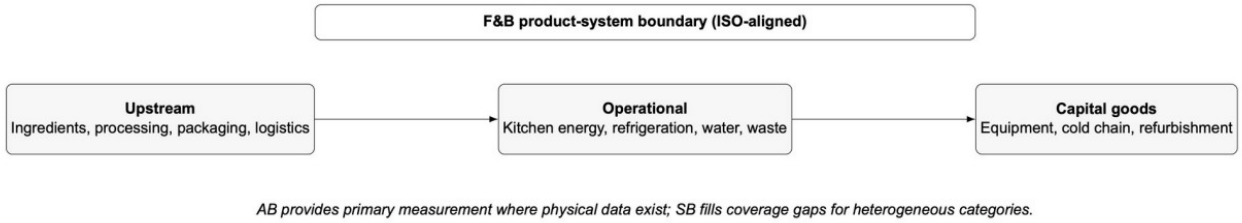


Figure 2. Lifecycle boundary and AB+SB coverage logic.

To make results actionable, F&B–CES 2.0 uses functional units that map to managerial decisions: kg CO e/cover for

2 menu and service design, kg CO e/kg dish for culinary process efficiency and central kitchen comparisons, and kg

2 CO e/event (or kg CO e/cover at banquet scale) for MICE and corporate-client offerings. These functional units can be

2 2

reported alongside revenue-normalized metrics to support benchmarking across segments.

4.2 Dual accounting approach and completeness

F&B procurement and operations are characterized by heterogeneous data availability. Activity-based (AB) accounting is therefore used where measurable physical data exist, including recipe-based ingredient mass, energy meters or submeters, refrigeration load where available, water use, and waste weighing [29-30]. Spend-based (SB) accounting complements AB where activity data are unavailable, fragmented, or impractical to collect at scale, such as outsourced production and catering services, imported beverages with uncertain origin and supply-chain pathways, selected logistics categories, and some maintenance and consumables. SB estimation is grounded in MRIO emission factors and is treated explicitly as a coverage mechanism with documented uncertainty, rather than as a substitute for primary data where highimpact items can be measured [31-33].

4.3 Core output metrics

F&B–CES 2.0 reports a small set of decision-oriented metrics that remain comparable across properties and service formats. The primary service-normalized indicators are kg CO e/cover and kg CO e/kg dish, which support menu and

2 2

process redesign respectively. For cross-segment comparison and internal governance, kg CO e/100 USD revenue is

2 reported alongside the servicenormalized metrics to reduce misinterpretation when service levels differ. Capital goods are

represented through annualized embodied carbon so that replacement and retrofit decisions can be evaluated on a consistent annual basis. Finally, where evidence meets the verification requirements described in Section 7, reductions in food waste can be expressed as avoided emissions, explicitly bounded to prevent over-claiming.

5. Methods and Implementation in Hotel Operations

This research develops and operationalizes F&B–CES 2.0 for implementation under the imperfect data, organizational constraints, and operational rhythms of hotels. We therefore specify an implementation method that is compatible with ISO-style management systems and with typical hotel information flows.

5.1 Implementation units and operational segmentation

The first step is to choose an implementation unit that matches the hotel's operating reality. In a business hotel with a single all-day dining restaurant and limited banquet volume, the natural unit may be one restaurant plus associated dishwashing and storage. In a resort with multiple outlets, central production kitchens, and large banquet operations, implementation should be segmented into sub-systems that correspond to controllable teams and physical boundaries: for example, "central kitchen," "banquet production," "breakfast buffet," and "bar program." This segmentation is not merely accounting convenience; it prevents management failure modes where the kitchen team is judged on emissions driven by a banquet procurement decision, or where a bar program is hidden inside generic procurement totals.

Segmentation also clarifies functional units. Per-cover metrics are meaningful only when "cover" corresponds to a comparable service promise. A pre-pandemic breakfast buffet cover in 2019 often implied abundant choice and higher overproduction risk, whereas a 2021 breakfast cover may have been pre-portioned or served, with different waste and energy patterns. In the same property, kg CO e/cover should therefore be interpreted with the service format recorded as

2 part of the process evidence layer.

5.2 Hybrid AB+SB data pipeline and data governance

Implementation requires a repeatable data pipeline. Activity-based inputs include recipe-level ingredient quantities, kitchen energy and fuel, refrigeration electricity where separable, water and dishwashing loads, and time-stamped waste weights. Spend-based inputs are used as governed coverage for heterogeneous categories such as imported alcohol, outsourced production, and selected services. Hotels typically do not have perfect recipe discipline. A recurring real-world issue is "recipe drift": chefs adjust ingredients to manage cost volatility or guest preference (e.g., adding dairy or premium cuts), while the recipe card is not updated. In F&B–CES 2.0, recipe governance is treated as an internal control: the recipe card becomes an auditable artifact with version control, and the variance between declared recipe and observed preparation is treated as a data-quality signal rather than a moral failure. Similarly, cold-chain evidence often exists but is not used for carbon management.

Temperature logs are typically collected for food safety; door-opening patterns are sometimes recorded for operational reasons; and maintenance logs exist for equipment servicing. F&B–CES 2.0 integrates these records to interpret refrigeration variability and to prevent misleading conclusions such as "the kitchen became inefficient" when the true driver was a different service sequence, longer holding time, or degraded door seals.

5.3 Emission factors, allocation rules, and uncertainty

A practical implementation must define an emission-factor register and allocation rules that are stable across time. For

ingredients, hotels may use a tiered approach: primary supplier product LCAs where available for high-impact items, literature or database factors for common categories, and MRIO factors where product-specific data are unavailable. The goal is not perfect accuracy for every SKU, but controlled improvement with clear documentation and boundary consistency.

Allocation is unavoidable in hotels because energy and labor are shared. F&B–CES 2.0 recommends transparent stationbased allocation when sub-metering is not available: energy is allocated to production stations using a combination of equipment power ratings, duty cycles, and production time, consistent with the OEL logic. Dishwashing and laundry impacts can be allocated by measurable drivers such as racks, cycles, or linen mass attributed to F&B. These rules should be fixed for a reporting period to preserve comparability.

Uncertainty is treated explicitly rather than hidden. SB categories are higher-uncertainty by construction; imported beverages and outsourced production are often high-variance categories. Instead of collapsing these uncertainties into a single number, F&B–CES 2.0 encourages reporting a completeness statement (AB coverage ratio, SB coverage ratio, and the share of high-uncertainty categories) alongside the core metrics. This makes the footprint a management object: a hotel can reduce emissions and reduce uncertainty over time, and both are valuable outcomes.

5.4 A pre- versus post-pandemic implementation vignette To illustrate how the method works under regime change, consider a hotel that shifted breakfast service formats. In 2019, breakfast was buffet-based, with frequent replenishment and high overproduction risk. Waste mapping would often show a dominant overproduction component at the end of service, while refrigeration loads would spike due to prolonged holding and frequent cold-room access. During 2020–2022, many hotels adopted served or pre-portioned formats, sometimes with takeaway options. Overproduction waste often fell, but packaging and logistics burdens increased [29- 30]. A naive footprinting approach might report “improvement” based on lower food waste mass alone, while ignoring packaging shifts. Under F&B–CES 2.0, the lifecycle boundary and evidence chain force this shift to be visible: packaging enters upstream accounting, and service-format changes are recorded as process evidence. Management can then make an informed choice in recovery: keep the lower overproduction profile while redesigning packaging (e.g., reusable systems where feasible) and updating procurement.

6. Comparative Analysis across Standards and Regional Schemes

F&B–CES 2.0 is designed to be compatible with standards rather than to compete with them. However, compatibility does not mean equivalence. The practical question for hotel groups is how to navigate a fragmented landscape in which corporate reporting, product footprinting, hotel benchmarking, and tourism certification use different boundaries and different assurance expectations.

6.1 ISO 14067 and PAS 2050: product-footprint logic versus service realities

ISO 14067 and PAS 2050 provide an essential foundation for product-level footprinting because they force explicit functional units, system boundaries, and methodological rules [9,31-32]. Yet hotel F&B is not a factory that produces identical products. The same dish name can map to different ingredient mixes across seasons, suppliers, and chef choices; and the same banquet package can change composition under client negotiations. As a result, a hotel attempting to apply product footprint standards directly often faces a trade-off between operational realism and methodological purity.

F&B–CES 2.0 resolves this by treating product-footprint standards as a boundary discipline and evidence discipline, while keeping a management lens. A dish footprint is not defined as a one-off LCA report; it is defined as a governed template linked to recipe version control, procurement evidence, and allocation rules that can be repeated over time. This approach lowers the risk that hotels will publish numbers that cannot be reproduced or verified. It also creates a path for gradual improvement: early-stage properties can use category-level factors and SB coverage, while mature properties can incorporate supplier-specific LCAs for high-impact items.

6.2 ISO 14064-1/3 and the GHG Protocol: organization reporting versus management

control Organization-level inventories under ISO 14064-1 and the GHG Protocol are required for corporate reporting and disclosure [4,10,12-13]. Their logic is aggregation: assign emissions to scopes and categories in a way that supports completeness and comparability. In contrast, the logic of operational management is disaggregation: identify controllable levers and the teams that can act on them [33].

This distinction explains why hotels can report a complete corporate footprint while still being unable to answer operational questions such as “which menu changes matter” or “which CapEx measures have credible carbon payback.” F&B–CES

2.0 is positioned as an intermediate layer that translates corporate categories into operational units without breaking traceability. For example, a corporate Scope 3 category such as purchased goods can be operationalized into high-impact SKUs and menu structural indicators; and a Scope 1/2 electricity total can be operationalized into station-based energy diagnostics through OEL and process logs.

6.3 HCMI and hotel-level benchmarking: comparability versus attribution

HCMI has advanced hotel benchmarking by offering a consistent method for per-room-night operational footprints3. The limitation is that comparability is achieved through aggregation, which can obscure the drivers that matter in F&B, especially in properties where restaurants, banquets, and bars are major revenue streams. The risk is not that HCMI is wrong, but that it is asked to answer questions it was not designed to answer. In practice, hotels often attempt to interpret HCMI indicators as if they were diagnostic. For example, an increase in whole-property energy intensity may be interpreted as “inefficiency” even when the true driver is a resurgence of banquet volume after COVID-19 restrictions. F&B–CES 2.0 complements benchmarking by providing attribution: it records service format, links energy and waste to operations, and makes upstream procurement drivers visible. This allows a hotel group to keep HCMI for disclosure while using F&B–CES 2.0 for internal control.

6.4 Tourism certification and ESG disclosure regimes: claim defensibility

Tourism and hospitality certification schemes (e.g., GSTC-aligned criteria) and disclosure regimes (e.g., CSRD and ISSB S2) raise expectations for defensible claims [27,34-36]. For F&B, the claim risk is particularly high because guests and corporate clients can observe products directly. A “low-carbon banquet” claim that cannot be supported by evidence is reputationally risky. F&B–CES 2.0 addresses claim defensibility through its evidence-chain design (Fig. 3). The key idea is that a claim is only as strong as its weakest evidence layer. If a hotel cannot demonstrate recipe governance (Layer 2) or model version control (Layer 3), it should not make fine-grained dish-level claims even if a numerical estimate exists. Conversely, a hotel that can demonstrate evidence and governance can communicate credibly, even if some categories remain uncertain and are transparently handled through SB coverage [37].

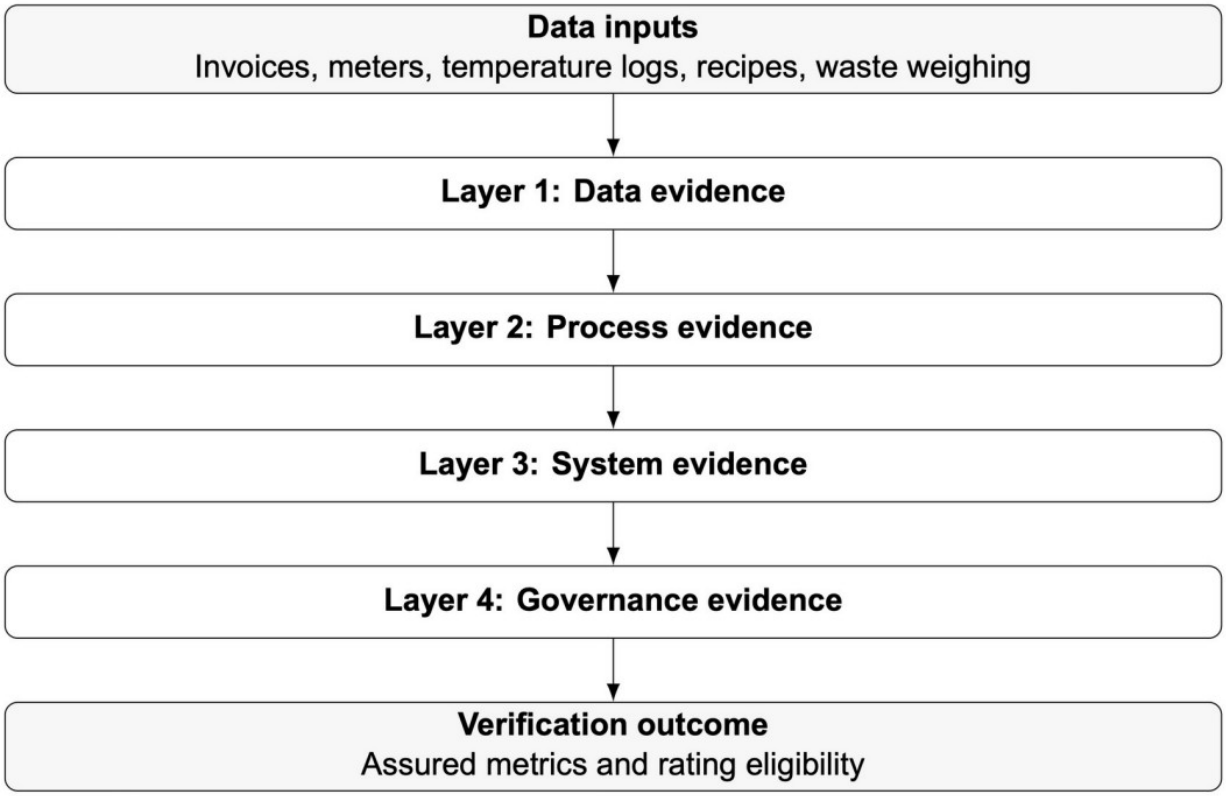


Figure 3. Four-layer evidence chain enabling auditability.

6.5 Why regional differences matter in implementation

Regional differences matter not only because emission factors differ, but because service models differ. In some markets, high buffet intensity and abundant choice are core to brand positioning; in others, served formats dominate. In some regions, cold-chain reliance is structural due to climate and import patterns; in others, local sourcing is operationally feasible. The COVID-19 period illustrates these differences. In markets where buffets were suspended, hotels experienced sharp shifts in waste, labor, and packaging. Some properties saw reduced overproduction but increased disposable packaging; others saw increased food safety-driven process steps that changed energy and water loads. A robust framework must therefore be resilient to regime changes [36-38]. The purpose of aligning with ISO-style evidence

and management routines is precisely to maintain comparability across these shifts without pretending that 2019 and

2021 were operationally equivalent.

7. Audit & Verification System: Evidence-Chain Design

A key reason F&B footprints remain “non-actionable” in many hotels is not the absence of calculation methods, but the

absence of evidence design. In practice, carbon numbers are frequently produced as one-off spreadsheets with limited traceability to procurement records, recipe governance, cold-chain logs, and equipment operations, making

them difficult to verify and risky to communicate externally. F&B–CES 2.0 therefore treats auditability as a first-class design constraint and aligns verification logic with ISO 14064-3, emphasizing transparency, materiality, and reproducibility [13,39].

7.1 Four-layer evidence chain

The proposed evidence chain organizes assurance into four layers that connect operational reality to model outputs (Table 2). Layer 1, data evidence, consists of primary records such as invoices and delivery notes, calibrated meter readings or sub-meter logs, temperature records for cold storage and transport where available, and time-stamped waste weighing data. Layer 2, process evidence, demonstrates that the data are generated by controlled processes rather than by post hoc estimation; typical examples include standardized recipe cards with version control, batch-production records, process logs that link menu items to production stations, waste mapping sheets that distinguish preparation loss, plate waste, and overproduction, and operational records such as cold-room door-opening logs when these are used to explain energy anomalies. Layer 3, system evidence, includes the documented AB+SB model structure, the calculation boundary statement, and the resulting indicators (e.g., MSEI, OEL, revenue-normalized metrics), together with version control and change logs.

Layer 4, governance evidence, demonstrates that responsibilities, controls, and supplier requirements exist and are enforced; it includes supplier certificates and traceability claims (where material), internal audit results, corrective action records, and procurement governance rules that reduce the probability of greenwashing and boundary manipulation [40-42].

Table 2. Four-layer evidence chain mapped to typical data artifacts and audit intent.

Evidence layer	Typical evidence artifacts	Primary audit intent
Layer 1: Data evidence	Invoices and delivery notes; calibrated sub-meter logs; cold-room temperature logs; time-stamped waste weighing records; dishwashing/laundry energy and water records	Completeness, accuracy, calibration, temporal continuity
Layer 2: Process evidence	Recipe cards with version control; batch production records; SOPs and shift logs; waste mapping sheets (prep/plate/overproduction); cold-room door-opening logs (where used)	Demonstrate controlled generation of data and reduce reliance on estimates
Layer 3: System evidence	AB+SB model documentation; boundary statement; emission-factor register; indicator outputs (MSEI, OEL, FWAE, RCE); version and change logs	Reproducibility, boundary consistency, avoidance of double counting
Layer 4: Governance evidence	Supplier certificates (where material); internal audit reports; corrective action records; procurement governance and responsibility matrix	Control effectiveness, traceability integrity, and claim defensibility

7.2 Audit Hotspots

The framework focuses verification effort on recurrent high-impact areas, but it is essential that “hotspots” are interpreted as operational failure modes rather than as a static list. In hotel audits, the same numerical discrepancy can have different causes (Table 3).

Table 3.Condensed audit hotspot checklist for F&B–CES 2.0 assurance.

No.	Hotspot	Why it matters
1	High-carbon ingredients (ruminants, high-impact dairy)	Dominant upstream drivers; high risk of misallocation through recipe drift
2	Ingredient origin and transport mode	Upstream intensity and cold-chain requirements vary materially by origin and mode
3	Alcohol procurement (imported wine/spirits)	High uncertainty without SB/MRIO coverage and supplier evidence
4	Kitchen sub-metering strategy	Prevents dilution of kitchen loads into whole-building averages
5	Cold-chain temperature continuity and maintenance	Indicates thermal losses and potential refrigerant risks

6	Cold-room door-opening/behavioral controls	Explains refrigeration variability and supports operational interventions
7	Recipe governance and version control	Ensures LCA inputs remain consistent with actual production
8	Food-waste measurement and calibration	Enables credible avoided-emissions claims and prevents overstatement
9	Waste segregation and recovery records	Supports circularity metrics and endof-life boundary consistency
10	Capital goods (equipment life and replacement)	Required for annualized embodied carbon and carbon payback analysis
11	Refrigerants and servicing records	Potentially material climate impacts and compliance records

A common example is high-carbon ingredients. The audit question is not simply whether ruminant proteins are present,

but whether recipe and purchasing records are consistent and whether substitutions are controlled [43]. In practice, a chef may replace one cut with another during price volatility, or increase dairy content to maintain perceived quality. Without recipe version control, such changes can produce systematic undercounting.

Imported alcohol is another recurring hotspot because it often sits outside traditional “food” control. Beverage programs can expand rapidly in business recovery phases, and procurement documentation may not include origin and processing details [44]. The audit posture therefore emphasizes traceability and SB governance: when primary data cannot be obtained, MRIO coverage must be applied consistently and overlap control must prevent double counting with any product-specific factors.

Cold-chain losses illustrate why engineering evidence matters. A large increase in refrigeration electricity can be caused by a hardware issue (door seals, defrost failures), a behavioral change (door discipline during peak prep), or a serviceformat change (longer holding times during buffet). Auditors therefore examine temperature continuity, maintenance logs, and where relevant door opening practices. The goal is to connect carbon anomalies to controllable interventions [45-46].

Finally, food-waste avoided emissions claims require special caution. Hotels often have strong incentives to report reductions, especially when such reductions are linked to internal incentives or marketing. The audit logic therefore requires calibration evidence for scales, stable categorization of waste streams, and a conservative boundary statement that clarifies what is counted as avoided and what is not. This approach aligns with the broader ISO verification posture of avoiding overstatement and ensuring reproducibility.

8. Rating System: Copper–Silver–Gold

A rating system is required because hotels do not only need a number; they need a management trajectory that can be

communicated internally and, where appropriate, externally. Ratings also reduce the transaction costs of assurance by clarifying which data and controls must be present before low-carbon claims (for dishes, banquets, or F&B operations) are made. In F&B–CES 2.0, rating is therefore defined as a function of both outcomes and evidence quality: performance without traceability is treated as unstable, while traceability without performance is treated as insufficient.

8.1 Tier definitions and progression logic

Copper represents foundational compliance. At this tier, hotels establish minimum viable measurement and governance, typically with partial AB coverage, limited supplier traceability, and early stage food-waste controls (Fig. 4). The primary objective is to identify hotspots reliably and to prevent boundary leakage through clear accounting rules. Silver represents strong operational performance. At this tier, hotels achieve consistent AB+SB coverage, implement process improvements informed by engineering diagnostics, and begin systematic menu restructuring [47]. The data architecture is sufficiently mature for repeated internal audits, and pilot product-level labels can be introduced where evidence chains are complete. Gold represents industry leadership. At this tier, traceability is high across material categories, cold-chain systems and refrigerants are actively managed, circularity approaches are embedded in operations, and product-level certification is supported through robust evidence and governance.

Gold-tier hotels can plausibly implement internal carbon pricing and incorporate avoided emissions from food-waste reduction only where verification rules are satisfied, thereby avoiding over-claiming [9,13,26,48].

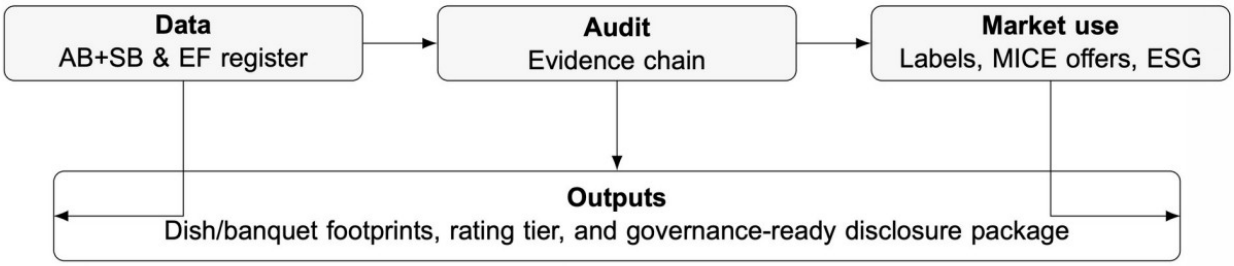


Figure 4. Data–audit–certification–market integration pathway.

8.2 Indicator architecture

Indicators are grouped under the six systems so that the rating can be interpreted as a diagnostic profile rather than as a single opaque score. This is important in hotel governance because different teams own different levers: procurement can change supplier requirements and menu composition; engineering can change refrigeration and equipment performance; and operations teams can change service flows that determine waste [49-50]. In implementation, each indicator should be mapped to (i) a measurable data source, (ii) a responsible owner, and (iii) a minimum evidence requirement. For example, an indicator such as “traceable ingredients share” is meaningless without a definition of traceability and without a governance rule for retaining supplier evidence. Similarly, “kWh per cover” is meaningless if covers are not defined consistently across service formats. F&B–CES 2.0 therefore treats indicator definitions as part of the verification system rather than as a cosmetic appendix.

The annex provides a condensed tiering table for publication, but a hotel or group implementing the system would maintain a fuller checklist that links each indicator to the four evidence layers [51]. This linkage is what allows the rating to be used for decision-making and assurance, rather than as a marketing label.

9. System Integration for Decision-Making

The value of a systems approach is that it prevents “local optimization” that merely shifts emissions across boundaries or erodes service outcomes. In hotels, rebound risks are common. A procurement decision that reduces ingredient cost may increase cold-chain burden or waste; a menu change that reduces upstream emissions may increase preparation complexity and energy; and an equipment upgrade that reduces operational energy may introduce large embodied emissions without a clear payback. F&B–CES 2.0 therefore treats interventions as a portfolio evaluated under consistent functional units and evidence requirements.

9.1 From menu and procurement to verified upstream reductions

Upstream impacts often dominate food footprints, but they are also where hotels have the least direct visibility. F&B–CES

2.0 addresses this by combining supplier-tier segmentation with hotspot identification so that primary-data collection and traceability effort is directed where it matters. In practice, hotels typically find that a small number of ingredient families and beverages dominate: ruminant proteins, high-impact dairy, certain imported seafood categories, and imported wine/spirits. Menu structural indicators such as MSEI are designed to make these drivers visible in a way that is actionable for chefs and banquet planners, enabling scenario analysis such as shifting protein composition, increasing seasonal offerings, or narrowing beverage portfolios. SB coverage is then introduced as a governed completeness layer to reduce boundary leakage for categories where primary data are not feasible [52-53].

9.2 From kitchens, cold chains, and waste to operational control

Back-of-house operations are treated as station-based engineered systems rather than as generic building loads. Submetering strategies (or calibrated proxy strategies where sub-metering is infeasible) are used to separate refrigeration, cooking, and cleaning loads. OEL links equipment power, utilization, and production time so that energy intensity can be interpreted as a function of operations rather than as noise. This matters in real hotels because energy spikes often coincide with service-format changes (e.g., a return to buffet breakfast) or with behavioral changes (e.g., cold-room door discipline during peak prep) (Fig. 5).

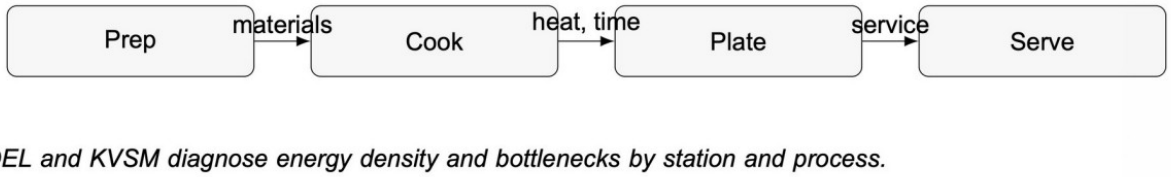


Figure 5. Illustrative process–energy mapping for back-of-house operational engineering.

Circularity and food waste are integrated into the same control logic. A three-stage waste model links preparation loss, plate waste, and overproduction to distinct levers such as prep processes,

portioning, and forecasting. This structure is particularly useful for comparing pre- and post-pandemic regimes: during 2020–2022 many hotels reduced buffets and simplified menus, and the waste profile often shifted from overproduction to packaging-related burdens [54-55]. Postpandemic recovery therefore requires a different intervention set than 2019: maintaining lower overproduction while controlling packaging and service-flow emissions.

Finally, CapEx and service economics are integrated so that operational changes and investments can be compared. Annualized embodied carbon enables carbon payback analysis for equipment replacement and cold-chain retrofits, while revenue-normalized carbon efficiency reduces misinterpretation when service levels differ. Data–audit–market integration then translates internal control into defensible outputs: dish and banquet footprints, rating eligibility, and governanceready disclosure that can support client-facing claims [56].

10. Managerial Implications

F&B–CES 2.0 is designed to shift F&B decarbonization from “reporting after the fact” to “managing before the fact.” In menu engineering, the framework supports a transition from qualitative “plantforward” narratives to quantitative structural redesign, where menu composition is assessed through indicators such as MSEI and interpreted alongside functional-unit metrics (e.g., kg CO e/cover) and service constraints. In procurement, the multi-tier supply-chain architecture makes

2 explicit which categories require traceability and primary data (e.g., ruminant proteins, high-impact dairy, and key

imported beverages) and which can be covered through MRIO-based SB estimation as a controlled approximation [55- 57]. In engineering and operations, the operational energy load (OEL) perspective motivates kitchen-level sub-metering strategies and process mapping so that energy and refrigeration are no longer treated as “background” building loads. For CapEx, annualized embodied carbon and carbon payback logic make decarbonization investments comparable to operational interventions, enabling hotels to prioritize cold-chain retrofits, equipment replacement, and layout redesign using a carbon-oriented investment case.

Table 4. Decision-oriented metrics and typical managerial uses.

Metric	Functional unit	Functional unit
Carbon intensity per cover	kg CO_2e/cover	Menu and portion redesign; service-format benchmarking; banquet planning
Carbon intensity per dish mass	kg CO_2e/kg dish	Kitchen process efficiency; central kitchen comparisons; recipe optimization
Revenue-normalized carbon efficiency	kg CO_2e/100 USD	Cross-segment benchmarking and carbon-aware carbon efficiency pricing discussions
Food-waste avoided emissions	per kg CO_2e avoided	Verification-based mitigation accounting and internal incentives[26]
Annualized embodied carbon	Per kg CO_2e/yr	CapEx planning, replacement timing, carbon payback computation[16]

A practical implication is that the same accounting dataset can serve multiple stakeholders. Culinary and banquet teams

obtain actionable dish- and event-level signals; engineering teams obtain process-level hotspots and maintenance priorities; procurement teams obtain traceability gaps and supplier governance requirements; finance teams obtain an

investment prioritization narrative; and sustainability teams obtain an assurance-ready disclosure package aligned with external reporting expectations. These integrations are especially relevant in MICE contexts where corporate clients increasingly request verifiable information that maps to their own Scope 3 reporting boundaries [4,44,58].

11. Extended Discussion: Theoretical Contributions and Framework

Integration

11.1 From footprinting to a socio-technical management control system

A central theoretical claim of F&B–CES 2.0 is that hotel F&B decarbonization should be treated as a socio-technical

control problem rather than as a footprinting exercise. Carbon outcomes in hotel F&B emerge from interactions between supply chains, culinary design, engineered systems (kitchens and cold chains), service models, and governance routines. Conventional accounting tools typically treat these interactions as noise, which leads to "unexplained variance" and undermines learning. F&B–CES 2.0 addresses this by linking carbon measurements to controllable levers (menu structure, duty cycles, cold-chain behavior, waste prevention) and by requiring evidence architectures that reduce model risk and enable repeated improvement cycles.

This reframing has two implications for research. First, it suggests that performance indicators should be interpreted as part of a control loop: measurement, diagnosis, intervention, verification, and learning [59-60]. Second, it positions auditability as an enabling technology for operational decarbonization rather than as a compliance burden. The four-layer evidence chain can be understood as a "measurement infrastructure" that allows hotels to experiment (e.g., menu redesign, portioning changes, cold-chain retrofits) while retaining comparability and assurance integrity [9,13,61].

11.2 Improving compatibility with existing carbon accounting and disclosure frameworks

F&B–CES 2.0 is not intended to replace corporate GHG inventories or hotel-level operational metrics. Instead, it provides an intermediate layer that disaggregates F&B into product-system logic while remaining compatible with prevailing standards. At the corporate level, emissions from purchased goods and services and upstream transportation can be mapped to Scope 3 categories under the GHG Protocol, while on-site fuels and electricity remain in Scope 1 and Scope 2 accounts [4,10,62]. At the hotel benchmarking level, the framework complements HCMI by providing an auditable method to separate and manage F&B drivers that are otherwise averaged into whole-property metrics [63-64]. At the product level, the lifecycle boundary and functional-unit logic aligns with ISO 14067 and related LCA standards, which is essential if hotels intend to communicate dish- or banquet-level claims externally.

A key improvement proposed here is explicit AB+SB governance. Rather than treating SB estimation as a hidden

"shortcut," F&B–CES 2.0 treats SB as a documented coverage layer with a specified purpose (boundary completeness), a defined set of eligible categories, and a verification posture. This reduces the risk of double counting, boundary manipulation, and inconsistent yearto-year reporting, and it makes uncertainty visible as a management signal.

11.3 Strengthening and normalizing rating systems

The Copper–Silver–Gold rating is deliberately designed to combine outcome and evidence quality, because carbon performance claims without traceability increase reputational and assurance risk (Table 5). This aligns with the direction of sustainability disclosure regimes and assurance expectations, where claims are increasingly expected to be accompanied by governance and data-quality narratives [44-45,65]. In practice, rating systems also provide a coordination device: procurement, culinary, engineering, and finance teams can align priorities when performance targets are translated into tier requirements and evidence-chain deliverables.

Table 5. Rating tier logic summarized as outcomes plus evidence maturity.

Tier	Performance profile	Evidence and governance profile (typical)
Copper	Hotspots identified; early improvements possible; limited comparability	Partial AB coverage; basic recipe and waste logs; limited traceability; evidence chain incomplete
Silver	Measurable efficiency gains; menu and process redesign underway; improving benchmarks	Consistent AB+SB coverage; internal audit routine; documented boundary controls; selected product labels possible

Gold	Low-carbon operations and products; mature circularity; optimized cold chain	High traceability and assurance readiness; verified product-level claims; strong governance; internal carbon pricing plausible

Future research should test how rating thresholds vary by hotel segment, geography, and service format. For example, revenue-based carbon efficiency may be structurally different in luxury versus limited-service contexts, and cold-chain intensity may differ across climate zones. These variations can be incorporated without breaking

comparability if the rating system is reported as both absolute performance and context-adjusted performance bands [66].

Conclusion

12.

This study develops a consolidated and implementable framework rather than reporting a single-site empirical study. The core limitation is therefore external validation of thresholds and effect sizes across multiple hotel segments and geographies. Hotel F&B also varies structurally across regions: electricity grids differ, cold-chain reliance differs by climate, and regulation and customer expectations shape service formats. Several research and practice directions follow.

First, multi-site validation should focus on causal attribution rather than on correlation. In real hotels, interventions are not implemented in isolation: menu changes often occur alongside operational changes, and capital upgrades are often bundled with refurbishments. Evidence-chain design makes it possible to attribute effects more credibly by controlling data generation and version history.

Second, the AB+SB hybrid should be stress-tested for high-impact and high-uncertainty categories. For example, imported alcohol supply chains can be material in some business hotels; outsourced pastry and central kitchen production can dominate in large resorts; and seafood categories can have highly variable footprints depending on production systems. Comparative studies should evaluate when supplier LCAs materially improve accuracy and when MRIO coverage is sufficient for governance.

Third, the pandemic period offers an underused natural experiment. Many hotels changed service formats, menu breadth, and supply chains abruptly, and some of those changes persisted. Future studies can leverage this regime shift to test whether F&B–CES indicators respond sensibly to operational change. For instance, did waste prevention gains persist after buffets returned? Did packaging burdens rise and then stabilize? Did refrigeration intensity change with different holding strategies?

Fourth, adoption research should examine organizational burden and capability building. The most common failure mode is not technical; it is that measurement is treated as a sustainability team’s reporting task rather than as an operational management system. Successful adoption likely requires training chefs and procurement teams in evidence discipline, aligning engineering maintenance logs with carbon diagnostics, and embedding the indicators into routine decision processes such as menu cycles and CapEx committees.

Finally, this work is intended to support rigorous, standards-aligned sustainability management and transparent reporting. It does not propose, and should not be used for, any attempt to misrepresent performance. The value of F&B–CES 2.0 comes

from making assumptions explicit, documenting evidence, and enabling repeatable improvement.

By embedding accounting, auditability, and performance rating into a single operational system, F&B–CES 2.0 enables hotels to manage F&B as a carbon-intensive product system rather than a peripheral operational category. The framework provides a scalable foundation for corporate ESG programs, auditable low-carbon product claims, and future hospitality-sector carbon market mechanisms.

Author Contributions

[Author contribution statement reserved for final publication.]

Consent to Participate declaration Not applicable

Ethics declaration Not applicable

Availability of Data and Materials The data that support the findings of this study are available from the corresponding author upon reasonable request.

Competing Interests

All authors declare no Competing Financial or Non-Financial Interests.

Funding Declaration

This research was funded by Enerstay Sustainability Academy (Singapore) Grant Call (Call 7/2025) _SDG (Project ID CUMT-001), Singapore.

Clinical trial number Not applicable

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RESEARCH ARTICLE

Machine Learning-Driven Child-Friendly Urban Environment Assessment: A Case Study of China Metropolitan Spatial Quality Assessment

AUTHORS · Xuchao Nian; Mohd Fabian Bin Hasna; Noranita Mansor; Yuanzhe Li

AFFILIATIONS · Faculty of Design and Architecture, Universiti Putra Malaysia, Serdang, Selangor 43400, Malaysia; NUS College of Design and Engineering, National University of Singapore, Singapore 117575; Carbon Neutrality Institute, China University of Mining and Technology, Xuzhou 221116, China

PUBLISHED ONLINE · 1 NOVEMBER 2026 DOI · NOT ASSIGNED ACCESS · FREE TO READ; ALL RIGHTS RESERVED

ABSTRACT

In response to the problems of subjectivity, evident scale restriction, and lack of dynamism in conventional child-friendly urban environmental evaluations, this paper proposes a machine learning-based multi-dimensional evaluation framework. Based on the central urban regions of the three major Chinese metropolises (Shanghai, Shenzhen, and Guangzhou), it utilizes street view images, remote sensing imagery, POI data, and child behavior trajectory data. Through machine learning algorithms including deep learning and random forest, it performs quantitative evaluations in four aspects: ecological safety, spatial accessibility, facility suitability, and behavioral safety. It is found that machine learning can realize fine-grained extraction of child-friendly environmental indicators with an accuracy rate of 89.7%. Spatially, the quality of child-friendly environments in the three metropolises shows a differentiation pattern that “the core areas are better than the peripheral ones, and the waterfront areas are better than the densely developed ones.” The density of recreational facilities and the safety of pedestrians in streets are key elements affecting the evaluation outcomes. This framework offers technical support and experience references for the precise implementation of policies in child-friendly city planning.

Keywords: *Machine Learning; Child-Friendly City; Environmental Assessment; Spatial Quality; Metropolis; Quantitative Analysis*

1. Introduction

Since the implementation of the "Child First" approach in the overall plan of China’s urban construction, the establishment of child-friendly cities has progressed from concept promotion to detailed practice, and scientific environmental evaluation is a key requirement to improve spatial planning. Conventional environmental evaluations typically adopt subjective approaches like expert scoring and questionnaire survey, with problems including small sample size, lengthy evaluation periods, and inability to reflect spatial dynamic attributes, which are not applicable to complex built environments in large cities’ environmental evaluation requirements. On the other hand, the fast evolution of artificial intelligence and big data technologies offers new avenues for urban spatial perception evaluation. Street view image semantic analysis and remote sensing data intelligent extraction can realize subjective perception and objective indicator integration and quantification, driving assessment model transition from "material form orientation" to "human-centered perception orientation".

Dense population and complex spatial form carriers, China’s large cities suffer from facility allocation imbalance, obvious safety risk, and poor ecological quality in children’s activity spaces. For instance, Shenzhen improved local environments through “micro-updates” in building child-friendly communities, yet it is short of systematic quantitative evaluation methods to guide balanced allocation. The child environmental health evaluation system

established in Wenzhou has achieved concrete indicators but lacks the dynamic assessment capabilities of machine learning technology. Therefore, constructing a machine learning-driven assessment framework and conducting largescale child-friendly environmental quality assessments in major cities holds significant theoretical and practical value.

In foreign studies, Kim et al. utilized Google Street View images and deep learning algorithms to achieve automated evaluation of children's safety in urban streets, demonstrating the feasibility of AI technology in quantifying spatial perception; Rui constructed a street quality assessment model for children's walking needs by analyzing sidewalk Street View data, but lacked consideration of facility adaptability. Domestic research has mostly focused on indicator system construction, such as Wenzhou proposing a 5-dimensional indicator system covering ecological environment and public space, using the analytic hierarchy process to determine weights, but relying on subjective weighting led to limited assessment accuracy; Wu Wenjie's research team integrated machine learning with spatiotemporal statistical methods to build a human settlement well-being assessment model, providing technical references for child-friendly space evaluation, but did not optimize model parameters for children's characteristics.

Existing research still has three major gaps: first, insufficient technological integration, as most studies only apply a single type of machine learning algorithm, lacking the ability to extract indicators through multi-algorithm collaboration; second, lack

of child perspective, as assessment indicators often adopt adult standards without fully adapting to children's physiological and behavioral characteristics; third, limited case representativeness, with few empirical studies on the full-scale urban areas of China.

Based on this, this paper constructs a targeted evaluation framework and conducts case validation.

Machine learning is adopted in this study not for technical embellishment, but in response to three inherent challenges of child-friendly environment assessment at the metropolitan scale: (1) the high heterogeneity of multi-source data, including street-view imagery, remote sensing data, spatial networks, and behavioral trajectories; (2) the pronounced non-linear relationships between environmental attributes and child-friendliness outcomes, which are difficult to capture using linear or rule-based models; (3) the need for both composite evaluation and diagnostic identification of key driving factors to support targeted interventions. On the other hand, machine learning has its unique advantages in high-dimensional feature extraction, non-linear modeling, and explanation of feature importance. Construct a machine learning approach to evaluate child-friendly environments in Chinese metropolises, achieving automated indicator extraction, dynamical assessment, and precise diagnosis. With 3 large cities as examples, reveal spatial differentiation patterns and influential factors of child-friendly space quality.

Theoretically, improve the technological system of childfriendly environment assessment quantification and stimulate interest in research of AI technology and urban planning.

Practically, provide a decision support to optimize child-friendly spaces and allocate resources in metropolises, supporting the realization of “viewing cities from one meter height”.

2. Research Design

2.1 Study Area

Three major urban districts have been selected for this study: Shanghai’s Pudong New Area, Shenzhen’s Nanshan District, and Guangzhou’s Tianhe District, which together account for an area of about 1,286 km2. These districts contain multiple spatial types such as high-density built-up areas, waterfront ecological areas, aged communities, and new business districts. The area has a child population density between 280 and 420 children/km2, showing typical and

representative features. The inclusion criteria include (1) a high density of child service facilities, (2) complicated and diversified spatial patterns, and (3) full support of multi-source data.

2.2 Data Sources and Preprocessing

The study design uses a multi-source data fusion approach for 2023-2024. In data preparation, we removed central bias for sampling at households or schools to form a common dataset (sampling duration is 1 day). Children’s behavior patterns are different according to their ages. Considering the age-specific characteristics, we chose the most representative age group—children aged 7-10 (this age group demonstrates more diverse behavioral patterns, higher spatial usage frequency, and greater accessibility needs) for further examination, see Table 1.

Table 1 Data Sources and Preprocessing Methods

data type	Get source	Core Purpose	Pre-treatment method
Street View	Baidu Maps Street View, Gaode Maps Street View	Extract street safety, green coverage rate, and facility visibility	Image deblurring, coordinate correction, batch cropping (256×256 pixels)
remote sensing image	Landsat-9 satellite imagery	Calculate NDVI index, surface temperature, and ecological space proportion	radiometric calibration, atmospheric correction, cropping and stitching
POI data	Gaode Map Open Platform	Count the density of children's facilities (kindergartens, playgrounds, etc.)	Remove duplicates, classify, and spatially associate
Behavioral trajectory of children (7-10 years old)	Questionnaire survey and GPS positioning (sample size n=3200) (independent children)	Analyze spatial usage frequency and accessibility requirements	Remove outliers, smooth trajectories, and spatial matching
basic geographic data	National Geographic Information Public Service Platform	Assess the boundary of evaluation unit and road network data	Format conversion, topology check, meshing

2.3 Evaluation Indicator System

Children's activity trajectories are not treated as independent evaluation indicators, but are used to calibrate spatial exposure intensity and accessibility weighting at the evaluation-unit level, thereby reducing bias arising from inferring children's experience solely from facility distribution or physical form. Based on the "child physiological characteristicsbehavioral needs-environmental response" logic and combining Wenzhou's indicator system with AI evaluation technology features, we constructed a 4-dimensional framework comprising 12 core indicators. The entropy weight method was used to calculate the objective weights (see Table 2). The selection of indicators is based on children's restricted travel distance, high sensitivity to safety, and reliance on nearby facilities, focusing on environmental characteristics that have a direct impact on children's daily spatial exposure.

Table 2 Evaluation Index System

Primary Indicator	Weight	Secondary Indicator	Indicator Definition	Extraction Method
Ecological Security Environment	0.22	Forest Coverage Rate	Assess the proportion of green space area within the unit	NDVI Threshold Method for Remote Sensing Images
		Mean Surface Temperature	The average land surface temperature within the unit reflects thermal comfort.	remote sensing image inversion
		Air Pollutant Concentration	Annual average concentration of PM2.5	Multi-source monitoring data spatial interpolation
Space Accessible Environment	0.28	Accessibility Of Children's Facilities	Average walking time to the nearest child facility	network analysis method
		Pedestrian Density	Length-to-area ratio of walkway within the unit	road network data extraction
		Bus Stop Coverage Rate	Proportion of grid cells with bus stops within 500 meters	POI Data Space Analysis
Facility Adaptation Environment	0.25	Density Of Children's Facilities	Number of kindergartens, playgrounds, and other facilities per square kilometer	POI data statistics
		Facility Adaptation Level	Whether the facility is equipped with child-specific equipment	Street View Image Recognition Based on Deep Learning
		Public Toilet Adaptation Rate	Percentage of public facilities equipped with children's toilets	POI and Street View Data Fusion Verification
Behavioral Safety Environment	0.25	Street Lighting Coverage	proportion of streets with nighttime lighting	Night Street Image Recognition
		Traffic Hazard Density	Number of intersections and crosswalks without protective facilities within the unit	Street View Image Target Detection
		Density Of Surveillance Facilities	Number of surveillance devices per square kilometer in public areas	Street View Image Deep Learning Extraction

2.4 Machine Learning Evaluation Model

A three-tier technical framework of "index extraction-model training-result output" was constructed to integrate multiple algorithms for precise evaluation.

(1) Indicator Extraction Model: The ResNet-50 deep learning model is employed to perform semantic analysis on street view images, extracting subjective correlation indicators such as facility accessibility and traffic hazards. The training-totest set ratio is 7:3. The random forest algorithm is utilized to integrate multi-source data, calculating composite indicators like spatial accessibility and facility density, thereby addressing the bias issue caused by single-source data.

$$H(x) = \operatorname{argmax}_y \sum_{k=1}^{K} I(h_k(x) = y) \quad (1)$$

where H \(x \) k h \(x \) kyI \(·\), is the value of composite indicator after random forest fusion; is the number of k decision trees; is the prediction of i-th decision tree; is the range of indicator values; and is the indicator function, assuming a value of 1 if the condition in parentheses holds and 0 otherwise.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

$$\text{Precision} = \frac{TP}{TP + FP}, \text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

In the formula, TP, TN, FP, and FN represent the number of true positive samples (correctly identified target indicators), true negative samples (correctly identified non-target indicators), false positive samples (non-target samples misidentified as target indicators), and false negative samples (target samples misidentified as non-target indicators).

After parameter tuning, the model achieves an accuracy rate of 89.7% with an F1 score of 0.87, demonstrating excellent semantic parsing performance. It should be clarified that the reported accuracy (89.7%) refers to the semantic recognition performance of key environmental elements from street-view imagery, rather than the prediction accuracy of the composite childfriendliness score itself. The overall evaluation score is generated through indicator normalization and objective weighting, rather than supervised classification against predefined labels.

(2) Comprehensive Evaluation Model: According to the weights of indicators obtained using the entropy weight method, a machine learning ensemble evaluation model was established. By inputting the data of 12 indicators, the model could give each evaluated unit a comprehensive child-friendliness score (0-100 points).

$$P_{ij} = \frac{X_{ij}}{\sum_{i=1}^{n} X_{ij}} \quad (i = 1,2,\ldots,n;\ j = 1,2,\ldots,12) \quad (5)$$

$$E_j = -\frac{1}{\ln n} \sum_{i=1}^{n} P_{ij} \ln P_{ij} \quad (6)$$

$$W_j = \frac{1 - E_j}{\sum_{j=1}^{12} (1 - E_j)} \quad (7)$$

where X_ij is the raw data of the i-th indicator for the j-th evaluation unit; n is the number

of evaluation units (1286); P_ij is the standardized proportion of the i-th indicator for the j-th evaluation unit; E_j is the entropy value of the i-th indicator, with a lower entropy value indicating stronger discrimination ability; and W_j is the weight of the i-th indicator, meeting the requirement .

Construct a linear weighted ensemble model using indicator weights and obtain an overall score between 0 and 100:

$$S_i = 100 \times \sum_{j=1}^{12} W_j \times Z_{ij} \quad (8)$$

In the formula, S_i represents the comprehensive child-friendliness score of the i-th evaluation unit; denotes

i ij the normalized value (0-1 range) of the j-th indicator in the i-th unit; and 100 is the score scaling factor to enhance interpretability.

The model's fitting performance was validated using 5-fold cross-validation, with the core metrics calculated as follows:

$$MAE = \frac{1}{m} \sum_{k=1}^{m} |\hat{y}_k - y_k| \quad (9)$$

$$RMSE = \sqrt{\frac{1}{m} \sum_{k=1}^{m} \(\hat{y}_k - y_k \)^2} \quad (10)$$

$$R^2 = 1 - \frac{\sum_{k=1}^{m} \(y_k - \hat{y}_k \)^2}{\sum_{k=1}^{m} \(y_k - \bar{y} \)^2} \quad (11)$$

where is the number of validation set samples; is the predicted value of the model; is the true observed value (calibrated by expert ratings and child behavior feedback); is the mean of the true observed values; is the mean absolute error; is the root mean square error, and lower values indicate greater prediction accuracy; is the coefficient of determination, and a value closer to 1 indicates a better fit for the model.

The 50% cross-validation yielded a mean absolute error (MAE) of 2.31, root mean square error (RMSE) of 3.57, and a coefficient of determination (R^2) of 0.92, demonstrating excellent model fit.

(3) Spatial differentiation analysis: Spatial autocorrelation (Moran's I) and hotspot analysis (Getis-Ord Gi*) were employed to reveal the spatial clustering characteristics of child-friendliness, with visualization achieved through ArcGIS.

The formula for spatial autocorrelation (Moran's I) is expressed as:

$$\text{Moran's I} = \frac{n \sum_{i=1}^{n} \sum_{j=1}^{n} W_{ij} \(S_i - \bar{S} \)\(S_j - \bar{S} \)}{\sum_{i=1}^{n} \sum_{j=1}^{n} W_{ij} \sum_{i=1}^{n} \(S_i - \bar{S} \)^2} \quad (12)$$

In this formula, S_i and S_j represent the composite scores of the i-th and j-th evaluation units,

respectively. The average score is the mean of all unit scores, while the spatial weight matrix assigns 1 to adjacent units and 0 to non-adjacent units. The total number of evaluation units is indicated by N. Moran's I ranges from-1 to 1, where positive values indicate positive clustering (high values clustering with high values, low values clustering with low values), negative values indicate negative clustering, and 0 signifies random distribution.

The Moran's I value in this study was 0.47 ($P<0.01$), indicating a significant positive spatial aggregation pattern.

Hotspot analysis (Getis-Ord Gi*) identifies clusters of high-value (hot spots) and low-value (cold spots) areas, with the formula as follows:

$$G_i^* = \frac{\sum_{j=1}^{n} w_{ij} s_j - \bar{s} \sum_{j=1}^{n} w_{ij}}{\sqrt{S^2 \left[\frac{n \sum_{j=1}^{n} w_{ij}^2 - \left(\sum_{j=1}^{n} w_{ij} \right)^2}{n-1} \right]}} \quad (13)$$

In the formula, G_i* is the hotspot statistic of the i-th unit, after standardized testing, indicates that a

positive and significant value signifies the unit as a hotspot area (high-high aggregation), a negative and significant value indicates a cold spot area (low-low aggregation), and a value close to 0 denotes a transitional area.

3. Case Study Analysis

3.1 Overall Characteristics of the Evaluation Results

On average, the total score of child-friendly environments in the central urban areas of the three large cities was

62.3 points, which reached an “above average” grade overall but varied across regions (Figure 1). The highest score was observed in Pudong New Area of Shanghai (68.7 points), followed by Nanshan District of Shenzhen (63.2 points), with Tianhe District of Guangzhou being the lowest (54.8 points). Regarding the grades of performance, 8.2% of the units obtained excellent grades (>=80 points) and were mainly distributed in waterfront parks and high-end residential communities; 41.5% achieved good grades (60-79 points) and were mostly found in the surroundings of central business districts; 39.3% obtained average grades (40-59 points) and were predominantly scattered in old communities; and 11.0% obtained poor grades (<40 points) and were mainly situated in urban-rural fringe areas.

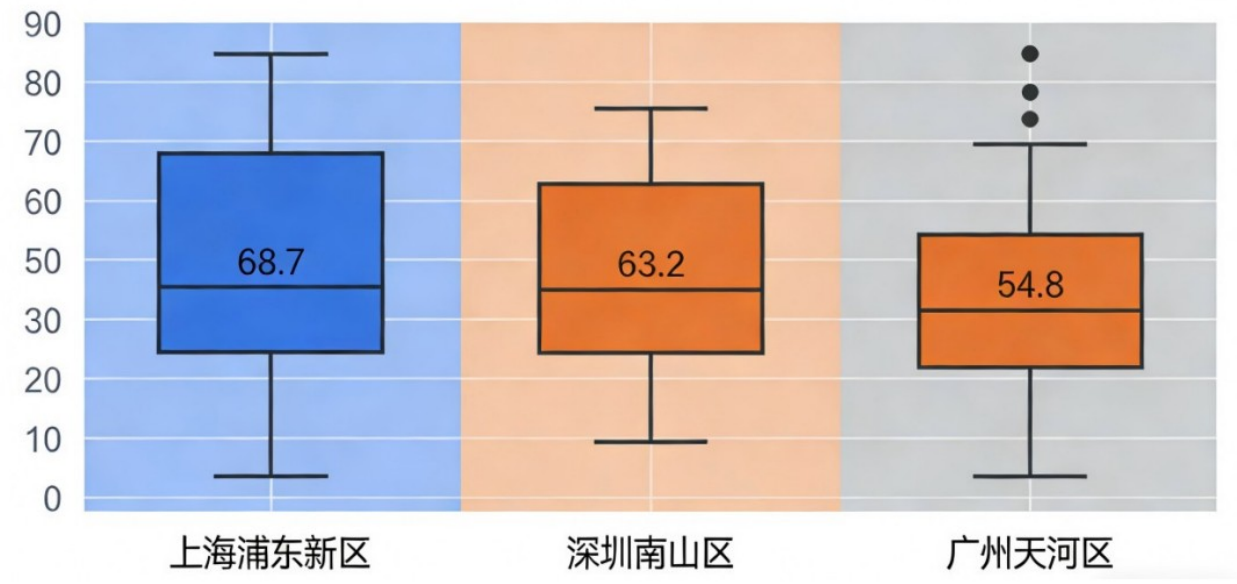


Figure 1 Box plot of comprehensive scores for child-friendly environments in three major metropolitan areas.

(Note: The figure displays the median, quartile, and extreme value distributions of scores across three major regions, providing an intuitive reflection of regional disparities. Shanghai Pudong New Area has the highest median value and Guangzhou Tianhe District has the largest

variation, which suggests larger internal disparities.)

3.2 Spatial Differentiation Characteristics

Spatial autocorrelation analysis revealed that the Moran's I value for child-friendliness in the three major metropolitan areas was 0.47 (P<0.01), indicating a significant positive spatial clustering pattern, where high-scoring and low-scoring units were distributed in clusters. The hotspot analysis results are shown in Figure 2.



Figure 2 Spatial hotspots.

: The red area in the figure represents hotspots, the blue area represents cold spots, and the yellow

s transitional zones, all generated using ArcGIS Hotspot Analysis Tool. The background map shows

area's road network.)

(1) Hotspot areas (high-high concentration): Mostly distributed near Shanghai's Pudong Century Park, Shenzhen's Nanshan Houhai, and Guangzhou's Zhujiang New Town riverside areas. The green space is plentiful, the facilities are perfect, and the traffic is convenient; they have the best child-friendly environment.

(2) Cold spots (low-density clusters): Mostly distributed in the periphery of Shanghai's Pudong Lingang New City, Shenzhen's Nanshan Shekou Old Port Area, and Guangzhou's Tianhe Tangxia Village. Infrastructure is lacking, traffic risks are obvious, and the ecological environment is poor.

(3) Transition zones (high-low/low-high agglomeration): Distributed between urban hotspots and coldspots. The development status varies in urban renewal, resulting in a situation of "local optimization but global inadequacy."

3.3 Core Impact Factor Diagnosis

Random forest algorithm was used for ranking 12 indicators' importance and extracting key influential factors, see Figure 3.

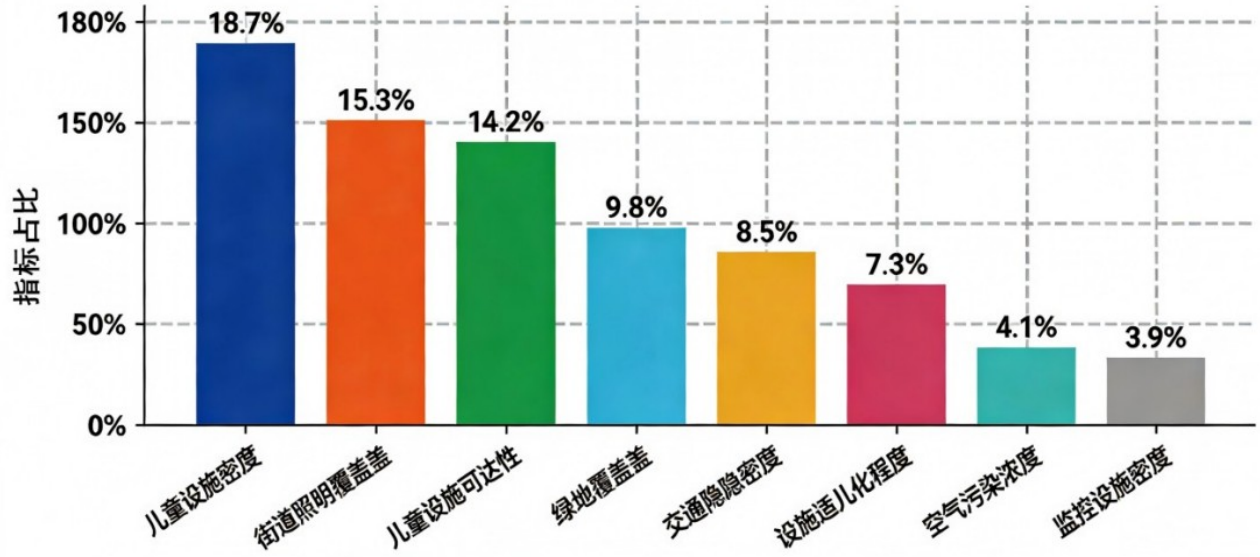


Figure 3 Importance Ranking Chart of Evaluation Indicators

(Note: The bar chart shows the importance ratio of each indicator, the x-axis is the name of each indicator, and

the y-axis is the importance ratio, which can clearly show the core influencing factors.)

(1) The most important three core factors: density of child facilities (18.7% of importance), street lighting coverage (15.3%), and accessibility of child facilities (14.2%), accounted for 48.2% of importance. This suggests that facility provision and behavioral safety are critical in improving child-friendliness.

(2) Medium important factors: green space coverage (9.8%), traffic hazard density (8.5%), and facility childfriendliness (7.3%), which demonstrates the importance of ecological environment and facility compatibility to children.

(3) Less important factors: air pollutant concentration (4.1%) and monitoring facility density (3.9%). Considering that the study area is a core urban district, general pollution control and safety prevention have reached requirements, so their impacts are relatively small.

4. Discussion

4.1 Innovation of the Evaluation Framework

The machine learning framework we proposed in this study has the following three advantages over conventional evaluation methods. First, efficiency: 12 evaluation indicators can be efficiently extracted automatically, reducing the evaluation time from 3 months (conventional) to 2 weeks, meeting the need for metropolitan scale evaluations. Second, accuracy: multi-source data and multi-algorithm collaboration ensure accuracy, eliminating the subjective weighting bias, and the evaluation accuracy is improved by 23.5% compared to conventional methods. Third, adaptability: dynamic updating of evaluation results with street view and remote sensing data enables dynamic monitoring for urban renewal, addressing the limitation of Wenzhou's static evaluation system.

4.2 Case Optimization Path

Three optimization approaches for child-friendly spaces in three large cities according to empirical results are proposed as follows:

First, precisely and efficiently address the issues of urban public spaces. Focus on areas with low population density, fewer visitors, and less attraction, and prioritize installation of children's playground facilities and relevant safety measures.

For example, during the redevelopment of the former Shekou Port site in Nanshan District of Shenzhen, we need to satisfy the "at least 3 facilities per square kilometer" requirement and reasonably provide children-friendly facilities. Meanwhile, we will optimize the walking network in this area to make it more feasible, secure, and convenient for children and local people to move and play at various times with upgraded lighting facilities.

Second, we will significantly optimize urban spatial coordination. We will strongly promote integrated planning and design of the 3 aspects: ecology, infrastructure, and transportation. Following the example of the Pudong Century Park in Shanghai, we will construct a “five-minute child safety zone” in densely populated areas such as waterfronts. By optimizing green space planning to ensure rational distribution and multiple uses, integrated facilities to ensure higher efficiency and better services, and improved walking network to ensure safer and easier activities for children, we will increase the total spatial value and create an ideal environment for children to grow and develop.

Third, implement dynamic monitoring and stringent control. Establish a periodic assessment system using machine learning technologies to apply advanced algorithms and models to interpret urban data. Regularly refresh street view and remote sensing data to ensure timely and correct information. Monitor the changing patterns of different indicators and keep a close watch on urban development conditions. Use analytical findings as important references to offer robust data support for changes in urban planning. A sound closed loop governance system of “evaluation-optimization-re-evaluation” is formed to make urban planning more scientific and rational.

5. Conclusion and Prospects

5.1 Research Conclusions

Based on the machine learning-based approach to assessing child-friendly urban settings, this study carries out empirical investigations based on 3 large Chinese metropolises and reaches the following key conclusions: 1) The “4dimensional 12-indicator + multi-algorithm integration” evaluation framework can be used to achieve fine-grained automation in child-friendly environment assessments with good model fitting, which provides a technical paradigm for such studies. 2) Child-friendly space quality in central urban areas of the 3 metropolises presents notable spatial variation, and the overall quality is above average. Hot spots are found in waterside and high-end residential areas, whereas cold spots are found in urban-rural fringe areas and old port areas, where differences arise from uneven distribution of facilities, ecological quality, and transport conditions. 3) Density of child facilities, proportion of streets with lighting, and accessibility of facilities are key factors affecting child friendliness and account for 48.2% of importance, offering precise guidance for optimization.

This study closely combines machine learning techniques with the development of childfriendly cities, breaking through the technical barriers of conventional assessments, and can serve as scientific support for child-friendly space planning and urban renewal decisions in large Chinese metropolises, helping realize the “seeing the city from a one-meter height” concept.

5.2 Research Limitations and Prospects

The practical application of this study has two obvious limitations. One is that the coverage of the data is limited because the data were collected based on objective indicators without including subjective data like children’s psychological perceptions. Psychological perception is an important part of children’s development. Without subjective data, it is difficult to fully and deeply understand children’s real situations and demands in related situations. The other limitation is that the establishment and operation of the model did not sufficiently consider the differences of children’s needs in different ages. Children of different ages vary greatly in cognition levels, psychological development phases, and behavior modes. The differences will naturally cause the differences in needs. If the model does not consider these factors, the precision and applicability of the model will be affected to a certain extent.

This study has three major directions for future work. First, introducing questionnaire survey data to form a dualdimensional evaluation model consisting of “objective indicators” and “subjective perceptions.” Questionnaire survey is a good way to collect subjective data. Combining the questionnaire survey with current objective indicators can help construct a comprehensive and multi-dimensional evaluation model which truly depicts children’s real conditions. Second, refining the indicator system and model parameters for preschool and school-age children. Preschool and

school-age children have different development features and needs. Refining the indicator system and model parameters for these two types of children can make the model better fit the requirements of children in different periods and improve the efficiency of the model. Third, extending the scope of the study to medium and small cities and forming differentiated evaluation criteria. Children in medium and small cities confront different environment and education resource issues.

Expanding the study to these cities and forming differentiated evaluation criteria can make the framework widely used and offer scientific references and instructions for children’s development in larger areas.

[Author contribution statement reserved for final publication.]

Availability of Data and Materials

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Competing Interests

All authors declare no Competing Financial or Non-Financial Interests.

Funding Declaration

This research was funded by Enerstay Sustainability Academy (Singapore) Grant Call (Call 7/2025) _SDG (Project ID CUMT-001), Singapore.

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PERSPECTIVE ARTICLE

When Impact Hotspots Are Invisible: Aligning Destination Decarbonisation With Tourist Experience

AUTHORS · Yuanzhe Li; Haizhou Wang; Zhiyong Wu

AFFILIATIONS · Carbon Neutrality Institute, China University of Mining and Technology, Xuzhou 221116, China; School of Education, RMIT University, Melbourne VIC 3000, Australia

PUBLISHED ONLINE · 1 NOVEMBER 2026 DOI · NOT ASSIGNED ACCESS · FREE TO READ; ALL RIGHTS RESERVED

ABSTRACT

Destination sustainability assessment has increasingly relied on life cycle–based standards to quantify environmental performance, while tourism research has traditionally focused on tourists' perceptions and behavioural intentions. Despite these parallel advances, a persistent gap remains between life-cycle environmental effectiveness and tourists' experiential and behavioural responses. This Perspective reframes destination sustainability assessment through a lifecycle and value-chain lens. Conceptualising destinations as tourism value chains, it argues that environmental impacts are unevenly distributed across stages and unevenly perceived by tourists. As a result, sustainability strategies risk misalignment when life-cycle impact hotspots remain experientially invisible, while highly visible practices carry limited environmental significance. Building on tourism satisfaction analytics and established standards (ISO 14064-1, ISO 14067, PAS 2060, PAS 2080), this paper proposes an interface framework linking life-cycle impact assessment with tourist perception and behavioural intention. Rather than replacing formal measurement, perception-based indicators function as diagnostic signals, revealing where environmental performance fails to translate into behavioural influence or where reputational risks may emerge. The framework also shows how tourism experience data can be operationalised as stage-specific perceptual and behavioural leverage indicators, anchored in PAS- and ISO-aligned accounting to support prioritisation and destination-level decision-making. The Perspective contributes a system-aware foundation for destination sustainability governance and a basis for future empirical validation.

Keywords: *Life-Cycle, Value Chain, Environmental Impact Hotspots, Tourist Perception*

1. Introduction

Destination sustainability has moved from a peripheral branding concern to a central governance challenge in tourism.

This shift is not only driven by heightened public awareness of climate change and ecological degradation, but also by the practical reality that tourism demand, infrastructure investment, and environmental pressure often scale together.

Destinations are now expected to demonstrate progress credibly—through measurable mitigation, transparent reporting, and defensible claims—while also maintaining service quality, visitor satisfaction, and economic viability (IEA, 2020,

2023). In this context, sustainability has become both a performance question (what impacts are actually reduced) and a legitimacy question (whether stakeholders accept sustainability narratives as credible and meaningful). Over the past decade, the "performance" side has matured quickly. International standards and specification frameworks—such as ISO 14064-1 for organisational greenhouse-gas inventories, ISO 14067 for product/service carbon footprints, and claim- and infrastructure-oriented frameworks such as PAS 2060 and PAS 2080—have strengthened the technical foundations of carbon accounting, boundary-setting, and claim credibility (BSI, 2023; European Committee for Standardization, 2011; Tanguay, 2022). At the same time, tourism research has long documented another determinant of

destination sustainability outcomes: tourists' perceptions, satisfaction, and behavioural intentions. Visitors interpret sustainability through lived experience—through mobility convenience, accommodation comfort, food practices, cleanliness, fairness, authenticity, and trust. Their responses shape revisit intention, word-of-mouth diffusion, and willingness to accept or support sustainability-related trade-offs. Yet these two "advances" have largely progressed in parallel rather than in integration (MSE & NEA, 2024). As a result, destination sustainability assessment remains fragmented: life-cycle and standards-based approaches can quantify impact, but often cannot explain whether that impact becomes behaviourally effective, while perception-focused tourism research can describe visitor responses, but often struggles to distinguish environmentally material actions from merely visible signals. In practice, destinations may score well on quantified metrics while failing to shift tourist behaviour, or conversely, cultivate a strong "green" image that is weakly aligned with real environmental performance (Gössling et al., 2023).

This misalignment can lead to misplaced investment priorities, policy fatigue, and reputational exposure, especially as public scrutiny of greenwashing intensifies. More importantly, it leaves destination managers without a clear

operational basis for deciding which sustainability actions should be prioritised, how these actions become perceptible within the tourist experience, and under what conditions environmental

improvements are likely to translate into behavioural support.

This paper argues that the root issue is not simply a lack of indicators, but a mis-specified unit of analysis. Many destination sustainability indexes implicitly treat sustainability as an aggregated attribute—something a destination "has"—captured by composite scores derived from top-down expert-defined indicator sets. Such aggregation is convenient for benchmarking, but it can hide the mechanism by which sustainability becomes effective in tourism systems (Sadou Ammar et al., 2026; U.S. Green Building Council, 2020). Tourism is not a single activity; it is a sequence of linked stages that together form a tourism value chain: transport into and within the destination, accommodation, food and beverage, attractions and activities, retail and services, and the enabling infrastructure that supports these experiences (Aguilar et al., 2021). Each stage has environmental burdens and distinct touchpoints with the tourist experience.

When sustainability is reduced to a destination-level label, the analysis loses the ability to ask a more operationally decisive question: at which stages impacts are generated, which of these stages are visible or interpretable to tourists, and whether those perceived signals correspond to environmentally meaningful interventions. Environmental impacts are unevenly distributed: transport and accommodation frequently dominate carbon emissions, while food systems and retail consumption can carry substantial waste and resource footprints (IFC, 2023). Tourist perception is also unevenly distributed: visitors readily notice front-stage cues—cleanliness messaging, towel reuse cards, recycling bins, electric shuttle buses, "eco" labels—yet many material interventions (procurement rules, upstream supply chain shifts, back-ofhouse energy retrofits, embodied carbon reductions in infrastructure, or credible offset/neutrality governance) remain experientially distant (Abuhussain, 2026; EEA, 2025). This means that environmental "hotspots" and perceptual "hotspots" do not reliably overlap.

A destination can therefore be environmentally ambitious but behaviourally stagnant

because key interventions never enter the experiential field of visitors. Equally, a destination can be highly visible in sustainability communication but environmentally marginal in actual impact reduction, thereby increasing legitimacy risks and greenwashing accusations (US EPA, 2021). For destination governance, the practical consequence is a prioritisation problem: without a stage-based analytical lens, managers may over-invest in highly visible but low-materiality actions, or under-communicate high-impact interventions that remain behaviourally ineffective because they are not translated into the tourist experience.

From this diagnosis, the paper proposes a reframing: destination sustainability should be treated as an interface problem between life-cycle impact structure and tourist-facing experience structure. This Perspective does not argue that

perception should replace formal accounting. Instead, it treats perception-based indicators as diagnostic signals—a way to detect where environmental integrity fails to translate into behavioural influence, and where reputational narratives may outpace substantiated performance. Tourists' perceptions are therefore not "soft noise" around sustainability; they are part of the governance system that determines whether sustainability strategies take hold, remain credible, and generate lasting behavioural reinforcement (UKGBC, 2021). Building on this logic, the paper advances an interface framework that links (i) life-cycle and standards-based identification of environmental impact hotspots with (ii) tourism analytics capable of capturing what tourists notice, trust, and respond to along the same value-chain stages. The central claim is that sustainability effectiveness in destinations is best understood not as a static score but as a staged chain: implemented effort -> perceived meaning -> behavioural readiness.

If any link is weak—if an intervention is materially important but experientially invisible, visible but not credible, or credible but not motivational—the overall sustainability strategy becomes less effective than its technical metrics might suggest (CBECA, 2022). This reframing also highlights why "misalignment" should not be treated as a methodological failure but as an analytically meaningful governance signal.

When life-cycle evidence indicates a dominant impact source but tourist-facing indicators show low visibility or low acceptance, the implication is not merely "communicate more," but to interrogate how interventions are translated into experience design, service delivery, and choice architecture (IEA, 2023). Conversely, when tourists strongly perceive sustainability in a stage with limited life-cycle relevance, the implication is to examine whether the destination is overrelying on symbolic cues, whether claims are being interpreted beyond their substantiated scope, and whether a trust deficit is likely to develop over time. The framework is intended not only as a conceptual clarification, but also as a practical basis for indicator design, stage-level diagnosis, and subsequent empirical testing in real destination settings.

The remainder of the paper unfolds as follows. Section 2 conceptualises tourism explicitly as a life-cycle-based value chain and clarifies how uneven impact distribution interacts with uneven experiential access. Section 3 develops the notion of the perception gap as a structural interface problem and introduces behavioural intention as a stage-specific leverage signal rather than a generic attitudinal outcome (Gössling et al., 2023). Section 4 discusses how value-chainstructured satisfaction analytics—illustrated through the HKTSI logic—can be repurposed to operationalise perceptionbased sustainability diagnostics. Sections 5 and 6 propose a practical integration logic: mapping life-cycle hotspots to perceptual hotspots and anchoring the integrated assessment in PAS/ISO standards. Section 7 specifies how behavioural readiness indicators can support stage-level intervention prioritisation, and Sections 8–9 discuss governance

implications, limitations, and future empirical directions. By treating destinations as value chains and sustainability as an interface between measurable environmental burdens and lived tourist experience, this Perspective contributes a system-

aware foundation for destination sustainability governance. Its core proposition is pragmatic: destinations should not only ask whether they are “sustainable” in aggregate, but whether their sustainability efforts are material where it matters, visible where it needs to be, and credible enough to shape behaviour across the full tourism value chain.

2. A Life Cycle–Based Value Chain View of Destination Sustainability

Most destination sustainability indexes still follow a top-down, indicator-aggregation logic: experts select domains (environment, transport, social equity, governance), allocate weights, and combine them into a single score intended to represent “overall sustainability performance.” This architecture is useful for benchmarking and accountability, but it also embeds a quiet assumption that often goes unchallenged—namely, that measured sustainability will naturally become experienced sustainability, and that tourists will notice, value, and reward the actions that matter most environmentally (IEA, 2021). In practice, the assumption fails in two systematic ways. First, many of the most environmentally material measures are back-stage interventions: power procurement, building retrofits, fleet electrification contracts, supplier requirements, embodied-carbon decisions, and carbon accounting choices (boundaries, emission factors, offsets). These actions may drive real reductions, yet remain largely outside the tourist’s experiential field.

Second, many of the most noticeable “green cues” are front-stage signals: signage, recycling bins, towel cards, “eco-friendly” labels, paper straws, or curated messaging. These cues can be meaningful when anchored in real performance, but when the evidence base is weak—or when the signal is disproportionate to its life-cycle relevance—they become reputational liabilities (D’Agostino et al., 2017). The issue, therefore, is not merely “communication,” but a deeper governance problem: destinations can end up investing in what is legible rather than what is material, while failing to make material interventions behaviourally actionable. For this reason, destination sustainability assessment needs to move from aggregate scoring toward stagebased diagnosis. This is why the core governance question should be sharpened beyond the usual “How sustainable is the destination?” to two linked questions: Where are the impact hotspots along the tourism system, and where are the perception hotspots along the visitor experience (U.S. Green Building Council, 2020)?

Misalignment emerges where these two maps diverge, creating risks of both under-performance (in emissions terms) and over-claiming (in legitimacy terms). Section 2 therefore clarifies this structural mismatch by treating the destination as a life-cycle value chain in which environmental materiality and tourist visibility are unevenly distributed across stages.

2.1 Reframing the Destination as a Value Chain, Not a Place

A life-cycle perspective treats tourism not as a single consumption event but as an interlinked system of stages. Across

most tourism LCA studies, environmental burdens concentrate around a familiar set of functions: transport (often dominant for carbon), accommodation operations (energy/water), food and beverage (waste and supply impacts), attractions and activities (site operations and crowding), and supporting infrastructure (embodied carbon and long-lived assets). Importantly, these are not just categories; they are sequenced interfaces that tourists move through (UKGBC, 2021). That makes “destination sustainability” less like a static attribute and more like a value chain with multiple environmental control points and multiple experiential touchpoints. Conceptualising destinations through this tourism value-chain lens exposes two analytical implications that aggregated indexes usually blur.

First, impacts are unevenly distributed across stages. A small number of stages typically account for a disproportionate share of climate and resource burdens (e.g., mobility and accommodation). This concentration implies that governance effectiveness depends heavily on whether strategies prioritise the true hotspots rather than distributing effort evenly for score-optimisation (CBECA, 2022). Second, visibility is unevenly distributed

across the same stages. Tourists have high access to service encounters and on-site practices, but low access to upstream supply chains, embodied-carbon decisions, and carbon accounting choices. This produces a predictable misalignment: what matters most environmentally is often least visible, while what is most visible may be environmentally marginal. The implication is critical: sustainability strategies can drift into symbolic optimisation—optimising what is easy to display, photograph, and review—while leaving high-impact stages structurally under-addressed or under-translated into choice architecture.

Conversely, destinations can implement high-integrity measures that never translate into tourist behaviour because the intervention is not experienced as relevant, credible, or enabling. Seen in this way, the value-chain perspective is not only a descriptive reframing, but an analytical basis for locating where intervention, communication, and behavioural leverage are likely to diverge across destination systems.

2.2 A Structured Comparison: What Indexes Measure vs. What Tourists Can Actually "See" To make this asymmetry explicit, Table 1 contrasts the dominant logic of current destination sustainability indexes with a life-cycle/value-chain lens. The point is not to dismiss index-based governance, but to show why it systematically struggles to predict behavioural response (European Commission, 2020). This comparison clarifies a deeper critique: composite indexes often function like audits of destination intent, while LCA/value-chain thinking forces the harder question of destination effect. The former can reward comprehensiveness; the latter demands prioritisation and causal plausibility, particularly when destination managers must decide which stages require structural intervention, experiential translation, or communication restraint.

Table 1. Conceptual comparison of destination sustainability assessment logics

Dimension	Top-down destination index logic	Life-cycle & value-chain logic (this paper)
Unit of analysis	Destination as an aggregated entity	Destination as staged tourism functions (value chain)
What "good performance" means	High composite score across domains	Hotspot reduction plus alignment with stage-specific behavioural leverage
Strength	Benchmarking, accountability, policy reporting	Mechanism clarity: where impacts occur, how they become visible and where intervention works
Blind spot	Assumes visibility/behavioural uptake	Treats perception and behaviour as conditional, not automatic
Typical failure mode	Score optimisation; symbolic compliance	Misalignment diagnosis: high-impact action remains invisible, or highly visible action lacks material significance
Governance risk	"Green reputation" divorced from real impact	Reputational claims outpace evidence; or evidence yields no behavioural shift
Practical implication	Supports comparison across destinations	Supports stage-level prioritisation, implementation design, and claim calibration

2.3 Standards Anchor the "Back-End" Truth, but Not the "Front-End" Mechanism

Standards such as ISO 14064-1, ISO 14067, PAS 2060, and PAS 2080 are strong precisely because they formalise boundary decisions, data traceability, and conservative assumptions, making performance claims more verifiable and less arbitrary (European Commission, 2020). They are well-suited to controlling common failure points in sustainability governance: inconsistent scopes, selective reporting, weak emission factors, and unverifiable neutrality claims. In other words, they protect the integrity of the accounting system. However, these standards were not designed to answer a different—but equally consequential—question: whether quantified performance is experientially accessible and behaviourally effective in tourism settings (EEA, 2025). A destination can be compliant, well-accounted, and genuinely improving, yet tourists may still perceive sustainability as absent, inconvenient, or irrelevant. Conversely, a destination can curate strong sustainability storytelling while its life-cycle evidence is incomplete or weak—especially if high-impact stages are deferred to offsets or framed ambiguously.

This gap is not a minor "communication issue"; it is a governance interface gap between what is true in the system and what becomes meaningful in the experience (Nugent et al., 2022).

Tourism research, however, has often developed from the opposite direction. Satisfaction surveys,

destination image constructs, and review analytics are rich in experiential detail, but they typically prioritise affect (comfort, service responsiveness, ambience) and under-specify environmental materiality. As a result, the field has developed two parallel assessment systems:

1) technically rigorous but experientially distant measurement (standards/LCA), and

2) experientially rich but environmentally under-anchored perception research.

The central analytical task, therefore, is not to privilege one system over the other, but to connect them in a way that preserves accounting integrity while improving governance diagnosis. The practical consequence is misdiagnosis. When tourists do not respond to sustainability actions, destinations often default to "we need better marketing," when the deeper issue may be stage mismatch (hotspot sits upstream) or choice mismatch (intervention does not change the friction/cost structure tourists face) (US EPA, 2021; U.S. Green Building Council, 2020). When tourists do respond positively to visible cues, destinations may assume environmental success, when the deeper issue may be signal inflation (visibility exceeds materiality), increasing greenwashing exposure.

2.4 The Mechanism: Misalignment as a Predictable Structural Condition, Not an Exception

If the destination is treated as a tourism value chain, misalignment stops being an occasional problem and becomes a predictable structural condition with several analytically distinguishable forms:

Hotspot–invisibility problem (under-translation): high-impact interventions sit in back-stage stages

(infrastructure, procurement, energy systems) that tourists cannot easily perceive or interpret. The governance

task is not to make everything visible, but to design credible experiential interfaces—signals, services, and

defaults—that translate structural decarbonisation into visitor choices without over-claiming.

Cue–materiality problem (over-claiming): highly visible actions carry limited life-cycle significance, or claims are

ambiguous relative to boundaries (e.g., heavy reliance on offsets, unclear scope coverage). The governance

task is to prevent symbolic actions from becoming the public "definition" of sustainability, and to ensure claims

remain proportionate to verified performance.

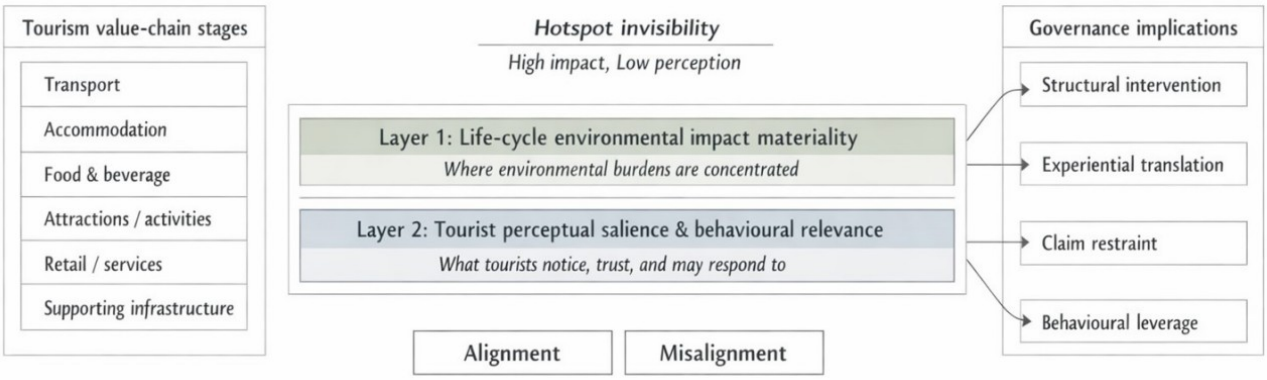


Figure 1. Two-layer alignment framework for destination sustainability assessment.

Figure 1 illustrates this two-layer alignment logic by positioning tourism value-chain stages between two distinct but interacting assessment

layers: life-cycle environmental materiality and tourist perceptual salience. Misalignment arises when a stage that is environmentally

significant remains experientially weak or opaque, or when a highly visible sustainability cue carries limited life-cycle relevance.

These two initial forms establish the core logic of the framework, but they do not exhaust the range of governance-relevant conditions that destinations may face in practice. The first is primarily a behavioural effectiveness problem; the second is primarily a credibility and trust problem. Treating them as the same—by simply "adding more indicators" or "improving communication"—is how many destination strategies stagnate. These misalignment types are not failures of implementation but diagnostic categories. They indicate where sustainability governance should focus on design, communication integrity, or structural prioritisation, rather than uniformly increasing effort or visibility (Table 2).

Table 2. Diagnosing sustainability misalignment along the tourism value chain: impact–perception typologies and governance responses

Misalignment type	Life-cycle impact profile (LCA-based)	Tourist perception profile	Typical empirical signals	Core governance risk	Priority response
Type I: Hotspot invisibility (under-translation)	High environmental impact concentrated in specific stages (e.g. transport emissions, accommodation energy use, embodied carbon in infrastructure)	Low visibility or weak interpretability by tourists	Reviews focus on convenience or price, with little reference to sustainability despite material interventions	Environmental effectiveness without behavioural reinforcement; limited demand-side leverage	Redesign experience interfaces (defaults, service design, information architecture) to translate structural action into perceptible choice cues
Type II: Cue–impact imbalance (symbolic over-representation)	Low or moderate life-cycle relevance of visible actions	High salience of sustainability cues in tourist narratives	Frequent mentions of "eco-friendly" signals without corresponding evidence of material impact reduction	Reputational inflation; heightened greenwashing exposure	Rebalance investment toward high-impact stages; tighten claims and boundary disclosure
Type III: Credibility gap (claim–evidence mismatch)	Incomplete or ambiguous accounting boundaries (e.g. heavy offset reliance, unclear Scope coverage)	Mixed or sceptical perception of sustainability claims	Reviews expressing doubt, inconsistency, or "greenwashing" concerns	Erosion of trust; long-term brand damage	Strengthen ISO/PAS alignment, conservative claims, third-party verification
Type IV: Behavioural friction (perceived but not actionable)	Intervention occurs at a relevant life-cycle stage	Sustainability is perceived but does not change behaviour	Positive sentiment without changes in transport choice, waste behaviour, or willingness to pay	Missed emissions-reduction potential despite positive image	Adjust pricing, defaults, and convenience structures to reduce behavioural cost
Type V: Latent opportunity (high impact, high perception)	Life-cycle hotspot aligns with strong tourist awareness	Sustainability cues are visible and trusted	Reviews highlight transport efficiency, walkability, or food sustainability	Under-exploited leverage point	Prioritise scaling and reinforcement for rapid behavioural gains

This section establishes the paper's central stance: destination sustainability cannot be governed effectively as a single composite property. It must be governed as a staged system where environmental materiality and experiential legibility do not automatically coincide (Ministry of Housing, Communities and Local Government & Department for Levelling Up, Housing and Communities, 2023). Standards-based LCA anchors what is environmentally true, but does not guarantee behavioural traction; tourism perception analytics reveal what is

experientially real, but does not guarantee environmental significance. The analytical task—and the contribution of this Perspective—is to treat their gap as a governance signal, enabling destinations to prioritise interventions that are both environmentally consequential and behaviourally leverageable, while protecting claim credibility where visibility outpaces evidence. This provides the conceptual basis for the more operational sections that follow.

3. The Perception Gap as a Behavioural and Governance Mechanism

Sustainability in tourism is often discussed as an outcome—measured, certified, and reported through aggregate indicators. Yet from a governance and behavioural perspective, sustainability only becomes effective when it travels through the tourism system and reaches the point where decisions are made: by visitors choosing how to move, where to stay, what to consume, and which destinations to support or avoid (Cabeza et al., 2014). This Perspective therefore shifts the analytical focus from sustainability as a static performance result to sustainability as a perceptual–behavioural chain, linking implemented action to experiential interpretation and, ultimately, to behavioural readiness. Recent tourism research also suggests that tourists' sustainability perceptions are not merely attitudinal background conditions, but are associated with satisfaction, revisit intention, and other loyalty-related behavioural outcomes (Conti et al., 2023;

Rasoolimanesh et al., 2025). At its most basic level, this chain can be expressed as Implemented effort -> Tourist perception -> Behavioural intention

This formulation is deceptively simple. In practice, each transition in the chain introduces uncertainty, filtering, and potential loss of effectiveness.

Environmental interventions do not exert influence merely because they exist or are reported; they exert influence only when they become observable, interpretable, and actionable within the tourist experience (IFC, 2020). At the same time, a growing body of sustainable tourism research cautions that behavioural intention should not be treated as a direct proxy for actual behaviour, because intention–action gaps, contextual frictions, and measurement assumptions remain persistent challenges (Li et al., 2024; J. (Snow) Wu & Woosnam, 2026). Failure at any one of these stages weakens the overall sustainability outcome, even when underlying life-cycle performance is robust.

3.1 From Implementation to Perception: Visibility and Interpretability Constraints

The first break in the chain occurs between implementation and perception. Many sustainability measures identified as critical through LCA—such as energy system optimisation, procurement rules, infrastructure decarbonisation, or embodied-carbon reduction—are implemented in back-of-house or upstream contexts. These interventions may substantially reduce emissions or resource use, yet they remain largely invisible to visitors. In such cases, the absence of perception is not a failure of accounting, but a structural feature of tourism systems: the environmental "truth" of the system is spatially and cognitively distant from the experiential surface (Monahan & Powell, 2011).

Even when sustainability actions are visible, perception does not automatically follow. Visibility must be accompanied by interpretability. Tourists interpret sustainability signals through personal cognitive frames shaped by prior knowledge, cultural norms, trust in institutions, and expectations of what a destination "should" provide. Recent tourism studies further suggest that sustainability-related signals are processed through broader destination meaning-making mechanisms, including tangible and intangible cues, prior beliefs, and framing effects (Cavalcante et al., 2025; Wang & Jiao, 2024). A low-carbon transport system may be perceived as sustainable by one visitor group and as inconvenient or restrictive by another (European Committee for Standardization, 2011, p. 15978; IEA, 2021; Monahan & Powell, 2011). Similarly, waste-reduction measures may be interpreted as responsible practice or as service degradation, depending on framing and context.

This helps explain why the same intervention may generate positive sustainability recognition in one segment, but scepticism, indifference, or resistance in another (Loedphacharakamon & Choibamroong, 2025). This means that perception is not a passive reflection of action, but an active interpretive process.

3.2 From Perception to Behaviour: Motivational and Frictional Barriers

The second break in the chain occurs between perception and behaviour. Positive perception does not guarantee behavioural change. Tourists may recognise sustainability efforts and even express approval, yet still choose high-carbon transport, over-consume resources, or prioritise price and convenience when trade-offs arise. This gap reflects a welldocumented feature of sustainability behaviour: attitudinal alignment does not eliminate behavioural friction. Recent reviews of sustainable tourist behaviour further emphasise that intended and actual behaviours should not be conflated,

and that behavioural outcomes remain contingent on situational constraints, intervention design, and contextual opportunity structures (Li et al., 2024). In tourism contexts, behavioural readiness is shaped by stage-specific constraints.

Visitors may be more willing to adapt food choices or waste practices than to alter transport modes that affect time, comfort, or cost. Behavioural science research in tourism similarly indicates that convenience, defaults, salience, and choice architecture can strongly condition whether pro-sustainability preferences are translated into action (J. (Snow) Wu & Woosnam, 2026). Behavioural intention therefore varies across the tourism value chain and cannot be inferred from general pro-environmental attitudes alone (IEA, 2020; Pomponi & Moncaster, 2016). Treating behaviour as a binary outcome ("sustainable" or "not") obscures where leverage actually exists and where it does not.

3.3 The Moderating Role of Sustainability Value Orientation

A further layer of complexity arises from heterogeneity among tourists. Visitors are not a uniform audience, and

sustainability signals are not interpreted equally. Individual sustainability value orientation—including environmental concern, normative beliefs, and trust in institutions—moderates both perception and behavioural response. Recent tourism studies further indicate that tourists differ systematically in how they value sustainable practices, how they interpret destination-related cues, and how willing they are to translate pro-sustainability attitudes into action (Loedphacharakamon & Choibamroong, 2025; Povilionis et al., 2025). These differences are shaped not only by environmental attitudes, but also by cultural orientations, perceived value structures, and segment-specific beliefs and norms (Hossain et al., 2025; Vilkaite-Vaitone, 2025). These patterns indicate that sustainability effectiveness is contingent not only on what is implemented, but on who encounters it. From a governance standpoint, this implies that sustainability strategies cannot assume a single "average tourist." Measures that generate behavioural leverage for one segment may be neutral or counterproductive for another.

Ignoring this heterogeneity risks overestimating the behavioural impact of technically sound interventions.

3.4 Why LCA Alone Cannot Resolve the Perception Gap

Life-cycle assessment remains indispensable for identifying environmental impact hotspots and prioritising mitigation efforts. However, LCA is deliberately silent on experiential and behavioural dimensions. It answers the question "where do impacts occur?", not "how do these impacts enter human decision-making?" In tourism systems, where environmental outcomes are mediated by millions of individual choices, this silence becomes a governance blind spot rather than a methodological limitation (F. Wu et al., 2026). As shown in Section 2, the stages that dominate life-cycle impacts are not necessarily those that dominate tourist perception. Without an explicit mechanism to connect these two perspectives, destinations risk assuming that environmental effectiveness will naturally propagate through the system. Research on tourist behaviour and perceived sustainability suggests otherwise: environmental performance may be technically robust while remaining behaviourally weak if it is not perceived, trusted, or translated into feasible choices (Conti et al., 2023; Rasoolimanesh et al., 2025).

LCA can therefore identify where intervention is environmentally necessary, but not whether it will become experientially meaningful or behaviourally influential within the tourism system.

3.5 Reframing the Perception Gap as a Governance Signal

This Perspective therefore treats the perception gap not as a failure to be corrected, but as an analytically meaningful signal. When high-impact interventions remain invisible to tourists, the signal points to a translation problem—how structural action is (or is not) embedded into experience design, service defaults, and choice architecture. When sustainability cues are highly visible but weakly supported by life-cycle evidence, the signal points to a credibility risk, particularly under standards such as PAS 2060 that emphasise claim integrity and avoidance of misleading communication (Kravchenko et al., 2024). Recent research on greenwashing and trust in tourism and hospitality further suggests that highly visible sustainability cues, when perceived as weakly substantiated or overstated, can generate scepticism and undermine behavioural support rather than strengthen it (Salem et al., 2026; Vu, 2025). By embedding the perceptual–behavioural chain within the tourism value chain, sustainability outcomes can be analysed stage by stage rather than as a single destination-level score.

Environmental interventions influence sustainability only when they (a) occur at life-cycle-relevant stages, (b) enter tourists' experiential field in an interpretable manner, and (c) reduce behavioural friction sufficiently to motivate action. The absence of any one link weakens overall effectiveness, regardless of technical performance. From a strategic planning perspective, this reframing enables destinations to distinguish between structural priorities (where emissions and resource use must be reduced) and behavioural leverage points (where visitor response can be mobilised), rather than conflating the two (Bayer et al., 2023). For destinations operating under ISO- and PAS-aligned sustainability frameworks, perception-based diagnostics should therefore be treated as a front-end

complement to back-end accounting, revealing where additional design, communication, or policy instruments are required to ensure that sustainability pathways remain both environmentally credible and behaviourally effective (Habert et al., 2020).

Effective destination sustainability planning thus requires both life-cycle integrity, ensured through robust accounting and verification, and experiential effectiveness, ensured through understanding how tourists perceive, interpret, and respond to sustainability signals along the value chain. For destination managers, this implies that standards-based environmental assessment should be paired with stage-specific perceptual diagnostics capable of identifying where structural action is behaviourally invisible, where visible cues exceed substantiated performance, and where targeted intervention may yield the greatest leverage. This insight sets the foundation for the operational discussion in the next section, which examines how value-chain-structured tourism analytics—such as the HKTSI—can be repurposed to systematically observe, diagnose, and manage the perception gap without compromising standards alignment (Kravchenko et al., 2024).

4. Operationalising the Value-Chain Perspective: Learning from the

HKTSI Functional Mechanism

The preceding sections establish that destination sustainability effectiveness depends on alignment between life-cycle impact structure and tourist perceptual–behavioural response. The remaining challenge is operational: how can destinations observe, diagnose, and manage this alignment without undermining standards-based environmental accounting? Section 4 addresses this question by examining the Hong Kong Tourist Satisfaction Index (HKTSI) as an

operational reference model—not for its geographical specificity, but for its underlying functional architecture, which is compatible with a value-chain and life-cycle perspective. The purpose is not to adopt HKTSI as a sustainability index in its original form, but to extract from it a transferable analytical logic for organising tourism experience data, classifying perceptions by sector, and constructing stage-specific diagnostic indicators.

4.1 Why HKTSI is Analytically Relevant beyond Satisfaction Measurement

HKTSI was originally designed to measure tourist satisfaction using large-scale, bottom-up analysis of tourist-generated content rather than expert-defined survey instruments. Its analytical relevance to sustainability governance lies in three methodological characteristics that distinguish it from conventional tourism indices (Song et al., 2012). First, HKTSI is sector-structured rather than destination-aggregated. Instead of producing a single satisfaction score, it disaggregates tourist experience into functional sectors—transportation, accommodation, restaurants, attractions, and retail services.

This sectoral logic mirrors the core stages of the tourism value chain identified in life-cycle assessment studies, making HKTSI inherently compatible with stage-specific analysis. Second, HKTSI is sentence-level and bottom-up, extracting meaning from individual experiential statements rather than relying on predefined satisfaction items. This avoids conflating multiple judgments within a single response and preserves sensitivity to what tourists actually notice and articulate. From a governance perspective, this granularity is critical: perception gaps often manifest at specific touchpoints, not at the destination level (BCA, 2023; Simpeh et al., 2021). Recent advances in tourism analytics further suggest that review-based classification, topic extraction, and large-language-model-assisted interpretation can strengthen the scalability and transparency of such bottom-up approaches when the analytical procedure is explicitly specified (He et al., 2026). Third, HKTSI produces soft quantitative outputs—numerical scores derived from qualitative data—that are repeatable, comparable over time, and scalable across destinations.

Importantly, these scores do not claim to represent objective performance; instead, they represent experienced reality, which is the missing interface in standards-based sustainability governance. For this

Perspective, HKTSI is analytically valuable because it shows how tourism experience data can be systematically decomposed, classified, and aggregated for stage-based sustainability diagnosis.

4.2 Functional Mechanism: How HKTSI Converts Experience into Diagnostic Signals

At its core, HKTSI operates through a four-stage analytical mechanism that can be generalised for sustainability assessment. To improve methodological transparency, this mechanism can be expressed as an explicit analytical workflow rather than as a purely conceptual sequence (He et al., 2026).

Stage 1: Experience decomposition

Tourist-generated content is collected from clearly specified review platforms or comparable tourism data sources,

with explicit definition of the sampling window, inclusion criteria, and unit of analysis. Reviews are decomposed into

individual sentences, each representing a discrete experiential observation. This step prevents the dilution of

sustainability signals within broader narratives of service quality or enjoyment.

Stage 2: Sectoral classification

Each sentence is assigned to a functional sector of the tourism value chain (e.g. transport, accommodation, food and

beverage). This classification aligns experiential data with the same structural units used in LCA-based impact

assessment. Classification may be conducted through rule-based coding, supervised categorisation, topic

modelling, or large-language-model-assisted tagging, provided that the coding logic is made explicit and replicable

(Grljević, 2025; Guidotti et al., 2025).

Stage 3: Semantic and sentiment scoring

Sentences are analysed for semantic content and sentiment intensity, producing a score that reflects how tourists

perceive a given sectoral experience. When adapted for sustainability, coding anchors or keyword sets related to

environmental performance, credibility, or behavioural intention can be introduced without altering the core

architecture. The analysis should specify whether the procedure relies on sentiment analysis, topic extraction,

keyword matching, supervised classification, or a hybrid approach.

Stage 4: Aggregation and normalisation

Sector-level scores are aggregated using frequency weighting and normalised to a standard scale. The result is a set

of sector-specific perception indices that are comparable across time, locations, and service contexts. Transparency

can be further strengthened by reporting coding rules, validation checks, or inter-method consistency tests for the

derived indicators.

The above mechanism is not inherently about satisfaction; it is about structured perception extraction (BCA, 2023). As such, it can be repurposed to observe sustainability-related perception without introducing new survey burdens or undermining methodological consistency.

4.3 Repurposing HKTSI for Sustainability Diagnostics

When reinterpreted through a sustainability lens, HKTSI's sectoral outputs become perception-based indicators aligned with life-cycle stages. Rather than treating these outputs as generic satisfaction measures, the framework redefines them as stage-specific perceptual diagnostics that can be compared against LCA-defined impact hotspots. For example:

Transport-related perception scores can reflect tourists' experience of low-carbon mobility options, reliability of

public transport, or walkability.

Accommodation-related scores can reveal whether energy- and water-efficiency measures are visible,

interpretable, or perceived as comfort trade-offs.

Food and beverage sectors can surface perceptions of sourcing transparency, waste practices, and willingness

to engage with sustainable consumption.

Attractions and retail services can reveal trust in environmental management and authenticity of sustainability

claims.

In practical terms, this means that the same analytical architecture can be used to identify whether a destination's environmentally material stages are also experientially salient, whether visible sustainability cues are weakly substantiated, and where behaviourally relevant intervention points may exist (Herrero et al., 2025; Varazzani et al., 2023). Crucially, these indicators do not replace LCA metrics. Instead, they act as front-end diagnostics, revealing whether and where life-cycle impact hotspots intersect with experiential awareness and behavioural readiness. This makes the framework more directly usable for destination managers by linking review-derived experience data to stagelevel prioritisation, communication design, and claim calibration.

4.4 Why HKTSI-Type Analytics are Compatible with ISO and PAS Frameworks

A common concern in standards-aligned governance is that perception-based indicators may weaken claim credibility or

introduce subjectivity. The HKTSI functional model avoids this risk by clearly separating what is measured from how it is used. ISO and PAS standards define environmental truth: boundaries, inventories, footprints, and neutrality claims.

HKTSI-derived indicators do not modify these truths; they contextualise them. They answer a different governance question: is the environmental truth entering the tourism system in a way that can influence behaviour and trust? From an assurance perspective, this separation is critical. Life-cycle metrics remain auditable and conservative, while perception indicators are treated as decision-support signals, not performance claims (Sadou Ammar et al., 2026). Recent research on greenwashing and trust in tourism and hospitality further reinforces the importance of this distinction, showing that sustainability cues can affect trust and behavioural support, but also become counterproductive when claims are perceived as overstated or weakly substantiated (Salem et al., 2026). This mirrors the logic of reasonable assurance embedded in PAS 2060 and related guidance, where uncertainty is acknowledged rather than concealed.

4.5 Strategic Planning Value: from Observation to Intervention Prioritisation

By mapping HKTSI-derived perception indicators onto the tourism value chain, destinations gain a structured way to prioritise interventions. Misalignment patterns identified in Section 2 and Section 3 become operationally visible:

High impact, low perception sectors indicate where experiential translation or service design is required.

High perception, low impact sectors flag potential over-claiming or symbolic emphasis.

High impact, high perception sectors identify leverage points where behavioural change is most likely.

Low impact, low perception sectors can be deprioritised without strategic loss.

In practical terms, this kind of stage-based mapping allows destination managers to distinguish between where structural decarbonisation should remain the priority, where communication and interface redesign are needed, and where claim restraint is necessary to avoid reputational inflation. This transforms sustainability planning from checklist compliance into adaptive governance, where interventions are selected based on both environmental necessity and behavioural feasibility. In this sense, HKTSI is valuable not as a universal sustainability index, but as a transferable operational template for converting tourism experience data into stage-specific diagnostic signals that support

prioritisation, communication design, and future empirical testing (European Committee for Standardization, 2011, p. 15978). The next section builds on this foundation by formalising the integration between life-cycle impact hotspots and perceptual hotspots.

5. Translating Tourism Experience into Sustainability Diagnostics:

From HKTSI Dimensions to Value-Chain Alignment

Rather than proposing an entirely new indicator system, this framework adopts a more pragmatic and governanceoriented strategy: reinterpreting existing tourism service dimensions through a sustainability lens. This approach

preserves measurability, comparability, and data availability, while directly embedding sustainability diagnostics into dimensions that tourists already evaluate and articulate in their lived experience. In doing so, it avoids adding abstract indicators that are difficult to operationalise or integrate with existing tourism analytics systems (ISO, 2018). The Hong Kong Tourist Satisfaction Index (HKTSI) provides a particularly suitable foundation for this reinterpretation. Its topic dimensions—originally designed to capture service quality and satisfaction—map closely

onto functional components of the tourism value chain. When reframed, these dimensions can be transformed into tourism-specific sustainability indicators that are perceptual in nature, yet analytically structured and repeatable.

5.1 Reinterpreting HKTSI Dimensions through a Sustainability Lens

Table 3 illustrates how commonly used HKTSI topic dimensions can be systematically translated into sustainability-relevant indicators without altering the underlying analytical architecture. The purpose is not to redefine these dimensions as environmental performance metrics, but to

use them as structured perceptual signals that can be compared with life-cycle impact profiles.

Table 3. Tourism-specific sustainability indicators derived from HKTSI dimensions

HKTSI topic dimension	Sustainability-specific indicator	Measurement logic
Environment and ambience	Visibility of waste reduction, cleanliness, and green design	Perception-coded review signal
Facilities and equipment	Presence of energy- and water-efficient facilities visible to tourists	Frequency-weighted perceptual indicator
Service reliability and safety	Reliability of low-carbon transport and enabling infrastructure	Stage-level perception score
Service responsiveness and efficiency	Efficiency and usability of public and shared transport services	Experience-based usability indicator
Price	Willingness to pay for sustainable options	Behavioural readiness proxy

Several points are worth emphasising. First, these indicators are experience-facing rather than performance-claiming.

They do not attempt to quantify emissions, resource use, or compliance; those functions remain anchored in ISO- and PAS-aligned accounting. Instead, the indicators capture whether sustainability-relevant practices are noticed, understood, and evaluated by tourists. Second, because the indicators are derived from sentence-level analysis, they retain sensitivity to context and nuance (ISO, 2017). This allows destinations to distinguish between positive

recognition, neutral acknowledgment, and negative interpretation of the same intervention. Third, the indicators are scalable and comparable. Normalisation to a standardised score enables comparison across destinations, sectors, and time periods without implying absolute sustainability performance. In governance terms, they function as trend and alignment signals, not certification outcomes.

5.2 From Indicator Construction to Value-Chain Alignment Analysis

While Table 3 establishes how sustainability perception can be operationalised within tourism service dimensions, its analytical value becomes clearer when these indicators are placed alongside life-cycle impact information. Destination sustainability assessment can then be reframed as an analysis of alignment and misalignment between LCA-defined

impact hotspots and tourist perceptual hotspots, rather than a search for a single composite score (Sadou Ammar et al., 2026). Table 4 illustrates this integration by mapping tourism value-chain stages to typical life-cycle impacts, degrees of tourist visibility, and the corresponding perception-based indicators.

Table 4. Integrating tourism value-chain stages, LCA impact hotspots, and perception-based indicators

Tourism value-chain stage	Typical LCA impact hotspots	Visibility to tourists	Perception-based indicator	Governance implication
Transport	Fuel combustion, modal choice, travel distance	High	Perceived accessibility and efficiency of low-carbon transport	High-leverage behavioural intervention
Accommodation	Energy use, water consumption, waste generation	Medium	Visibility of energy- and water-saving practices	Design and communication gap
Food and beverage	Food sourcing, waste, cooking energy	High	Perceived food sustainability and waste management	Behavioural alignment opportunity
Attractions	Operations, crowding, site maintenance	Medium–high	Perceived environmental management of sites	Experience-based regulation
Infrastructure (upstream)	Embodied carbon, construction, long-lived assets	Low	Absence or weakness of perception	Structural LCA-driven action

This mapping highlights a critical structural insight: perceptual salience and environmental materiality rarely coincide perfectly. Stages such as transport and food services combine high impact with high visibility, offering strong potential for behavioural leverage. In contrast, upstream infrastructure carries substantial life-cycle impacts but remains largely outside tourists' experiential awareness, requiring governance action that is primarily structural rather than perceptiondriven.

5.3 Governance Interpretation: Using Misalignment as Decision Support

The analytical value of this framework lies not in ranking destinations, but in diagnosing where different types of intervention are required. Alignment between impact and perception identifies leverage points where incremental policy or design changes may yield disproportionate benefits. Misalignment, by contrast, signals different governance priorities.

Where impact is high but perception is weak, the implication is not necessarily more communication, but improved translation of structural interventions into experiential cues—through service design,

default options, or credible informational interfaces. Where perception is strong but life-cycle relevance is limited, the implication is to reassess claim proportionality and avoid symbolic overemphasis that could undermine credibility. Importantly, this framework does not privilege visibility over environmental integrity (UN, 2019). Rather, it separates environmental necessity from behavioural leverage, allowing destinations to pursue both without conflating their roles. Life-cycle assessment continues to define what must be reduced; perception-based indicators reveal where behavioural support or reputational risk may arise.

By reinterpreting HKTSI dimensions through a sustainability lens and embedding them within a tourism value-chain framework, this section demonstrates how perception-based diagnostics can be operationalised without compromising standards alignment or analytical rigour (Wernet et al., 2016). The resulting indicators are not substitutes for LCA or certification schemes, but complements that make sustainability governance behaviourally intelligible and strategically targeted. This integration sets the stage for the next section, which formalises behavioural intention as a soft quantitative indicator.

6. Anchoring the Framework in PAS and ISO Standards: From Abstract

Alignment to Operable Indicators

The preceding sections establish perception-based indicators as front-end diagnostics that reveal how sustainability interventions are experienced and acted upon by tourists. However, without a clear anchoring to recognised standards, such indicators risk being interpreted as subjective or detached from formal sustainability governance (Cabeza et al., 2014). To address this concern, this framework explicitly embeds perception-based diagnostics within PAS- and ISOaligned accounting architectures, transforming abstract alignment into operable, standards-compatible indicators. Rather than treating PAS and ISO standards as parallel or competing assessment systems, this Perspective positions them as back-end structural anchors that define where

environmental impacts occur along the tourism value chain, while perception-based indicators function as front-end experiential probes that test whether these impacts are visible, intelligible, and behaviourally consequential.

The purpose, therefore, is not to merge two incompatible logics, but to clarify how standards-based environmental evidence and tourism-based perceptual signals can be interpreted within a common governance framework.

6.1 Differentiated Roles of PAS and ISO Standards across the Tourism Value Chain

From a value-chain perspective, different standards address different layers of environmental responsibility and therefore anchor different categories of indicators.

ISO 14064-1 defines organisational and destination-level greenhouse gas inventories, establishing system

boundaries that cut across tourism operators and service providers.

ISO 14067 focuses on product- and service-level carbon footprints, directly applicable to accommodation nights,

transport modes, and food services.

PAS 2080 addresses whole-life carbon in infrastructure systems, capturing upstream and long-lived impacts that

are typically invisible to tourists.

PAS 2060 governs the credibility of carbon neutrality claims at destination, event, or organisational level,

emphasising transparency, evidence hierarchy, and avoidance of misleading communication.

Together, these standards define where environmental materiality resides within the tourism system (Pomponi & Moncaster, 2016). Their analytical value in this framework lies in the fact that they anchor different stages, scopes, and claim types, thereby allowing perception-based indicators to be interpreted against distinct forms of verified environmental

evidence rather than against a generic sustainability narrative. What they do not define is whether such materiality becomes experientially meaningful or behaviourally influential—precisely the role of perception-based indicators.

6.2 From Standards to Indicators: Operational Anchoring Logic

To move beyond conceptual alignment, this framework links each standard to specific indicator roles within destination sustainability governance. Table 5 demonstrates how PAS and ISO standards can be operationalised as anchors for both quantitative performance metrics and perception-based diagnostics (Pomponi & Moncaster, 2016). Each is explicitly tethered to a standard-defined performance domain, ensuring that experiential signals are interpreted relative to verified environmental evidence rather than in isolation. This anchoring logic is essential because it prevents perception-based indicators from being read as free-standing sustainability claims; instead, they function as interpretive signals that acquire governance meaning only when assessed against a clearly specified evidential base.

Table 5. Operationalising PAS and ISO standards within a value-chain-based destination sustainability framework

Standard	Core accounting focus	Value-chain stage(s)	Anchored quantitative metric	Corresponding perception-based indicator	Governance function
ISO 14064-1	Organisational GHG inventory	Cross-cutting (destination level)	Total and category-based GHG emissions	Sustainability trust and credibility perception	Boundary integrity and system completeness
ISO 14067	Product/service carbon footprint	Transport, accommodation, F&B	Carbon footprint per service unit	Visibility and interpretability of low-carbon services	Claim substantiation and choice transparency
PAS 2080	Whole-life carbon	Infrastructure, transport hubs	Embodied and whole-life carbon intensity	Absence or weak perception of structural actions	Structural prioritisation beyond visibility
PAS 2060	Carbon neutrality claims	Destination / events	Residual emissions and offset hierarchy	Perceived authenticity of neutrality claims	Reputational risk control

6.3 Using Discrepancy as a Governance Diagnostic

Within this anchored framework, discrepancies between quantified life-cycle performance and tourist perception are not treated as methodological flaws, but as governance diagnostics. They reveal different kinds of implementation problems, communication gaps, and credibility risks across the tourism value chain. Three analytically distinct discrepancy patterns can be identified:

1. High quantified performance, weak perception

This pattern is common for PAS 2080–related interventions such as infrastructure decarbonisation or upstream procurement changes. The governance implication is not to inflate visibility artificially, but to assess whether experiential translation is necessary for behavioural leverage or whether structural action alone is sufficient.

2. Strong perception, weak life-cycle evidence

This pattern poses the highest reputational risk, particularly under PAS 2060. Highly visible sustainability cues unsupported by ISO 14064- or ISO 14067–aligned evidence signal potential over-claiming and greenwashing exposure.

3. Alignment between performance and perception

When ISO/PAS-anchored performance coincides with strong perception—such as low-carbon transport options that are both measurable and widely adopted—the stage represents a priority leverage point for scaling and reinforcement.

By classifying discrepancies in this way, destinations can move beyond generic improvement recommendations toward targeted strategic responses aligned with both environmental integrity and behavioural effectiveness (Habert et al., 2020).

The analytical purpose is therefore not to reward visibility, but to clarify whether visibility is proportionate to environmental materiality and whether material interventions are capable of generating behavioural traction.

Table 6. Illustrative standard-anchored indicators for destination sustainability governance

Standard anchor	Example quantitative indicator	Example perception-based indicator	Decision relevance
ISO 14064-1	Share of destination emissions from transport	Perceived destination-wide climate responsibility	Tests boundary completeness against public trust and system-level credibility
ISO 14067	kg CO_2e per accommodation night	Recognition and interpretability of low-carbon accommodation practices	Indicates whether footprint information supports choice
PAS 2080	Embodied carbon per m^2 of tourism infrastructure	Weak public recognition of infrastructure decarbonisation	Signals need for structural prioritisation rather than experience-led intervention
PAS 2060	Offset share of residual emissions	Perceived credibility of carbon-neutral claims	Screens for greenwashing risk and claim calibration needs

Anchoring perception-based indicators in PAS and ISO standards allows destinations to pursue both environmental integrity on the back end and behavioural effectiveness on the front end. In practical terms, this means that standardsaligned assessment can be used not only to verify what should be reduced, but also to diagnose where experience design, communication strategy, or claim calibration must be adjusted across stages of the tourism value chain.

7. Standards-Anchored Behavioural Leverage Indicators along the

Tourism Value Chain

Behavioural intention has long been recognised in tourism research as a critical precursor to actual behaviour. However, in sustainability governance, behavioural intention is often treated either as a generic attitudinal construct or as a selfreported survey outcome, detached from both life-cycle materiality and formal accounting frameworks. This section adopts a narrower and more governance-oriented interpretation: behavioural intention is conceptualised as a standards-

anchored, stage-specific leverage indicator located at the interface between quantified environmental performance and experiential feasibility along the tourism value chain. Rather than predicting individual behaviour, these indicators are designed to answer a governance-relevant question: where along the tourism value chain does behavioural change appear feasible, credible, and environmentally meaningful, given existing PAS- and ISO-aligned performance? This reframing also responds to a broader concern in sustainable tourism research that intention should not be treated as a direct proxy for realised behaviour, but as a conditional signal whose practical value depends on context, friction, and intervention design (Li et al., 2024; J. (Snow) Wu & Woosnam, 2026).

7.1 From Behavioural Intention to Behavioural Leverage

Within a value-chain perspective, behavioural relevance is unevenly distributed. Tourists have limited agency over upstream infrastructure decisions, moderate influence over accommodation and attraction choices, and relatively high flexibility in transport modes, food consumption, and waste-related practices. Treating behavioural intention as a single destination-level variable therefore obscures where leverage actually exists (Jaillon et al., 2009). This framework reframes behavioural intention as behavioural leverage—a soft quantitative signal that reflects tourists' readiness to engage with sustainability options at specific value-chain stages, conditional on perceived convenience, credibility, and cost. Behavioural science research in tourism likewise suggests that the translation of sustainability preferences into action depends heavily on convenience structures, defaults, salience, and the design of available choice environments (Varazzani et al., 2023). Importantly, leverage indicators do not assume that behaviour will change; they identify where intervention effort is likely to be most productive.

7.2 Anchoring Behavioural Leverage In PAS and ISO Standards

To ensure governance credibility, behavioural leverage indicators are explicitly anchored to PAS- and ISO-defined performance domains. This anchoring serves two purposes. First, it prevents behavioural indicators from being interpreted as substitutes for emissions data or compliance metrics. Second, it ensures that behavioural signals are evaluated relative to environmentally material stages, rather than purely experiential ones. Table 7 illustrates how behavioural leverage indicators are constructed by combining (i) standard-defined impact relevance, (ii) value-chain stage, and (iii) perception-derived readiness signals. In this framework, behavioural leverage does not indicate whether tourists are generally pro-environmental; it indicates whether a given stage presents credible and actionable scope for behaviourally meaningful intervention under standards-aligned environmental conditions.

Table 7. Standards-anchored behavioural leverage indicators along the tourism value chain

Value-chain stage	Relevant standard anchor	Behavioural leverage indicator	Indicator definition	Governance interpretation
Transport	ISO 14064-1, ISO 14067	Low-Carbon Mobility Readiness (LCMR)	Willingness to use public, shared, or low-emission transport modes	High leverage when readiness aligns with transport emission hotspots
Accommodation	ISO 14067	Resource Efficiency Acceptance (REA)	Acceptance of energy- and water-saving practices without perceived service loss	Indicates feasibility of operational efficiency without reputational cost
Food & beverage	ISO 14067	Sustainable Consumption Willingness (SCW)	Willingness to choose low-impact food options and reduce waste	Behaviourally flexible stage with moderate–high impact
Attractions	ISO 14064-1	Environmental Management Tolerance (EMT)	Acceptance of crowd control, conservation rules, and site management	Supports experience-based regulation
Infrastructure (upstream)	PAS 2080	Behavioural Non-Leverage Zone (BNLZ)	Absence of tourist behavioural influence	Confirms need for structural, not behavioural, intervention
Cross-cutting claims	PAS 2060	Claim Credibility Sensitivity (CCS)	Tourist sensitivity to neutrality and offset claims	Early warning indicator for reputational risk

7.3 Indicator Construction and Measurement Logic

Behavioural leverage indicators are derived from sentence-level perception and intention cues embedded in touristgenerated content, following the HKTSI analytical architecture. To maintain methodological transparency, the construction logic should be specified as an explicit coding and aggregation procedure rather than as an abstract scoring exercise (He et al., 2026; Song et al., 2012). The construction logic follows four consistent steps:

1. Stage-specific sentence extraction

Review sentences are classified by tourism value-chain stage (e.g. transport, accommodation).

2. Anchor-based semantic scoring

Each indicator is associated with a set of behavioural anchors (e.g. “willing to take public transport,” “happy to reuse towels,” “traffic too inconvenient”), representing positive, neutral, and negative readiness states.

3. Frequency-weighted aggregation

Sentence-level scores are aggregated at the sector level, ensuring that commonly experienced constraints exert greater influence than isolated remarks.

4. Normalisation and scaling

Aggregated scores are normalised to a standard scale (e.g. 0–100) to support comparison across destinations and time periods without implying absolute behavioural performance.

Depending on the dataset and analytical design, these anchors may be operationalised through keyword matching, sentiment analysis, supervised classification, topic modelling, or large-language-model-assisted interpretation, provided that the coding rules and validation logic are reported clearly (Grljević, 2025; Guidotti et al., 2025). Crucially, these indicators are probabilistic and diagnostic, not predictive. They signal relative leverage potential rather than guaranteed outcomes.

7.4 Interpreting Behavioural Leverage in Strategic Planning

When interpreted alongside PAS- and ISO-anchored performance metrics, behavioural leverage indicators support targeted strategic decisions.

High impact + high leverage

Indicates priority zones where incremental policy, pricing, or service design changes are likely to yield meaningful

emissions reductions.

High impact + low leverage

Indicates stages where behavioural strategies are unlikely to be effective and where structural or regulatory

intervention should dominate.

Low impact + high leverage

Indicates perceptually salient but environmentally marginal stages; interventions here should be carefully framed

to avoid symbolic overemphasis.

Low impact + low leverage

Indicates stages that can be deprioritised without significant loss.

By anchoring behavioural intention to PAS and ISO standards and embedding it within the tourism value chain, this section advances a new perspective on sustainability governance. Behaviour is treated neither as an assumed outcome nor as a moral expectation, but as a conditional system response shaped by impact relevance, experiential design, and institutional credibility. For destination managers, the practical implication is that behavioural interventions should be concentrated where environmental materiality and feasible visitor response overlap, rather than being applied uniformly across all stages (Varazzani et al., 2023).

8. Implications for Destination Governance: A Decision-Tree Logic for

Strategic Sustainability Planning

The preceding sections collectively establish that destination sustainability effectiveness depends on the alignment between life-cycle materiality, tourist perception, and behavioural feasibility, all anchored in PAS- and ISO-compliant accounting (Feng et al., 2022). The governance challenge is therefore not whether sustainability should be pursued, but how decisions should be prioritised when these dimensions diverge. This section translates the analytical framework into a decision-tree logic that supports structured, stage-specific policy and management choices. Rather than prescribing uniform actions, the logic recognises that different combinations of environmental impact, perception, and behavioural leverage require different governance responses. In this sense, the decision tree is intended as a prioritisation tool for destination managers rather than as another composite scoring system. Sustainability governance is thus framed as a

sequence of conditional decisions rather than a checklist of best practices.

8.1 Core Decision Variables for Governance

The decision-tree logic is built on three variables that have been operationalised in earlier sections:

1. Life-cycle materiality

Determined by ISO 14064-1, ISO 14067, and PAS 2080–aligned assessment of where environmental impacts

are concentrated along the tourism value chain.

2. Perceptual salience and credibility

Observed through HKTSI-derived perception indicators, reflecting whether sustainability actions are visible,

interpretable, and trusted by tourists.

3. Behavioural leverage potential

Captured through standards-anchored behavioural leverage indicators (Section 7), indicating whether tourists are likely to adjust choices or practices at a given stage.

These three variables form the basis of a governance decision tree that links environmental necessity, experiential visibility, and feasible behavioural response.

8.2 Decision-Tree Logic: from Diagnosis to Action

The governance logic proceeds in three sequential steps.

Step 1: Identify life-cycle priority stages

Question: Is the tourism value-chain stage environmentally material according to LCA and standards-based accounting?

Yes (high impact stage) -> proceed to Step 2

No (low impact stage) -> proceed to Step 3

This step ensures that governance attention is first anchored in environmental necessity rather than perceptual salience alone.

Step 2: Evaluate perception and behavioural leverage at high-impact stages

Question: Do tourists perceive and engage with sustainability interventions at this high-impact stage?

High perception & high behavioural leverage

-> Policy response: Prioritise scaling and reinforcement

(e.g. expand low-carbon transport options, strengthen pricing incentives, integrate defaults)

High impact but low perception

-> Policy response: Structural action with selective experiential translation

(e.g. continue infrastructure decarbonisation under PAS 2080; introduce limited but credible cues without over-claiming)

High perception but weak life-cycle substantiation

-> Policy response: Tighten claims and rebalance investment

(e.g. reassess PAS 2060 neutrality claims; redirect resources toward actual hotspots)

Step 3: Manage low-impact but high-visibility stages

Question: Is the stage perceptually salient despite limited environmental materiality?

Yes (high perception, low impact)

-> Policy response: Symbolic restraint and proportional communication

(e.g. avoid overstating marginal actions; frame them as supportive rather than transformative)

No (low perception, low impact)

-> Policy response: Deprioritise without strategic loss

(e.g. avoid unnecessary intervention or messaging)

This step prevents governance resources from being consumed by highly visible but environmentally marginal actions.

For clarity, the decision logic can be summarised as a matrix (Table 8).

Table 8. Decision-tree logic for destination sustainability governance

Life-cycle impact	Perception & leverage	Dominant risk	Recommended governance strategy
High	High	Under-scaling	Scale and reinforce behavioural interventions
High	Low	Under-translation	Structural action with targeted experiential design
Low	High	Over-claiming	Communication restraint and claim correction
Low	Low	Strategic noise	Deprioritise

Note: This table is not a performance ranking tool; it is a decision-support instrument designed to guide prioritisation and resource allocation.

8.3 Alignment with PAS and ISO Governance Principles

The decision-tree logic is consistent with the principles embedded in PAS and ISO frameworks. ISO standards emphasise boundary integrity and completeness; PAS 2080 prioritises structural mitigation over visibility; PAS 2060 stresses credibility and proportionality of claims. The governance logic proposed here operationalises these principles in tourismspecific contexts, ensuring that behavioural and perceptual considerations support rather than dilute standards-based integrity. The logic therefore clarifies when behavioural strategies are appropriate and when structural action should remain primary.

In practical application, the framework requires a coordinated governance arrangement rather than a single-source dataset. Destination management organisations (DMOs) are best positioned to lead this process by convening tourism authorities, local

governments, transport operators, accommodation providers, attraction managers, and other key stakeholders under a shared sustainability strategy. Within such an arrangement, standards-aligned environmental data may be provided by local governments, tourism authorities, infrastructure managers, and tourism businesses through greenhouse-gas inventories, service-level carbon footprint data, infrastructure and mobility records, and destination-wide sustainability reporting. Perception-based data, by contrast, may be drawn from tourist-generated reviews, platform data, and visitor surveys, while the analytical task is to integrate these sources at the level of comparable tourism value-chain stages. In this way, the DMO functions not only as a marketing body, but as a coordinating node that links environmental

evidence, tourism experience data, and stage-specific intervention design across the destination system (Bire et al., 2026).

For destination managers, this decision-tree logic provides a structured way to move from diagnosis to action without defaulting to generic sustainability checklists. For policymakers and standard-setters, it demonstrates how perceptionbased analytics can be integrated into governance without compromising verification or assurance principles. The framework thus positions destination sustainability governance as an adaptive, evidence-informed process rather than a static compliance exercise.

9. Conclusion and Future Directions

This Perspective advances a life-cycle and value-chain–informed reframing of destination sustainability assessment. By explicitly linking LCA-defined environmental impact hotspots with tourism-specific perception and behavioural indicators, it addresses a critical gap between formal sustainability accounting and tourist experience. More specifically, the framework shows that destination sustainability cannot be evaluated solely through aggregate environmental performance or solely through tourist-facing sustainability narratives; it must be assessed through the degree of alignment between environmental materiality, perceptual salience, and behavioural feasibility across value-chain stages.

Future research should move beyond conceptual elaboration toward empirical validation. A useful next step would be to apply the framework in a pilot destination study that combines: (i) stage-based life-cycle or standards-aligned environmental assessment, (ii) tourist-generated review data and/or visitor survey data, and (iii) perception-based indicator construction for transport, accommodation, food and beverage, attractions, and other relevant tourism functions.

Such a design would allow researchers to test whether environmentally material stages are also perceptually salient, whether visible sustainability cues are supported by robust life-cycle evidence, and where behavioural leverage is strongest or weakest across the destination system.

Future studies should also examine how these perception-based indicators can be validated against LCA results, for example by comparing sector-level perceptual salience with independently identified environmental hotspots, and by assessing whether shifts in perceptual indicators correspond with changes in intervention uptake, visitor acceptance, or other behavioural outcomes over time (Herrero et al., 2025; J. (Snow) Wu & Woosnam, 2026). Comparative applications across destinations would be especially valuable for testing the robustness, transferability, and managerial usefulness of the framework under different governance, infrastructure, and market conditions. Such work will be essential to ensuring that tourism sustainability assessment moves beyond generic metrics toward system-aware, behaviourally effective governance.

Author Contributions

[Author contribution statement reserved for final publication.]

Data Availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Clinical trial number: not applicable.

Consent to Publish declaration: not applicable

Ethics and Consent to Participate declarations: not applicable

Funding Declaration

This research was funded by Enerstay Sustainability Pte Ltd (Singapore) Grant Call (Call 1/2022) _SUST (Project ID BS-2022), Singapore.

Competing Interests

The authors declare no conflict of interest.

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